

Lithium Recovery Through Biohydrometallurgical Bioleaching

Submitted by:

Group E

Hekmat Kawas

Regan Huynh

Shivam Patel

Maxim Plotnik

Supervisor:

ChungHyuk Lee

Date:

March 16, 2026

Executive Summary

The global transition to electric vehicles demands a sustainable solution for the impending surge of spent lithium-ion batteries. This facility answers that call definitively. Designed to process exactly 15,000 tonnes of hazardous electronic waste annually, the plant transforms dangerous discarded cells into premium battery-grade raw materials. Biohydrometallurgy drives the entire operation by utilizing the metabolic activity of *Acidithiobacillus ferrooxidans*. This biological agent actively strips valuable metals from the black mass.

Traditional pyrometallurgical smelting requires massive energy inputs and generates toxic greenhouse gases. We completely abandoned those dangerous high-temperature furnaces. Operating securely at 27°C and standard atmospheric pressure, the biological circuit slashes energy demands entirely. Systems maintain absolute control over the highly acidic environment. They safely dissolve and extract copper, aluminum, manganese, cobalt, nickel, and lithium. A specialized precipitation and solvent extraction circuit then purifies these raw elements into high-value commercial products ready for market.

The financial projections are incredibly strong. Extracting lithium as lithium carbonate yields the highest economic return, ultimately determining the plant's overall profitability. Shenzhen provides the perfect operational base. Siting the facility in China integrates the operation directly into a massive battery manufacturing supply chain. It guarantees immediate physical proximity to abundant raw feedstock and eager industrial buyers.

This plant is definitively sustainable. Pairing biological extraction with automated batch control and passive gravity-driven flow systems secures critical supply chains. It strictly protects the local ecosystem. The design proves that next-generation battery recycling can be highly profitable and environmentally responsible.

Summary of Individual Contributions

Name of Student	List of Contributions	% *
Hekmat Kawas	- Mass and Energy Balance calculations - Equipment and Piping Sizing - Risk Assessment	25%
Regan Huynh	- Introduction - Drawing and Development of PFD along with P&ID - 3D Model Plant Development Creation - Process Safety & Loss Prevention Measures - Risk Assessment	25%
Shivam Patel	- Drawing and Development of PFD along with P&ID - Relevant Standards - Environmental Impact - Process Safety & Loss Prevention Measures - Risk Assessment	25%
Maxim Plotnik	- Mass and Energy Balance calculations - Economic Analysis - Equipment Sizing - Risk Assessment	25%

*: Percentage of total contributions made by all group members

The undersigned attest that the aforementioned contributions are accurate.

Name of Student	Signature
Hekmat Kawas	HK
Regan Huynh	RH
Shivam Patel	SP
Maxim Plotnik	MP

Table of Contents

Executive Summary.....	1
SUMMARY OF INDIVIDUAL CONTRIBUTIONS.....	2
LIST OF FIGURES.....	5
LIST OF TABLES.....	6
NOMENCLATURE.....	7
CHAPTER 1. INTRODUCTION.....	10
1.1 Technological Review of the selected process.....	10
1.2 Overview of Chemical Reactions.....	12
1.2.1 Biological Lixiviant Regeneration.....	12
1.2.2 Chemical Bioleaching (Batch Extraction).....	12
1.2.3 Copper Cementation.....	13
1.2.4 Aluminum and Manganese Precipitation.....	13
1.2.5 Cobalt Solvent Extraction.....	13
1.2.6 Nickel and Lithium Precipitation.....	14
CHAPTER 2. PROCESS FLOW DIAGRAM (PFD).....	15
2.1 Process Description.....	15
2.2 Process Flow Diagram.....	16
2.3 Mass and Energy Balances.....	20
CHAPTER 3. RELEVANT STANDARDS.....	28
3.1 Mechanical and Equipment Design Standards.....	28
3.2 Instrumentation and Process Control Standards.....	28
3.3 Plant Layout and Process Safety Standards.....	28
3.4 Environmental and Regulatory Standards.....	29
CHAPTER 4. DESIGN AND SIZING OF EQUIPMENT.....	30
4.1 Crusher.....	30
4.2 Conveyors.....	32
4.3 Thickeners.....	33
4.4 Filter Presses.....	34
4.5 Holding Tanks.....	37
4.6 Bio Reactors & Bioleaching.....	39
4.7 Brine Discharge Bath.....	41
4.8 Batch Reactors (Separators).....	41
4.9 Solvent Extractor.....	44
4.10 Pipes.....	45
4.11 Pumps.....	49
CHAPTER 5. PROCESS & INSTRUMENTATION DIAGRAM (P&ID).....	51
5.1 Instrumentation Philosophy.....	51
5.2 Piping and Instrumentation Diagram (P&ID).....	52

5.2.1 Zone 1: Pre-Treatment and Brine Discharge.....	53
5.2.2 Zone 2: Bioreactor Systems and Alarms.....	54
5.2.3 Zone 2: Bioleaching Systems and Alarms.....	55
5.2.4 Zone 3 Phase 1: Separation Systems and Alarms.....	56
5.2.5 Zone 3 Phase 2 & 3: Separation Systems and Alarms.....	57
5.3 Control Strategy Summary.....	58
Chapter 6. 3D Plant Layout.....	62
6.1 Layout Philosophy.....	62
6.2 3D Model.....	63
6.3 Detailed Zone Renderings.....	66
Chapter 7. Economic Evaluation.....	71
7.1.1 Capital Costs.....	71
7.1.2 Equipment Costs.....	71
7.1.3 Total Capital Investment.....	71
7.2 Profitability Analysis.....	72
CHAPTER 8. ENVIRONMENTAL IMPACT.....	73
8.1 Emissions and Effluents.....	73
8.2 Waste Generation and Disposal.....	75
8.3 Energy Efficiency.....	76
8.4 Life-cycle Considerations.....	78
8.5 Mitigation Strategies.....	79
8.6 Regulations.....	80
Chapter 9. Process Safety and Loss Prevention.....	82
9.1 Inherent Safety Considerations.....	82
9.2 Chemical Hazards.....	83
9.3 Hazard Identification (HAZOP Analysis).....	89
9.4.1 Pressure Relief Systems.....	92
9.4.2 Application in the LIB Bioleaching Plant.....	94
9.5 Fire and Explosion Prevention.....	95
9.6 Emergency Response Measures.....	96
Chapter 10. Risk Assessment.....	98
10.1 Risk Identification.....	98
10.2 Risk Analysis.....	99
10.3 Risk Mitigation.....	100
10.4 Final Recommendations to the Client.....	102
CHAPTER 11. CONCLUSIONS.....	104
REFERENCES.....	105

List of Figures

Figure 2.1.1. Left Half of the Final PFD.....	17
Figure 2.1.2. Right half of the Final PFD.....	18
Figure 4.1 Crusher (CR-101).....	30
Figure 4.2.1 Crusher conveyor (CO-101).....	32
Figure 4.2.2 Steel wool screw conveyor (CO-301).....	32
Figure 4.3.1 Bioreactor bacteria-thickener (TK-201) (Left).....	34
Figure 4.3.2 Separation phase 1 (TK-301) (Right).....	34
Figure 4.4.1 Filter press phase 1 Cu/Al/Mn extraction (FP-301 A/B).....	35
Figure 4.4.2 Filter press phase 2 Nickel extraction (FP-302).....	36
Figure 4.4.3 Filter press phase 3 Lithium extraction (FP-303).....	37
Figure 4.5.1 Brine solution Holding Vessel (V-100).....	37
Figure 4.5.2 Bioleached material Holding Vessel (V-202).....	38
Figure 4.5.3 Separations Phase 2 Holding Vessel (V-302) (Left).....	38
Figure 4.5.4 Separations Phase 3 Holding Vessel (V-303) (Right).....	38
Figure 4.6.1 Bioleaching Batch Reactor Vessel (V-201 A/B).....	39
Figure 4.6.2 Bioreactor Vessels (R-201 A/B).....	40
Figure 4.7.1 Brine Battery Discharge Bath (B-100).....	41
Figure 4.8.1 Separation Phase 1 (SP-301) (Left).....	42
Figure 4.8.2 Separation Phase 2 (SP-302) (Right).....	42
Figure 4.8.3 Separation Phase 3 (SP-303).....	43
Figure 4.9.1 Solvent Extraction Cobalt Stripping (SX-301).....	44
Figure 5.1.1. systems and alarms Legend for the P&ID.....	51
Figure 5.2.1. Zone 1 systems and alarms P&ID.....	53
Figure 5.2.2. Zone 2 Bioreactor systems and alarms P&ID.....	54
Figure 5.2.3. Zone 2 Bioleaching systems and alarms P&ID.....	55
Figure 5.2.4. Zone 3 phase 1 Separation systems and alarms P&ID.....	56
Figure 5.2.5. Zone 3 phase 2 & 3 Separation systems and alarms P&ID.....	57
Figure 6.1. overhead view of the plant.....	63
Figure 6.2. Image of relative scale for the plant.....	64
Figure 6.3. an isometric bird's-eye view of the entire facility from Zone 1 to Zone 3.....	65
Figure 6.4. isometric bird's-eye view, Zone 1 battery discharge & crushing.....	67
Figure 6.5. bird's-eye view, Zone 2 battery, Bacteria Bioleaching.....	68
Figure 6.6. bird's-eye view, Zone 3 battery, heavy metal separation recovery.....	69
Figure 6.7. Comprehensive overview of the integrated plant model.....	70

List of Tables

Table 2.1.1 PFD Equipment and respective codes.....	19
Table 2.3.1 Summary of the plant's output mass balances.....	22
Table 2.3.2 Summary of the plant's output Energy balances.....	23
Table 2.3.3 Summary of Compressor Power Calculations.....	25
Table 4.1: Crusher Parameter Summary.....	31
Table 4.10.1: Pipe Geometry and Pressure Drop.....	45
Table 4.10.2: Reynold's Number, Friction Factor, and Fittings.....	46
Table 4.11.1: Pump Hydra and NPSH.....	49
Table 5.3.1: Level Sensor Selection Guide.....	58
Table 5.3.2: Temperature Sensor Selection Guide.....	59
Table 5.3.3: Flow Sensor Selection Guide.....	59
Table 5.3.4: pH Sensor Selection Guide.....	60
Table 9.2.1. Health Effects of Hydrogen Fluoride resulting from acute and chronic exposure levels.....	84
Table 9.2.2. Health Effects of Sulfuric Acid resulting from acute and chronic exposure levels.....	85
Table 9.2.3. Health Effects of Organic Solvents resulting from acute and chronic exposure levels.....	86
Table 9.2.4. Health Effects of Heavy Metal Salts resulting from acute and chronic exposure levels.....	87
Table 9.2.5. Health Effects of Lithium Carbonate resulting from acute and chronic exposure levels.....	88
Table 9.3.1. HAZOP Analysis of Pre-Treatment Zone: Battery Crusher (CR-101).....	90
Table 9.3.2 HAZOP Analysis of Bioleaching Zone: Bioleach Reactors (V-201 A/B).....	91
Table 9.3.3. HAZOP Analysis of Recovery Zone: Solvent Extractors (SP-301/302).....	92
Table 9.4.1. Relief Valves Selection Guide (Crowl & Louvar, 2011).....	93
Table 10.2.1: Likelihood vs. Consequence Risk Matrix.....	99
Table 10.3.1: Comprehensive Risk Mitigation Strategy.....	101

Nomenclature

Symbol	Description
A	Cross-sectional or heat transfer area, m ²
C	Concentration of metal ions in solution, g/L or mol/L
c_p	Specific heat capacity, kJ/kg·K
D	Diameter (e.g., vessel, impeller, or pipe), m
f_r	Rolling resistance friction factor, dimensionless
g	Acceleration due to gravity, 9.81 m/s ²
H	Height of vessel or dynamic head for pumps, m
ΔH_{rxn}	Enthalpy of reaction, kJ/mol
ΔH_{vap}	Enthalpy of vaporization, kJ/kg
k	Reaction rate constant or mass transfer coefficient, units vary
L	Length (e.g., conveyor or pipe length), m
m	Mass, kg
$\dot{m}$	Mass flow rate, kg/h or kg/s
n	Empirical exponent or number of moles, units vary
N	Number of items (e.g., filter press plates, chambers), dimensionless
P	Pressure, Pa or bar
P_{lift}	Power required to lift material on conveyor, W or kW
P_{motor}	Total motor power supplied, W or kW
$P_{roll, belt}$	Power lost to belt rolling resistance, W or kW
$P_{roll, mat}$	Power lost to material rolling resistance, W or kW
P_{shaft}	Power transmitted through the shaft, W or kW

<i>Q</i>	Volumetric flow rate, m ³ /h or m ³ /s
<i>Q_{jacket}</i>	Heat duty required for reactor/separator cooling/heating jacket, kW
<i>R</i>	Reaction or precipitation rate, units vary
<i>S</i>	Supersaturation ratio, dimensionless
<i>t</i>	Time, s, min, or h
<i>T</i>	Temperature, °C or K
<i>U</i>	Overall heat transfer coefficient, kW/m ² ·K
<i>v</i>	Velocity (e.g., conveyor belt speed or fluid velocity), m/s
<i>V</i>	Volume, m ³ or L
<i>W</i>	Work or energy, J/kg or W
<i>X</i>	Conversion fraction or fractional recovery of metal, dimensionless
<i>Y</i>	Humidity ratio (mass of water vapour per mass of dry air), kg/kg
Greek Symbol	
<i>η (eta)</i>	Efficiency (e.g., pump, motor, or drivetrain efficiency), dimensionless
<i>μ (mu)</i>	Dynamic viscosity, Pa·s or kg/m·s
<i>ρ (rho)</i>	Density, kg/m ³
<i>τ (tau)</i>	Residence time, h or s
Acronym	
CAPEX	Capital Expenditure
EoL	End-of-Life
HEPA	High-Efficiency Particulate Air
ISD	Inherently Safer Design
LIB	Lithium-Ion Battery
LMTD	Log Mean Temperature Difference

NPSH	Net Positive Suction Head
OPEX	Operating Expenditure
ORP	Oxidation-Reduction Potential
P&ID	Piping and Instrumentation Diagram
PFD	Process Flow Diagram
PLS	Pregnant Leach Solution
SCADA	Supervisory Control and Data Acquisition
VOC	Volatile Organic Compound
PSM	Process Safety Management
XP	Explosion-Proof
ISD	Inherently Safer Design
PSE	Pressure Safety Elements
LOC	limiting oxygen concentration
PTW	Permit-to-Work
SCBA	self-contained breathing apparatus
PLC	Programmable Logic Controller
ESD	Emergency Shutdown
LAL	low-level alarms
LCV	Level Control Valves
LEV	local exhaust ventilation

Chapter 1. Introduction

1.1 Technological Review of the selected process

The recovery of valuable metals from end-of-life (EoL) lithium-ion batteries (LIBs) is traditionally dominated by two metallurgical pathways: pyrometallurgy and chemical hydrometallurgy. Pyrometallurgical processes utilize high-temperature smelting ($>1000^{\circ}\text{C}$) to recover transition metals (Co, Ni, Cu). However, this method suffers from excessive energy consumption, the generation of toxic off-gases, and the complete loss of critical elements, such as lithium and aluminum, to the slag phase (Velázquez-Martínez et al., 2019; Islam & Iyer-Raniga, 2022).

Conversely, conventional hydrometallurgy achieves high recovery rates for all metals but relies heavily on concentrated, hazardous inorganic acids (e.g., H_2SO_4 , HCl) and generates secondary toxic wastewater streams that require extensive downstream neutralization (Roy et al., 2021; Cerrillo-Gonzalez et al., 2024). Emerging alternatives, such as direct physical recycling, preserve the cathode's crystal structure; however, they are highly sensitive to impurities and require precise sorting of battery chemistries, which currently limit their industrial scalability (Roy et al., 2021).

To satisfy the 15,000 tonnes/year processing objective while strictly minimizing environmental impact and operating expenditures (OPEX), this plant utilizes a Biohydrometallurgical Bioleaching process.

Mechanism of the Selected Process

The selected biohydrometallurgical route relies on the metabolic activity of acidophilic bacteria. Instead of directly attacking the battery waste, the bacteria act as biological catalysts in a dedicated culture bioreactor, continuously oxidizing ferrous iron (Fe^{2+}) to ferric iron (Fe^{3+}). This biogenic Fe^{3+} serves as a highly reactive, self-regenerating lixiviant that chemically attacks the solid battery black mass (Garcia et al., 2025).

Process Parameters and Design Constraints

The primary advantage of this specific pathway is the ability to decouple the biological regeneration step from the chemical leaching step, allowing the primary Bioleaching Reactors to operate at exceptionally rapid kinetics without poisoning the microorganisms (Garcia et al., 2025).

The plant's design is governed by several fundamental operating parameters that ensure both process efficiency and microbial stability. The extraction process is conducted at a strict ambient temperature of 27°C. Because the bio-oxidation reaction is exothermic, a continuous cooling utility is required to manage the heat generated and protect the bacterial culture from thermal degradation.

Chemical stability is similarly regulated by a pH constraint, with the environment maintained at a highly acidic pH of 1.5. According to Garcia et al. (2025), this acidity is critical for preventing the precipitation of leached metals and ensuring the continued viability of the bacteria. Finally, the pulp density within the Reactor is limited to a 1% (w/v) solid-to-liquid ratio. This specific constraint is necessary to prevent heavy metal saturation, which could otherwise become toxic to the microbial lixiviant (Garcia et al., 2025).

Process Efficacy

By maintaining a 30-minute residence time within the Reactor under the aforementioned conditions, the biogenic Fe^{3+} lixiviant achieves highly efficient, selective dissolution. The process successfully transitions the EoL battery materials into a Pregnant Leach Solution (PLS) with the following proven recovery efficiencies: 98% Nickel, 97% Copper, 96% Aluminum, 92% Cobalt, 90% Lithium, and 80% Manganese (Garcia et al., 2025). This technological pathway ultimately eliminates the need for high-temperature smelting while achieving near-virgin purity targets required for a closed-loop circular economy (Roy et al., 2021; Velázquez-Martínez et al., 2019)

To ensure the safe and standardized execution of these operational parameters, the entirety of the plant's design is bound by rigorous international codes. The mechanical sizing of vessels and piping is governed by the American Society of Mechanical Engineers (ASME)

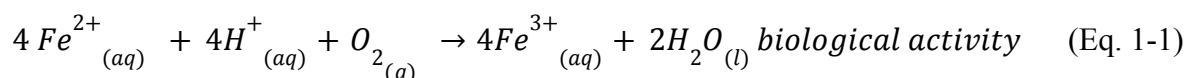
Boiler and Pressure Vessel codes, while process control follows the ANSI/ISA-S5-1 instrumentation standards. Furthermore, to mitigate the severe fire and explosion risks inherent to lithium-ion battery handling, the physical plant layout and pre-treatment spacing are strictly designed in accordance with established occupational safety and hazard distance regulations.

1.2 Overview of Chemical Reactions

The biohydrometallurgical recycling of spent lithium-ion batteries proceeds via a consecutive mechanism that incorporates biological lixiviant regeneration, reductive and acidic chemical leaching in batch reactors, and a sequential hydrometallurgical separation and precipitation train. The overall series of reactions is described below.

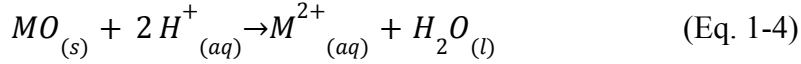
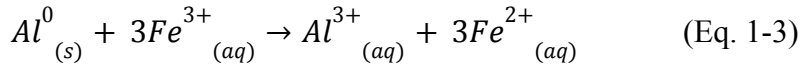
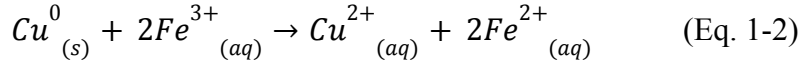
1.2.1 Biological Lixiviant Regeneration

The first critical step relies on the metabolic activity of *Acidithiobacillus ferrooxidans* to continuously oxidize ferrous iron (Fe^{2+}) into ferric iron (Fe^{3+}), which serves as the primary leaching agent. This biological oxidation requires an acidic environment, with a pH of 1.5 and continuous aeration. It is an exothermic reaction that requires strict temperature control to maintain the 27°C ambient temperature optimum (Garcia et al., 2025). The biological regeneration is expressed by the following chemical equation:



1.2.2 Chemical Bioleaching (Batch Extraction)

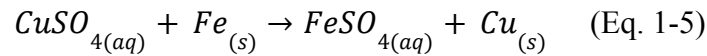
Inside the primary Batch Bioreactors, the biogenic Fe^{3+} acts as a strong oxidant to rapidly dissolve elemental metals (such as copper and aluminum fragments) from the suspended black mass at a 1% pulp density. This redox reaction simultaneously regenerates Fe^{2+} (Garcia et al., 2025). The regenerated Fe^{2+} subsequently reduces high-valent transition-metal oxides in the cathode materials (e.g., Co^{3+} , Ni^{3+} , Mn^{4+}) to more soluble divalent states. Concurrently, the metal oxides undergo proton-attack dissolution driven by the biogenically maintained sulfuric acid (Akbar et al., 2025; Roy et al., 2021):



(where M = Co, Ni, Mn, Li)

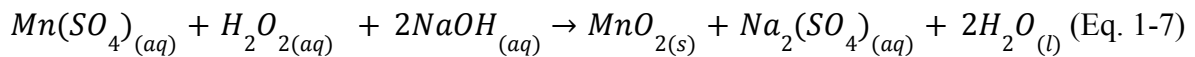
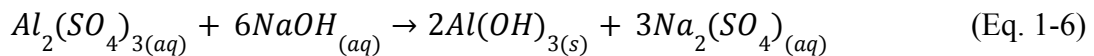
1.2.3 Copper Cementation

Following the batch extraction, the initial separation stage removes copper from the Pregnant Leach Solution (PLS) via an electrochemical redox reaction (cementation) using sacrificial scrap iron like steel wool (Fe^0) (Cerrillo-Gonzalez et al., 2024):



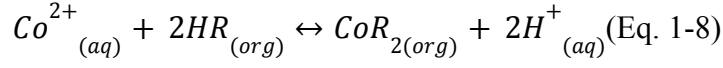
1.2.4 Aluminum and Manganese Precipitation

Aluminum is selectively precipitated as a solid hydroxide by raising the pH of the solution to 4.5–5.5 using sodium hydroxide. Following this, manganese is precipitated as insoluble manganese dioxide via chemical oxidation using hydrogen peroxide (H_2O_2) in an alkaline environment (Cerrillo-Gonzalez et al., 2024):



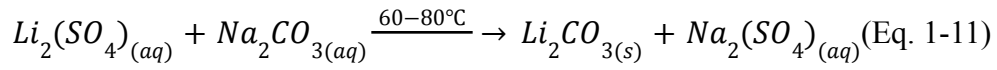
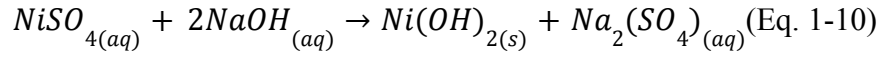
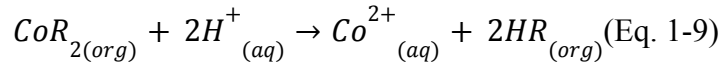
1.2.5 Cobalt Solvent Extraction

Cobalt is separated from the remaining nickel and lithium via liquid-liquid solvent extraction (SX) using the organic extractant Cyanex-272 (represented below as HR). The reversible ion-exchange reaction selectively pulls cobalt into the organic phase (Akbar et al., 2025):



1.2.6 Nickel and Lithium Precipitation

The aqueous raffinate from the SX circuit is further treated with sodium hydroxide to raise the pH above 8, precipitating nickel hydroxide. Finally, lithium carbonate, which is the primary target product, is precipitated from the remaining solution using sodium carbonate at elevated temperatures (60–80°C) to exploit its inverse solubility (Cerrillo-Gonzalez et al., 2024):



Chapter 2. Process Flow Diagram (PFD)

2.1 Process Description

The facility recycles 15,000 tonnes of spent lithium-ion batteries annually. The physical destruction of these high-energy cells presents a lethal fire risk. Operators plunge the raw batteries directly into a 10% sodium chloride brine bath (Torabian et al., 2022). This submersion lasts for 24 hours. A complete electrical discharge occurs safely within this conductive liquid. Mechanical shredding follows immediately. A heavy-duty twin-shaft macerator crushes the batteries while they remain fully submerged in brine. Wet crushing eliminates the threat of thermal runaway completely. Toxic hydrogen fluoride gases stay trapped in the liquid rather than escaping into the ambient air (Thompson et al., 2020). Plastic casings and separators float to the surface. Workers skim these light materials away, leaving behind a dense, highly concentrated black mass sludge.

The chemical attack relies on living organisms. A dedicated upstream bioreactor cultivates a massive colony of *Acidithiobacillus Ferrooxidans* bacteria. These acidophilic microbes survive by consuming ferrous iron. They oxidize ferrous sulphate into highly reactive ferric iron (Fe^{3+}) within a 6K nutrient medium (Garcia et al., 2025).

The core extraction shifts to a strictly controlled batch operation. Transfer pumps load a discrete volume of the black mass sludge into a massive rubber-lined batch reactor. The biological lixiviant floods into the vessel immediately after. The highly oxidative ferric iron aggressively attacks the solid battery particles. Copper, nickel, cobalt, aluminum, manganese, and lithium dissolve entirely into the aqueous phase. Bacterial metabolism produces large amounts of energy. The reaction throws off roughly 60 kW of exothermic heat. The microbes will die if the temperature spikes. A chilled water jacket surrounds the batch reactor to constantly strip away this heat, holding the internal environment at a strict 27°C (BRGM, 2021). Automated dosing pumps inject concentrated sulfuric acid to pin the pH exactly at 1.5. After the designated reaction time elapses, valves open to drain the saturated Pregnant Leach Solution (PLS) from the vessel. The reactor sits empty, ready to receive the next batch.

Pumps drive the saturated PLS directly into the primary chemical separation tank. Steel wool drops from a mechanical screw conveyor straight into the acidic pool. A violent electrochemical cementation reaction occurs instantly, forcing solid copper to precipitate entirely from the liquid phase (Velázquez-Martínez et al., 2019). Preparing for the next group of metals requires a massive shift in the chemical environment. Pumps transfer doses of hydrogen peroxide and sodium hydroxide. The pH pushes up to exactly 4.5. Aluminum and manganese immediately precipitate as solid hydroxides and oxides. Squeezing this thick slurry, a high-pressure membrane filter press successfully traps the solid copper, aluminum, and manganese cakes. This physical barrier prevents them from contaminating the sensitive downstream recovery circuits.

The purified liquid leachate moves into an organic solvent extraction circuit. Cyanex-272 enters the process. Upon violent mixing with the aqueous stream, an organic solvent containing this specific chemical extractant initiates the cobalt separation phase. It selectively captures cobalt ions from water, forming a valuable cobalt phosphinate product in the organic layer (Mantuano et al., 2006). The remaining aqueous raffinate is directed to a separate precipitation tank. Operators flood this new vessel with sodium hydroxide. The internal pH spikes to 10. Solid nickel hydroxide precipitates immediately as a heavy green powder. A dedicated filter press completely isolates this valuable material from the bulk liquid.

Lithium remains the only target left in the stream. The liquid enters the final precipitation unit. Automated systems dose sodium carbonate directly into the warm fluid. The dissolved lithium crystallizes rapidly into a fine white powder. The Lithium slurry is then fed into a high-pressure membrane filter press, which, in turn, squeezes this thick slurry and successfully traps the lithium cakes. Our plant safely transforms hazardous electronic waste into premium battery-grade raw materials.

2.2 Process Flow Diagram

The figures below present the final process flow diagram, detailing the sequence of operations within our system. This layout provides a clear visual breakdown of the entire workflow to support the technical analysis provided in this section.

BI-100	P-100	V-100	P-101	CO-101	CR-101	BI-101	P-102	P-103	P-200	P-201 A/B	P-202	V-200	R-201A/B	TK-201	C-201	C-202	AE-201 A/B	V-201 A/B	E-201	P-203 A/B
Brine Discharge Pool	Brine Recirculation Pump	Brine Solution Vessel	Brine Solution Pump	Battery Conveyor	Battery Crusher	Waste Plastic Bin	Black Mass Pump	Wastewater Pump	Nutrient Feed Pumps	Bioreactor Pumps	Cultivation Recycle Pumps	Bacteria Prep Tank	A. Ferro oxidans Bioreactor	Bacterial Recycle Thickener	Bioreactor Air Compressor	Batch Reactor Air Compressor	Aerator with sparger	Bleaching Vessel Batch Reactor	Heat Exchanger	Bioreactor Pumps

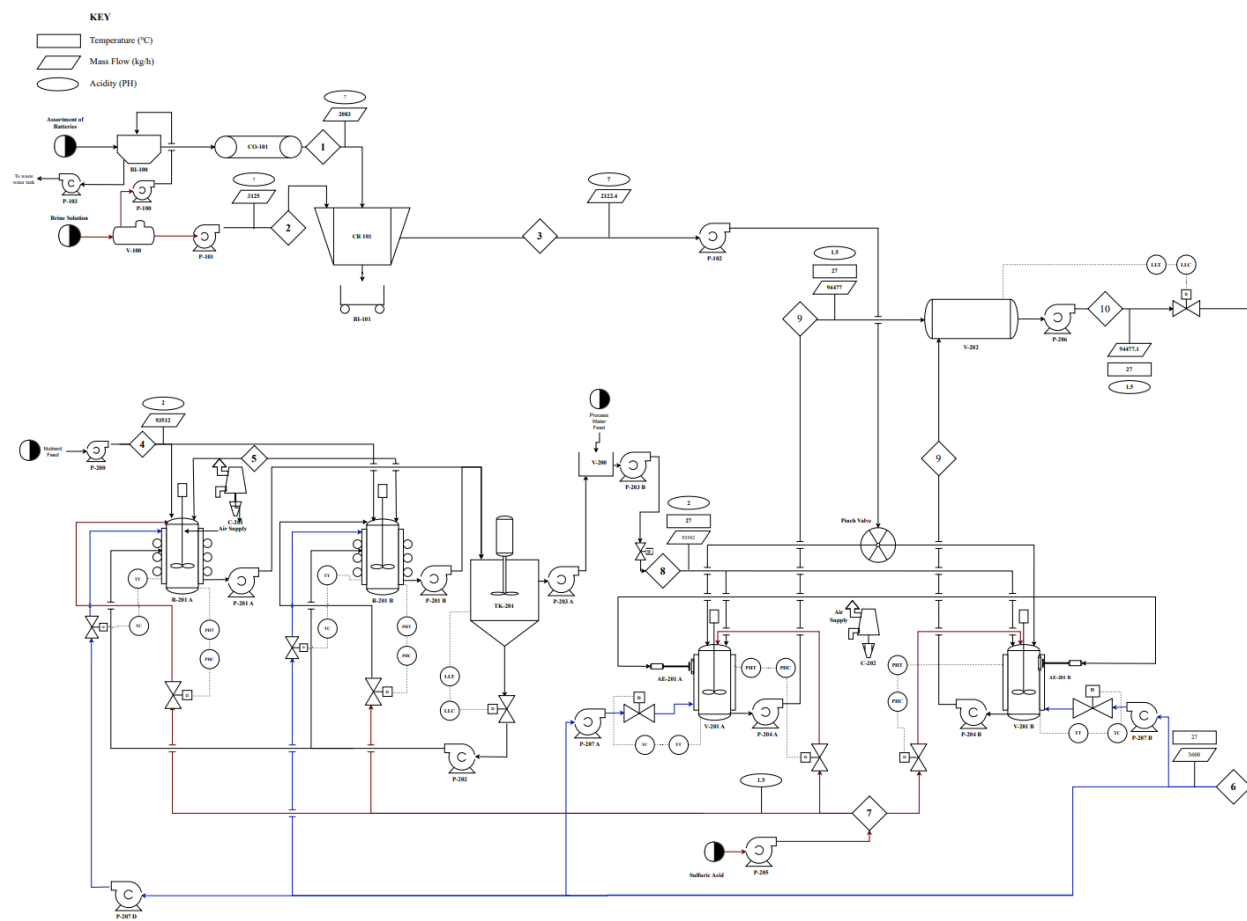


FIGURE 2.1.1. LEFT HALF OF THE FINAL PFD

P-204 A/B	V-202	P-205	P-206	P-207 A/B	V-203	E-300	P-300	CO-301	SP-301	TK-301	F-301	FP-301 A/B	BI-301	SX-301	P-305	SP-302	V-302	P-302	FP-302	BI-302	SP-303	V-303	P-303 A/B	FP-303	BI-303
Centrifugal Pump Feed In	Holding Vessel	Sulfuric Acid Dosing Pump	Holding Vessel Transfer Pump	Heated Water Pump	Water Reservoir	Heat Exchanger	Heat Exchanger Pump	Steel Wool Required Fe ³⁺ ions Conveyor	Impurity Precipitation Batch Reactor	High-Rate Thickener Unit	Thickener Sludge Removal Pumps	Filter Press of Mn, Al, Fe	Mn, Al, Fe Collection Bin	Solvent Extractor (Na ₂ CO ₃ & Cyanex -272)	Cobalt Sludge Removal Pumps	Nickel Precipitation Batch Reactor	Ni(OH) ₂ Sludge Buffer Tank	Ni(OH) ₂ Sludge Transfer Pump	Ni(OH) ₂ Filter Press	Ni(OH) ₂ Collection Bin	Lithium Precipitation Unit	Li ₂ (CO ₃) Sludge Buffer Tank	Li ₂ (CO ₃) Sludge Transfer Pump	Li ₂ (CO ₃) Filter Press	Li ₂ (CO ₃) Collection Bin

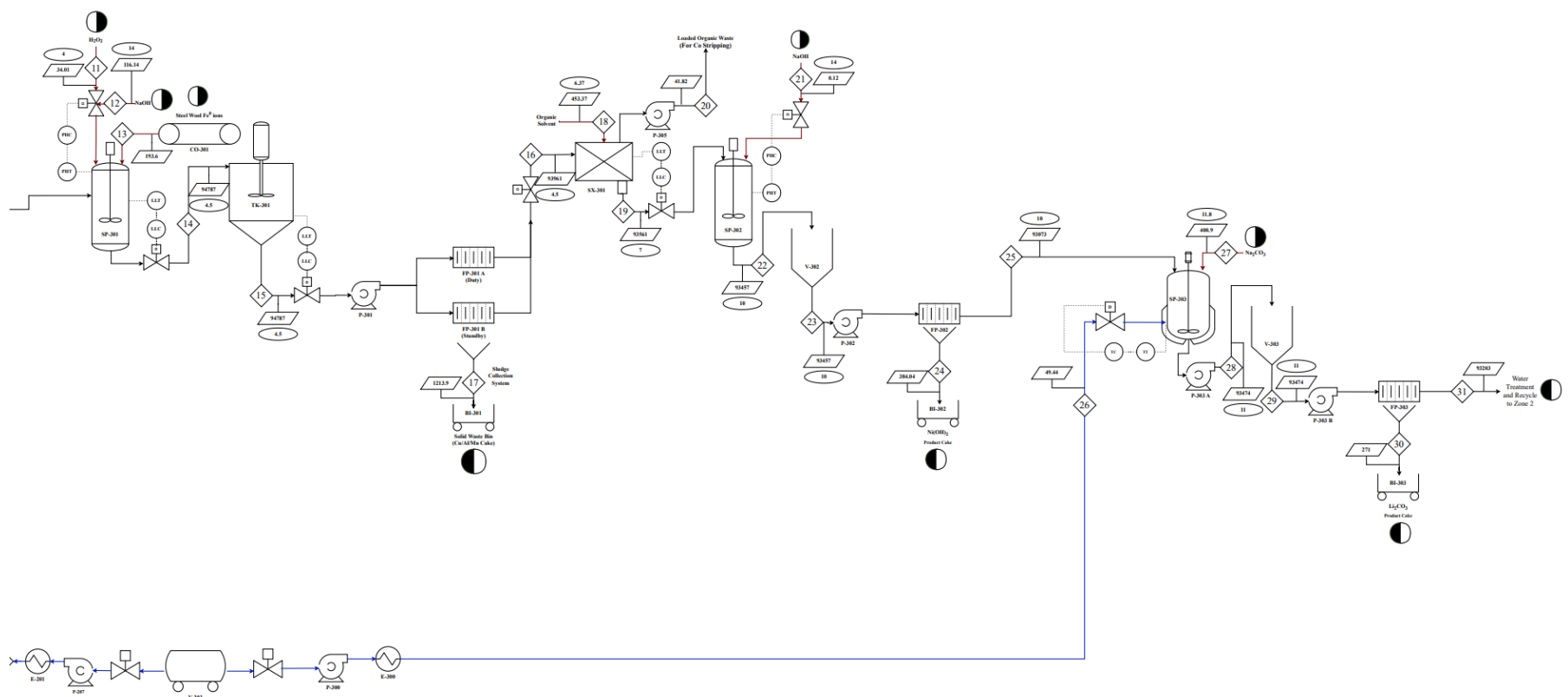


FIGURE 2.1.2. RIGHT HALF OF THE FINAL PFD

Table 2.1.1 below shows all the equipment used in the process, alongside their respective codes.

TABLE 2.1.1 PFD EQUIPMENT AND RESPECTIVE CODES

Code	Equipment Name	Code	Equipment Name	Code	Equipment Name
V-100	Brine Solution Vessel	P-202	Cultivation Recycle Pumps	P-200	Nutrient Feed Pumps
P-101	Brine Solution Pump	P-204 A/B	Centrifugal Pumps Feed in	P-205	Sulfuric Acid Dosing Pump
CO-101	Battery Feed Conveyor	TK-201	Nutrient Feed Buffer Tank	P-207	Heated Water Pump
CR-101	Battery Crusher	P-203 A/B	Culture Medium Pump	TK-301	High-Rate Thickener Unit
BI-101	Waste Plastic Bin	V-202	Holding Vessel	P-301	Thickener Sludge Removal Pumps
BI-100	Discharge Pool	P-206	Holding Vessel Transfer Pump	SP-302	Nickel Precipitation Batch Reactor
P-102	Black Mass Transfer Pump	E-201	Bioleaching Heat Exchanger	V-302	$Ni(OH)_2$ Sludge Buffer Tank
P-100	Brine Recirculation Pump	V-203	Water Reservoir	P-305	Cobalt Sludge Removal Pump
V-200	Bacteria Prep Tank	E-300	Recovery Circuit Heat Exchanger	P-302	$Ni(OH)_2$ Sludge Transfer Pump
P-103	Wastewater Pump	P-300	Heat Exchanger Pump	FP-302	$Ni(OH)_2$ Filter Press
R-201 A/B	<i>A. Ferrooxidans</i> Bioreactors	SP-301	Impurity Precipitation Reactor	BI-302	$Ni(OH)_2$ Collection Bin
P-201 A/B	Bacteria Transfer Pumps	FP-301 A/B	Filter Press - Mn, Al, Cu, Fe	SP-303	Lithium Precipitation Unit
AE-201 A/B	Aerators with Spargers	SX-301	Organic Solvent Extractor	V-303	Li_2CO_3 Sludge Buffer Tank
C-201	Bioreactor Air Compressors	CO-301	Filter Press Solid Waste Conveyor	P-303 A/B	Li_2CO_3 Sludge Transfer Pump
C-202	Batch Reactor Air Compressor	BI-301	Impurity Metal Collection Bin	FP-303	Li_2CO_3 Filter Press
V-201 A/B	Bioleaching Batch Reactors			BI-303	Li_2CO_3 Collection Bin

2.3 Mass and Energy Balances

2.3.1 Mass Balances

The plant is required to recycle 15,000 tons/year of batteries. It is assumed that the plant will operate 24 hours a day, 7 days a week, for 300 days a year. This would mean that the plant is required to process batteries at a rate of 2083.33 kg/h. Mass balance sample calculations: Conversion of mass flow for the battery input from Project 12 Statement:

$$m_{in} = \frac{15000 \text{ ton}}{\text{year}} * \frac{1000 \text{ kg}}{\text{ton}} * \frac{\text{year}}{300 \text{ day}} * \frac{\text{day}}{24 \text{ hr}} = 2083.33 \text{ kg/h of batteries processed every day}$$

Control Volume of Crusher Mass Balance:

For the crusher recovery of metal, based on Garcia et al. (2025), the percent recovery is as follows, obtained in an unsorted black mass of metal composition. Control Volume of CSTR Mass Balance: Given that the Lithium is 2.80% of the total battery weight,

$$m_{Li} = 2.80\% \times \text{Battery Mass} = 2.80\% \times 2083.33 \text{ kg/hr} = 58.33 \text{ kg/hr}$$

Determining the required mass flow of the Culture Medium needed, Using a 1% weight-to-volume ratio of 1kg to 100 L from the Reactor operation,

$$\text{sum of all material present in CSTR} * 100 = 935.4167 \text{ kg/hr} * 100 = 93541.67 \text{ kg/hr}$$

Mass Balance and Composition of the Culture Medium (CM) present to promote bio-reaction in the CSTR:

$$m_{CMcomp} = \text{Composition density of material} * \text{mass flow CM}$$
$$m_{(NH_4)_2SO_4} = 3 * 1000 \text{ kg/L} * 93541.64 \text{ kg/h} = 280.62 \text{ kg/h of } (NH_4)_2SO_4$$

Determination of waste/unreacted metals:

$$\text{Mass flow of Input metals} - \text{Mass flow of output metals} = \text{Waste of unreacted metals}$$

$$\text{Waste of unreacted metals} = 40.06 \text{ kg/h}$$

All other required chemicals are calculated using the same method for Separator 1 to obtain the following mass flow needed.

Determining Sodium Hydroxide based on the desired PH required to separate metals after the bioleaching process:

Sum of all mass flow in Separator 1 = 94477.1kg of black mass solution per batch

$$\text{Desired pH of 2} = (10^{-\text{Current PH}} - 10^{-\text{Desired PH}}) * 94477.1\text{kg}$$

$$\text{Desired pH of 2} = (10^{-1.5} - 10^{-2}) * 94477.1\text{kg} = 2042.86\text{mol of NaOH}$$

$$\text{Desired pH of 2} = 2042.86\text{mol of NaOH} * 40\text{g/mol} * 1000\text{kg/g} = 81.71\text{kg of NaOH}$$

81.71kg of NaOH is needed to raise the pH to 2, which precipitates out 100% of the Iron present in the separation vessel. Raising the pH to 3 precipitates Manganese out at 85% recovery, requiring 34.01kg of H₂O₂. Raising the pH to 4.5 precipitates Aluminum out at 95% recovery, requiring 0.412kg of NaOH. The table below shows a full summary of all Mass Balances that have been calculated for the plants operation.

Table 2.3.1 meticulously maps the output mass balances across the plant's entire piping network. It provides a definitive mathematical picture of the extraction circuit. The process initiates at pipe A-102, where raw battery feed enters at a constant 2083.3 kilograms per hour to establish a baseline dense mixture of metals. Flooding the system with over 93,000 kilograms of biological culture medium at A-203 B violently alters the chemistry. Extraction efficiencies immediately soar. The aggressive bio-acid captures 98% of the nickel and 97% of the copper directly from the raw slurry at A-204 A. Downstream sections systematically isolate the remaining individual elements. Copper drops out rapidly in the A-301 and A-302 streams. Cobalt isolation happens strictly at A-303. Pushing the remaining heavy liquid forward forces the final chemical precipitations in the A-305 and A-308 circuits. These terminal lines secure the facility's ultimate economic prize. They successfully force the final 99% precipitation of nickel and 99% crystallization of lithium, yielding continuous, premium-grade commercial products.

TABLE 2.3.1 SUMMARY OF THE PLANT'S OUTPUT MASS BALANCES

Pipe	A-102		A-203 B		A-204 A		A-301		A-302	
Metal	Recovery	Kg/hr	Recovery	Kg/hr	Recovery	Kg/hr	Recovery	Kg/hr	Recovery	Kg/hr
Culture Medium	-	-	-	93542	-	93542	-	92709	-	-
Li	2.80%	58.3	-	-	90%	52.5	-	-	-	-
Cu	10.90%	227.1	-	-	97%	220.3	99%	218.07	99%	215.89
Co	2.40%	50.0	-	-	92%	46.0	-	-	-	-
Al	7.60%	158.3	-	-	96%	152.0	95%	144.40	99%	142.96
Ni	19%	395.8	-	-	98%	387.9	-	-	-	-
Mn	2.20%	45.8	-	-	80%	36.7	85%	31.17	99%	30.86
Total	-	2083.3	-	93542	-	98077	-	94787	-	94787
	A-303		A-305 A		A-305 B		A-308 A		A-308 B	
	Recovery	Kg/hr	Recovery	Kg/hr	Recovery	Kg/hr	Recovery	Kg/hr	Recovery	Kg/hr
Culture Medium	-	-	-	92709	-	-	-	92709	-	92709
Li	-	-	-	-	-	-	98%	273.5	98%	268
Cu	-	-	-	-	-	-	-	-	-	-
Co	90.91%	41.82	-	-	-	-	-	-	-	-
Al	-	-	-	-	-	-	-	-	-	-
Ni	-	-	99%	384.04	99%	380.2	-	-	-	-
Mn	-	-	-	-	-	-	-	-	-	-
Total	-	94427	-	94430	-	94050	-	93735	-	93735

2.3.2 Energy Balances

An overall energy balance was performed for each major piece of equipment in the process to determine the energy input, energy output, heat duties, and mechanical work requirements. The results are summarised in Table 2.3.2 below.

TABLE 2.3.2 SUMMARY OF THE PLANT'S OUTPUT ENERGY BALANCES

Equipment	Name	Required Energy (kW)	Energy in (kJ/hr)	Energy out (kJ/hr)	Heat Duty (kJ/hr)	Work (kJ/hr)
Crusher	CR-101	266.2	39417	997748	-	958332
Conveyor	CO-101	0.021	36667	36741	-	74.7
Conveyor	CO-301	0.0012	2087	2091	-	4.44
Compressor	C-201	442.8	11471	1605708	-	1594237
Compressor	C-202	442.8	11471	1605708	-	1594237
Heat Exchanger	E-201	26.25	421344	515845	94501	-
Heat Exchanger	E-300	324.5	1489752	2657778	1168026	-
Separator	SP-301	5.54	7905334	7925290	-	19956
Separator	SP-302	5.54	7857108	7877064	-	19956
Separator	SP-303	5.54	7835135	7855091	-	19956
Stripper	SX-301	3.0	7842642	7853442	-	10800
Separator	FP-301 A/B	1.0	7924176	7927776	-	3600
Separator	FP-302	1.0	7857114	7860714	-	3600
Separator	FP-303	1.0	7858684	7862284	-	3600
Separator	TK-301	5.0	7924176	7942176	-	18000

For the Bioleaching Vessel the water temperature needed for optimal operation is at 27°C. To accomplish this we must find the total heat energy required by a jacket in the Reactor's energy balance. As such we can say that;

$$Q_{\text{jacket needed}} = \text{feed energy} + \text{air energy} + \text{evap(cooling)} + \text{agitator energy} + \text{rxn energy} + \text{wall loss/gain}$$

$$\text{Feed Energy} = \dot{m}_f c_{p,f} (T_{\text{feed}} - T_r) = 0.133 \frac{\text{kg}}{\text{s}} * 0.7 \frac{\text{kJ}}{\text{kg}^{\circ}\text{K}} * (15^{\circ}\text{C} - 27^{\circ}\text{C}) = -1.12 \text{ kW}$$

$$air\ energy = \dot{m}_{air} c_{p,air} (T_{air} - T_r) = 0.1 \frac{kg}{s} * 0.718 \frac{kJ}{kg \cdot K} * (15^\circ C - 27^\circ C) = -0.86 kW$$

$$evap(cooling) = \dot{m}_{evap} \Delta H_{vap} (T_r) = (10 \frac{kg}{s} * (Y_{out} - Y_{in}) \frac{1}{3600}) * 0.718 \frac{kJ}{kg \cdot K} * 15^\circ C = 0.015 kW$$

$$rxn\ energy = V * \sum_i r_i * (\Delta H_i) = 50 m^3 * 0.0004861 \frac{mol}{m^3 s} * 98 \frac{kJ}{mol} = 2.38189 kW$$

$$wall\ loss/gain = UA(T_a - T_r) = 3.7 \frac{kW}{K} (15^\circ C - 27^\circ C) = -44.4 kW$$

$$Q_{jacket\ needed} = -1.12 kW + -0.86 kW + 0.015 kW + 2.38189 kW + -44.4 kW = 23.98 kW$$

$$T_{water\ needed} = T_{needed} + \frac{Q}{UA} + \frac{Q}{2 * \dot{m} * c_p} = 27^\circ C + \frac{23.98 kW}{3.7 \frac{kW}{K}} + \frac{23.98 kW}{2 * 1 \frac{kg}{s} * 4.186 \frac{kJ}{kg \cdot K}} = 36.35^\circ C$$

The energy in calculations is done by multiplying mass flow rates with the material enthalpy, and the work is calculated by obtaining energy that is required in kW multiplied by 3600 to get the value in kJ/hr. Finally to get the energy out, the formula of energy in + work equals energy out is used.

To calculate the required energy in the crusher, the formula below is used:

$$P = \frac{\text{specific energy for shredding} * \text{capacity sizing}}{\text{efficiency}} = \frac{0.1 kWh/kg * 2395.83 kg/hr}{0.9} = 266.2 kW$$

The work would then be found using:

$$W = P * 3600 = 266.2 kW * 3600 = 958332 \frac{kJ}{h}$$

Now using information from the known mass balances and the enthalpies for each material, the energy in can be calculated by multiplying the mass flowrate with the specific enthalpy as shown below.

$$E_{in} = h_{battery} * \dot{m}_{battery} + h_{brine} * \dot{m}_{brine} = 17.6 \frac{kJ}{kg} * 20833.33 \frac{kg}{h} + 70.4 \frac{kJ}{kg} * 30.0625 \frac{kg}{h}$$

$$E_{in} = 39417 \frac{kJ}{h}$$

Now with both work and energy in, the energy out can be calculated by adding the two values.

$$E_{out} = E_{in} + W = 39417 \frac{kJ}{h} + 958332 \frac{kJ}{h}$$

The same process is used to calculate the energy in and out and work for all the equipment. The heat exchangers varied with heat duty being found instead of work; however it is the same calculation where it requires energy multiplied by 3600.

To calculate the required energy for the different equipment, the formula depends on the type of equipment and it's role.

The power required for the conveyors is calculated as shown below:

$$P = \rho * \dot{m} * \text{vertical lift}$$

$$P = 150 \frac{kg}{m^3} * 0.000036 \frac{m^3}{s} * 2.34m = 1.23W$$

The energy of the compressor required energy is calculated as shown in the table below

TABLE 2.3.3 SUMMARY OF COMPRESSOR POWER CALCULATIONS

Compressor C-201 and C-202		
intensity	25.00	NL/(hL)
vvm	0.01	vvm
vessel volume	12429.07	m ³
Airflow	248.58	m ³ /min
density	1.84	kg/m ³
height	28.36	m
hydrostatic pressure	511.91	kPa(g)
diameter	23.63	m
area	438.55	m ²
superficial gas velocity	0.01	m ³ /s
air flow (diff units)	4.14	m ³ /s

discharge	2.00	bar abs
power required	442843.55	W

Where the power required is

$$P = Q * P_1 * \ln\left(\frac{P_2}{P_1}\right) = 4.14 \frac{m^3}{s} * 101325 Pa * \ln\left(\frac{200000 Pa}{101325 Pa}\right) = 442843.55 W$$

Energy Required for heat exchanger is the heat needed as calculated from before divided by 3600.

$$P = heat * \frac{1}{3600} = 94501.44 kJ * \frac{1}{3600} = 26.2504 kW$$

The energy required for the separators is simply the energy of mixing which is the energy of the impeller, and because all the separators are extremely similar in size the calculation below will be the exact same for all of them.

$$P = Np * fluid\ density * (impeller\ speed)^3 * (impeller\ diameter)^5$$

$$P = 1.5 * 1600 * (2)^3 * (0.78)^5$$

$$P = 5543.37 W$$

Chapter 3. Relevant Standards

To ensure the safe operation of the plant, the structural integrity of the equipment, and environmental protection, the design of this 15,000 tonnes/year lithium-ion battery (LIB) recycling facility strictly adheres to established international engineering codes and industry standards. These regulations govern every phase of the project, from the mechanical sizing of the batch bioreactors to the physical layout of the plant.

3.1 Mechanical and Equipment Design Standards

The mechanical design and material selection for the plant's major and standard equipment are dictated by the American Society of Mechanical Engineers (ASME) and the American Society for Testing and Materials (ASTM). The ASME Boiler and Pressure Vessel Code (BPVC), Section VIII, is predominantly followed for the design and sizing of the pressurized and jacketed equipment, specifically dictating the hoop stress and minimum wall thickness calculations for the primary Batch Bioreactors, Heat Exchangers, and Solvent Extraction vessels. Furthermore, all stress analysis calculations for pipe design comply with ASME B31.3 Process Piping. This is particularly critical for the CPVC and Rubber-Lined Carbon Steel (RLCS) lines handling the highly abrasive black mass slurry and corrosive sulfuric acid at pH 1.5.

3.2 Instrumentation and Process Control Standards

To maintain the strict kinetic parameters required for the *Acidithiobacillus ferrooxidans* culture, including a temperature of 27°C, 1% pulp density, and pH 1.5, the plant requires a highly automated control architecture. The instrumentation symbols, tagging conventions, and logical control loop hierarchies developed for the Process and Instrumentation Diagrams (P&IDs) strictly follow the International Society of Automation (ISA) S5-1 standards, specifically ANSI/ISA-S5-1.

3.3 Plant Layout and Process Safety Standards

Handling end-of-life LIBs poses severe risks of thermal runaway, toxic gas release (such as Hydrogen Fluoride), and fire. The plant layout and safety interlocks are governed by several

safety codes to mitigate these risks. The physical separation distances between the pre-treatment mechanical crushing zone, the biological leaching zone, and the administrative buildings are based on the Center for Chemical Process Safety (CCPS) Siting and Layout Guidelines to prevent fire propagation and radiant heat exposure. To ensure the safety of plant employees, Occupational Safety and Health Administration (OSHA) regulations, specifically 29 CFR 1910.119 on Process Safety Management of Highly Hazardous Chemicals, are strictly followed when handling concentrated H_2SO_4 , NaOH , and flammable organic extractants such as Cyanex-272. Additionally, risk assessment and hazard operability across the design phase integrate ISO 10377 guidelines to ensure safety considerations are embedded against thermal, fire, and kinetic hazards.

3.4 Environmental and Regulatory Standards

The plant design complies with stringent environmental regulations to minimize its ecological footprint and validate its economic viability as a secondary materials provider. The process recovery targets were designed to meet or exceed the rigorous circular economy standards set by the EU Battery Regulation 2023, which mandate a minimum recovery yield of 90% for Cobalt, Copper, and Nickel, and 50% for Lithium by 2027. At the broader regulatory level, the plant adheres to the Clean Water Act (CWA) and the Clean Air Act (CAA). These programs, regulated by the Environmental Protection Agency (EPA) and equivalent international bodies, govern the required neutralization of the sodium sulphate (Na_2SO_4) wastewater effluents and the scrubbing of toxic off-gases generated during battery comminution.

Chapter 4. Design and Sizing of Equipment

This section will display the design and sizing considerations of the equipment in the process plant. Respective parameters and sample calculations will give an accurate numerical evaluation of the process.

4.1 Crusher

The twin shaft crusher (CR-101) serves as the primary mechanical pretreatment unit, reducing used lithium-ion batteries from ~10 cm chunks to a finer black mass sludge for bioleaching. This step liberates active cathode/anode materials (LiCoO₂, NMC, graphite) while separating casing metals (Al, Cu) through air classification.

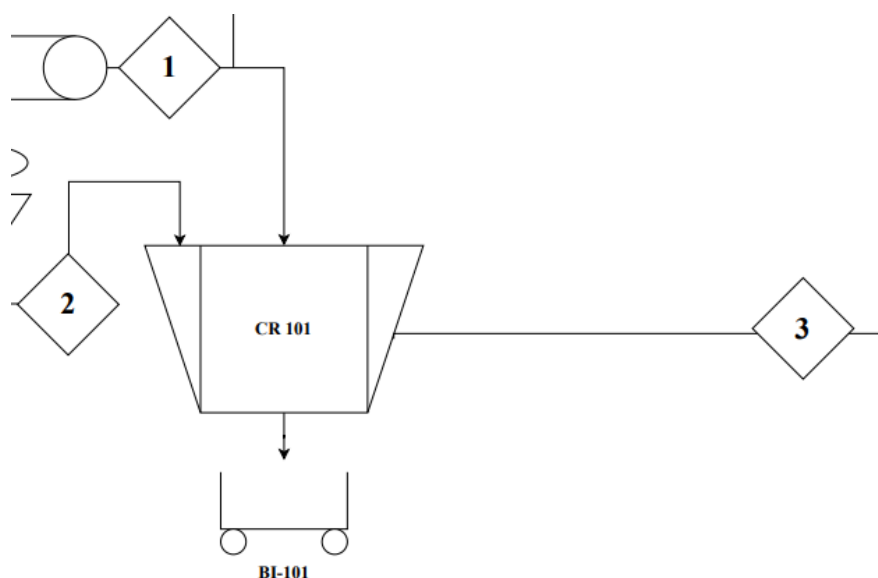


FIGURE 4.1 CRUSHER (CR-101)

The design follows ASME BPVC Section VIII standards with stainless steel construction for corrosion resistance against battery electrolytes. Operating at 50 rpm with a 150 mm feed opening, CR-101 processes 18.7 tonnes/hr of shredded battery input, yielding 8.38 tonnes/hr black mass sludge. CR-101 is sized for a 15,000-tonne/year capacity (7920 operating hours), handling various LIB designs/sizes/sources.

Capacity

$$Q = (0.6 \times W \times H \times S \times \rho \times 3600) / 1000$$

Where $W=1200$ mm, $H=150$ mm (feed), $S=20$ mm stroke, $\rho=2.8$ t/m³

$$Q = (0.6 \times 1.2 \times 0.15 \times 0.02 \times 2.8 \times 3600) / 1000 = 21.6 \text{ t/hr}$$

Wall Thickness

$$t = (P \times R) / (S \times E - 0.6P) + CA$$

P=0.15 MPa (set safety value), R=0.65 m, S=115 MPa (at 100°C), E=1.0, CA=3 mm

$$t = (0.15 \times 650) / (115 - 0.09) + 3 = 12.4 \text{ mm}$$

This wall thickness maintains safety requirements to contain the internal pressure of the machine. Following this standard for wall thickness the crusher can be safely run at the tremendous scale of the process plant. The CR-100 design adapts validated Metso Nordberg® C116™ parameters (1150×760 mm feed, 132 kW base) to our 18.7 t/hr LIB capacity using direct scale factor of 0.34 (45/132 kW). The feed opening scales proportionally (1200×150 mm vs 1150×760 mm) while still maintaining 50 rpm eccentric speed (Metso Outotec Corporation, 2025).

TABLE 4.1: CRUSHER PARAMETER SUMMARY

Parameter	Value
Feed Rate	18.7 t/hr
Product Size	<2 mm (80% passing)
Power Draw	45 kW
Crusher Ratio	6:1
Dimensions	1.2 m × 0.9 m × 1.1 m
Capacity	25 t/hr max

4.2 Conveyors

The screw conveyors in this process form the mechanical backbone between the crusher, the separation section, and downstream storage. Their design is driven directly by the crusher reject flowrate and steel wool recovery values established in the mass balance, and follows standard CEMA practice for screw conveyor sizing. Units are fully enclosed, and use stainless steel conveyors to ensure reliable handling of abrasive, brine-wetted solids while preventing dusting and cross-contamination with product streams.

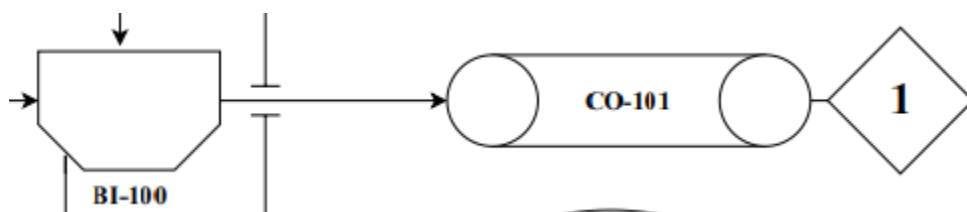


FIGURE 4.2.1 CRUSHER CONVEYOR (CO-101)

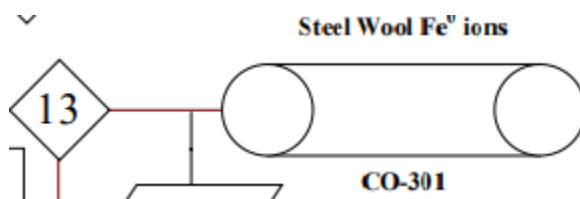


FIGURE 4.2.2 STEEL WOOL SCREW CONVEYOR (CO-301)

CO-101 moves the crusher reject stream to the separation section. The design capacity is 8.38 t/h, corresponding to the crusher waste flow from the mass balance. The conveyor is a horizontal/low-incline tubular screw (nominal diameter 200 mm, length 15 m), selected to balance capacity, footprint, and shaft stiffness for heavy duty continuous 24/7 operation.

Operating at 60 rpm and approximately 40–45% trough loading, CO-101 provides a calculated volumetric capacity slightly above the design solids rate, ensuring margin for irregular conditions and non-uniform feed. Stainless steel 316L is specified for the casing and center pipe to provide corrosion resistance to residual electrolyte and brine contamination from the crusher. While the flights are made of an abrasion-resistant alloy steel to withstand continuous contact with steel wool and metallic fines. A drive power of approximately 5.5 kW is adequate when

applying typical CEMA load factors for moderately abrasive materials and including a conservative service factor. The conveyor is fully enclosed and equipped with dust seals and inspection/clean-out ports to facilitate routine maintenance and flushing.

Capacity CO-101

$$Area = (\pi \times D^2 \times \phi) / 4 = (\pi \times 0.2^2 \times 0.45) / 4 = 0.0141 \text{ m}^2$$

$$Volumetric \text{ Capacity per Revolution} = A \times S = 0.0141 \text{ m}^2 \times 0.06 \text{ m} = 0.00085 \text{ m}^3/\text{rev}$$

$$Q = Vol/\text{rev} \times N \times \rho \times 60 \text{ min/hr} = 0.00085 \times 60 \times 3.2 \times 60 = 8.62 \text{ tonnes/hour}$$

CO-301 transfers the magnetically recovered steel wool fraction from the separation unit to storage tank TK-301. CO-301 is specified as a horizontal tubular screw conveyor with nominal diameter 150 mm and length 8 m. The operating speed is set at approximately 45 rpm, giving sufficient capacity at 40–45% trough loading while maintaining gentle, controlled conveying to limit dust and minimize mechanical degradation of the steel wool. As with CO-101, the casing and shaft are constructed of the same material and specifications. A motor rating of 2.2 kW, incorporates appropriate allowances for start-up torque and material factors. The fully enclosed configuration also aids maintenance and ensures that steel fibers are contained within the process.

Capacity CO-301

$$Area = (\pi \times D^2 \times \phi) / 4 = (\pi \times 0.15^2 \times 0.45) / 4 = 0.0080 \text{ m}^2$$

$$Volumetric \text{ Area} = A \times S = 0.0080 \text{ m}^2 \times 0.045 \text{ m} = 0.00036 \text{ m}^3/\text{rev}$$

$$Q = Vol/\text{rev} \times N \times \rho \times 60 \text{ min/hr} = 0.00036 \times 45 \times 2.8 \times 60 = 2.35 \text{ tonnes/hour}$$

4.3 Thickeners

Thickeners are used twice to increase the concentration of the metal-rich slurries at key process stages. TK-201 after the first reactor process 59.375 m³/h feed containing 1,235 kg/h solids, producing concentrated 65 kg/h liquid underflow while recycling clear overflow. TK-301 (post-bioreactors) handles 91.708 m³/h bioleach underflow, achieving required solids-liquid separation for the downstream filter press feed.

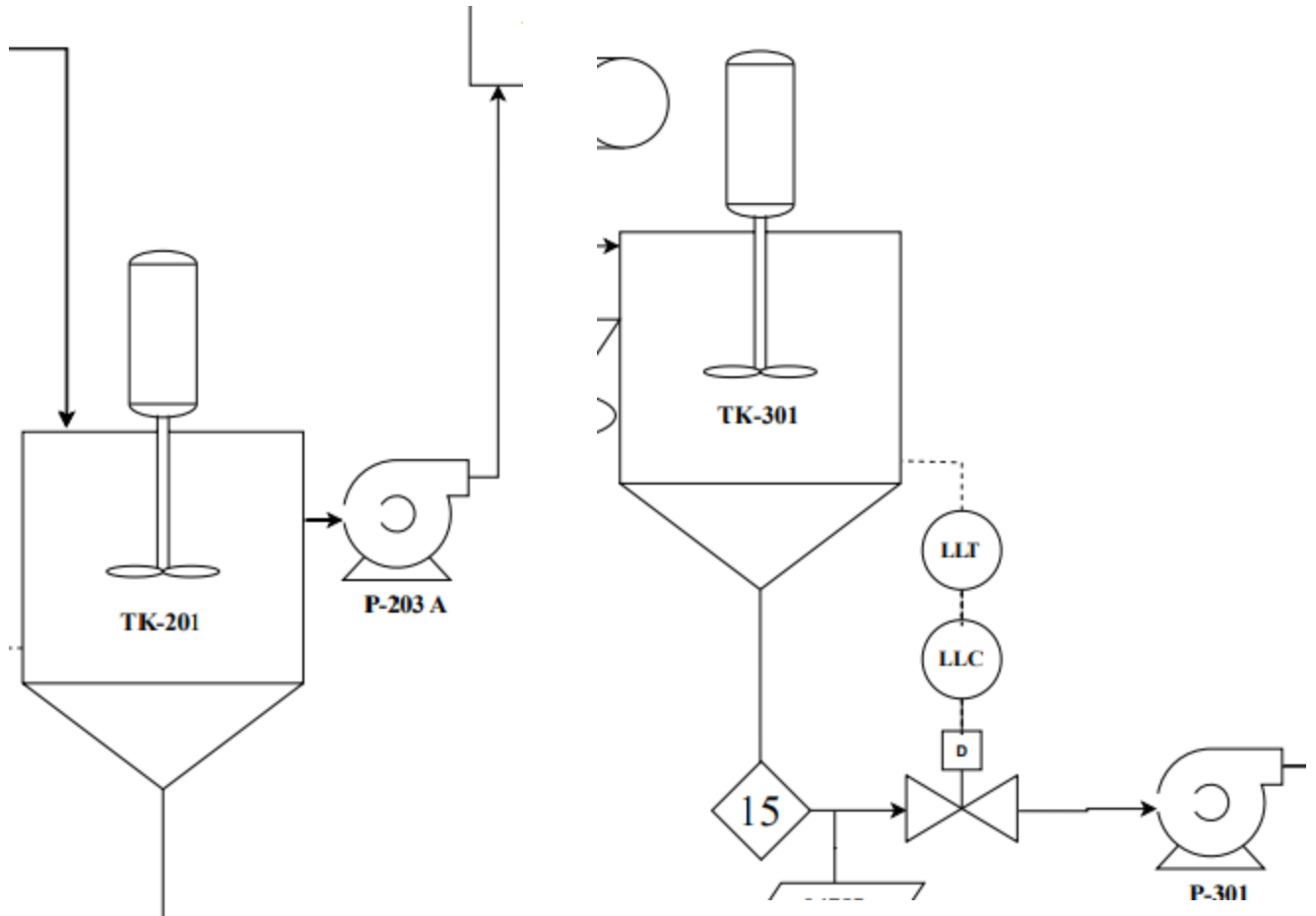


FIGURE 4.3.1 BIOREACTOR BACTERIA-THICKENER (TK-201) (LEFT)

FIGURE 4.3.2 SEPARATION PHASE 1 (TK-301) (RIGHT)

Thickener Scale-up Calculation

$$TK-201: \text{Scale-up Area} = 75.93 \, m^2 \times 1.2 \, \text{factor} = 91.11 \, m^2$$

$$\text{Diameter} = \sqrt{(4 \times 91.11/\pi)} = 10.77 \, m$$

$$TK-301 \text{ Scale-up Area} = 275.26 \, m^2 \times 1.2 \, \text{factor} = 660.29 \, m^2$$

$$\text{Diameter} = \sqrt{(4 \times 660.29/\pi)} = 28.99 \, m$$

4.4 Filter Presses

FP-301 A/B dual filter presses dewater 1235 kg/h bioleach solids from 0.8125 m³/h thickener underflow (1,600 kg/m³ slurry) post-TK-201, producing a dry filter cake (>25% solids)

for metal recovery while recycling liquid to the process water tank. Each 870 mm x 870 mm plate-frame unit (14 plates, 2.25 m²/plate) provides 59.28 m² total area with 1.2x safety factor, operating on 1-hour cycles (fill 15 min, press 30 min, discharge 15 min). PP plates (30 mm cake space) with CaSO₄ precoat will achieve 25 kg/m² loading at 10 bar hydraulic pressure (1.0 kW total), yielding 98% solids capture.

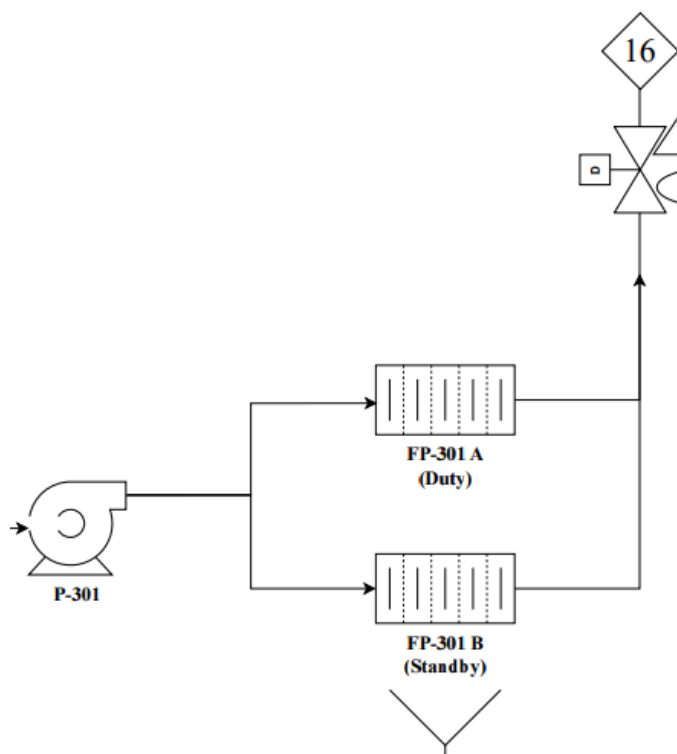


FIGURE 4.4.1 FILTER PRESS PHASE 1 Cu/Al/Mn EXTRACTION (FP-301 A/B)

FP-301 A/B Area Calculation

$$\text{Solids Loading} = 1,235 \text{ kg/h} \div 25 \text{ kg/m}^2 \times 1 \text{ hr/cycle} = 49.4 \text{ m}^2 \text{ required}$$

$$\text{Design Area} = 49.4 \text{ m}^2 \times 1.2 \text{ safety factor} = 59.28 \text{ m}^2 \text{ total}$$

$$\text{Plates Required} = 59.28 \text{ m}^2 \div 2.25 \text{ m}^2/\text{plate} = 14 \text{ plates}$$

FP-302 recovers 384 kg/h Ni/Co hydroxide precipitates from 85.73 m³/h slurry (1,100 kg/m³) following Cyanex-272 SX and pH adjustment, producing high-value mixed hydroxide precipitate cake for battery cathode recycling. The 7-plate unit (18.43 m² total, 2.63 m²/plate

oversized) handles intermittent batch feeds with 25 kg/m² loading at 12 bar, achieving near 100% metal recovery in 45 minute cycles. Membrane squeeze plates enhance cake washing, reducing Cyanex-272 in filtrate. The high recovery of metals is in large part due to the favorable black mass composition attained.

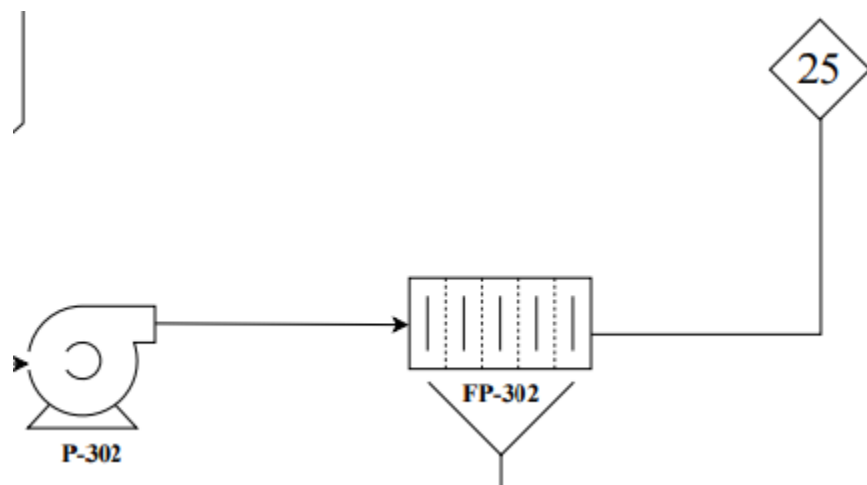


FIGURE 4.4.2 FILTER PRESS PHASE 2 NICKEL EXTRACTION (FP-302)

FP-302 Area Calculation

$$\text{Solids Loading} = 384 \text{ kg/h} \div 25 \text{ kg/m}^2 \times 1 \text{ hr/cycle} = 15.36 \text{ m}^2 \text{ required}$$

$$\text{Design Area} = 15.36 \text{ m}^2 \times 1.2 \text{ safety factor} = 18.43 \text{ m}^2 \text{ total}$$

$$\text{Plates Required} = 18.43 \text{ m}^2 \div 2.63 \text{ m}^2/\text{plate} = 7 \text{ plates}$$

FP-303 the final filter press dewateres 271 kg/h Li₂CO₃ crystals from 85.74 m³/h post-Na₂CO₃ precipitation slurry, producing battery-grade lithium carbonate cake (99.5% purity target) while recycling Na⁺-rich liquid. The compact 4-plate unit (13.01 m², 3.25 m²/plate oversized) operates continuously at 25 kg/m² loading with 30-min cycles, using recessed plates for maximum cake thickness.

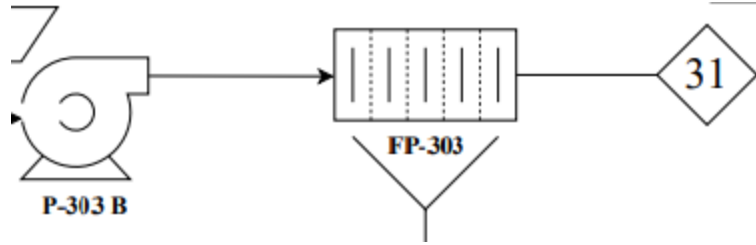


FIGURE 4.4.3 FILTER PRESS PHASE 3 LITHIUM EXTRACTION (FP-303)

FP-303 Area Calculation

$$\text{Solids Loading} = 271 \text{ kg/h} \div 25 \text{ kg/m}^2 \times 1 \text{ hr/cycle} = 10.84 \text{ m}^2 \text{ required}$$

$$\text{Design Area} = 10.84 \text{ m}^2 \times 1.2 \text{ safety factor} = 13.01 \text{ m}^2 \text{ total}$$

$$\text{Plates Required} = 13.01 \text{ m}^2 \div 3.25 \text{ m}^2/\text{plate} = 4 \text{ plates}$$

4.5 Holding Tanks

V-100 is the single holding vessel to store the required brine solution for the process. The volume of the tank is calculated based on the total brine required to process the feed. As per the mass balance the total brine volume required is 26.08m³. The tank was modeled as a vertical cylindrical vessel. Using the standard cylinder volume relationship

$$V = \frac{\pi D^2 H}{4} = 26.08 \text{ m}^3$$

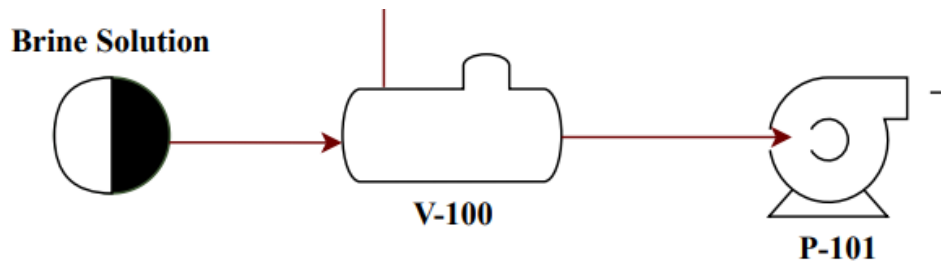


FIGURE 4.5.1 BRINE SOLUTION HOLDING VESSEL (V-100)

V-202 holding tank buffers 85,616 m³/h slurry flow from first reactor to separator SP-301, preventing process interruptions during batch operations. A standard vertical cylindrical SS316L tank provides working volume of 85,616 m³ with scale-up design volume 102,740 m³

(1.2x safety factor. Top-mounted 15 kW agitator (50 rpm) maintains solid suspension while high/low level alarms protect downstream pumps.

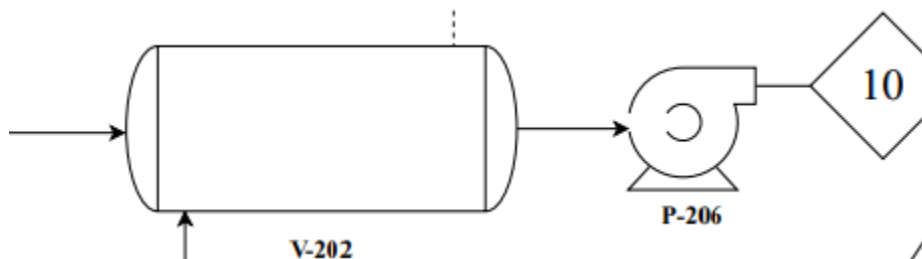


FIGURE 4.5.2 BIOLEACHED MATERIAL HOLDING VESSEL (V-202)

V-302 & V-303 holding tanks respectively manage 85,371.82 m³/h reactor slurry to FP-302 & FP-303 filter press, ensuring continuous solids-liquid separation feed. With working volume 85,371.82 m³ and scale-up 102,446.18 m³. Agitated construction prevents 1,235 kg/h solids settling during the long residence time required for process stability

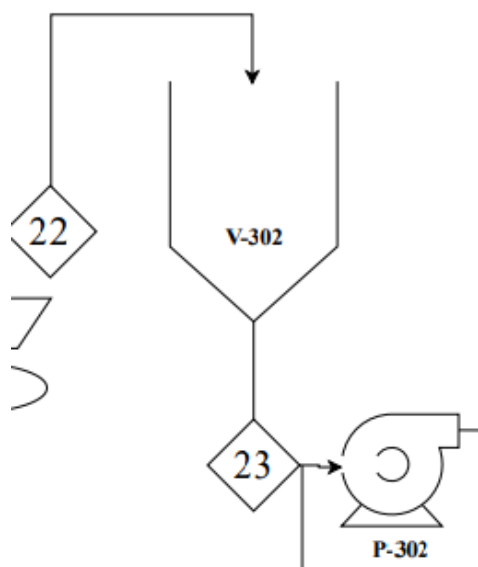


FIGURE 4.5.3 SEPARATIONS PHASE 2 HOLDING VESSEL (V-302) (LEFT)

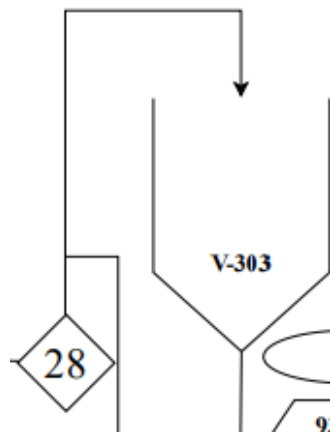


FIGURE 4.5.4 SEPARATIONS PHASTE 3 HOLDING VESSEL (V-303) (RIGHT)

4.6 Bio Reactors & Bioleaching

The bioleaching stage of the process is carried out using two identical batch stirred-tank bioreactors operating in parallel. These reactors are responsible for converting the incoming black mass feed through a biologically mediated dissolution reaction. Because both reactors are identical in geometry and operating conditions, the design calculations are shown for one. A kinetic verification was performed using the reaction rate of black mass dissolution. Based on the calculated dissolution rate of $-18.95 \text{ kg} \cdot \text{m}^{-3} \cdot \text{h}^{-1}$ and the amount of unreacted black mass entering the reactor (40.063 kg), the required reactor volume predicted by the kinetic model produced the same value of approximately 47.24 m^3 . The agreement between the hydraulic and kinetic approaches confirms that the reactor volume satisfies the reaction rate requirements. To provide operational flexibility and account for uncertainties in biological performance, mixing limitations, and potential scale-up effects, a design factor of 1.15 was applied to the calculated working volume. Conclusively, the design volume is 54.324 m^3 .

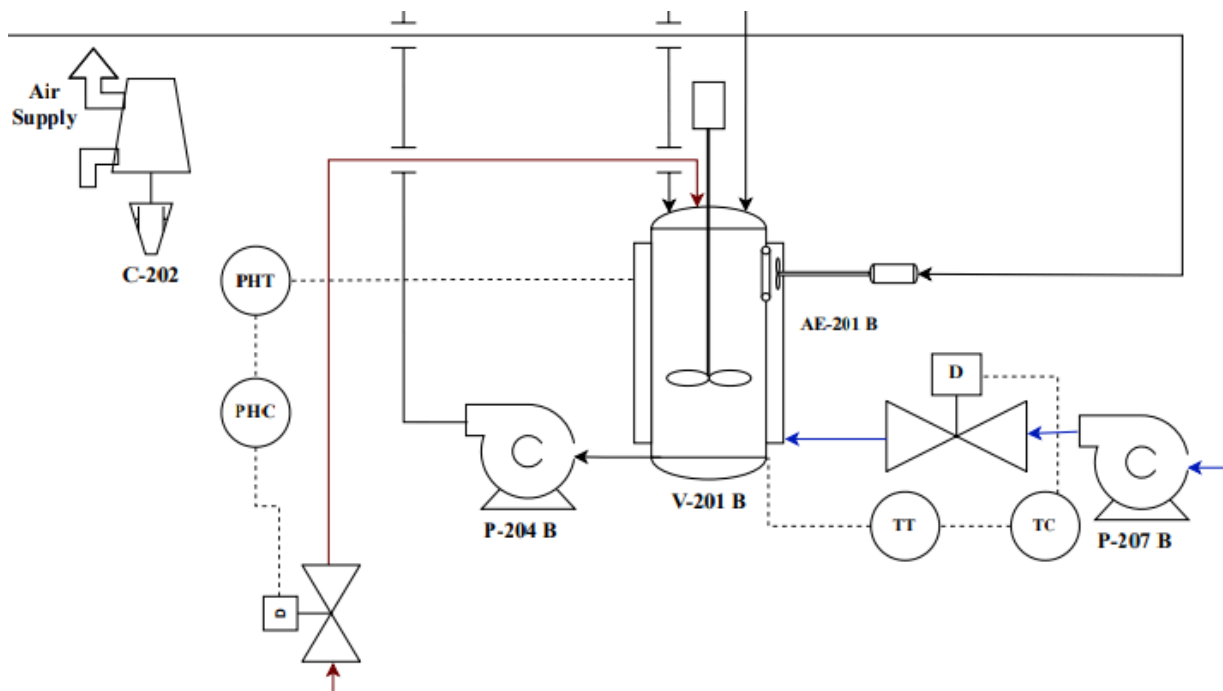


FIGURE 4.6.1 BIOLEACHING BATCH REACTOR VESSEL (V-201 A/B)

The reactor was modelled as a vertical cylindrical vessel. Using the standard cylinder volume relationship

$$V = \frac{\pi D^2 H}{4} = 2083.33 \text{ kg/h of batteries}$$

The reactor dimensions are determined to be 3.59m diameter and 5.38m height. Following the impeller common design guideline of $D_i \approx D/3$

$$D_i = \frac{3.59}{3} = 1.18m$$

As the reaction is temperature controlled at 28°C, the area of the heated jacket is 60.6m²

$$\begin{aligned} \text{Jacket Area} &= \pi DH \\ &= \pi(3.59)(5.38) \\ &= 60.6m^2 \end{aligned}$$

R-201A and R201B are the culture medium vessels that are the microbial inoculum for the bioreactors. R-201A receives nutrients, followed by sequential transfer to R-201B where additional bacteria growth occurs. The combined culture from R-201B feeds thickener TK-201, after which the thickened culture is combined with process water for delivery to the batch reactors (V-201A/B). The thickener will also recycle half of the bacteria back into the first vessel to be reused to make more bacteria. These vessels also have different dimensions due to their separate exact functions.

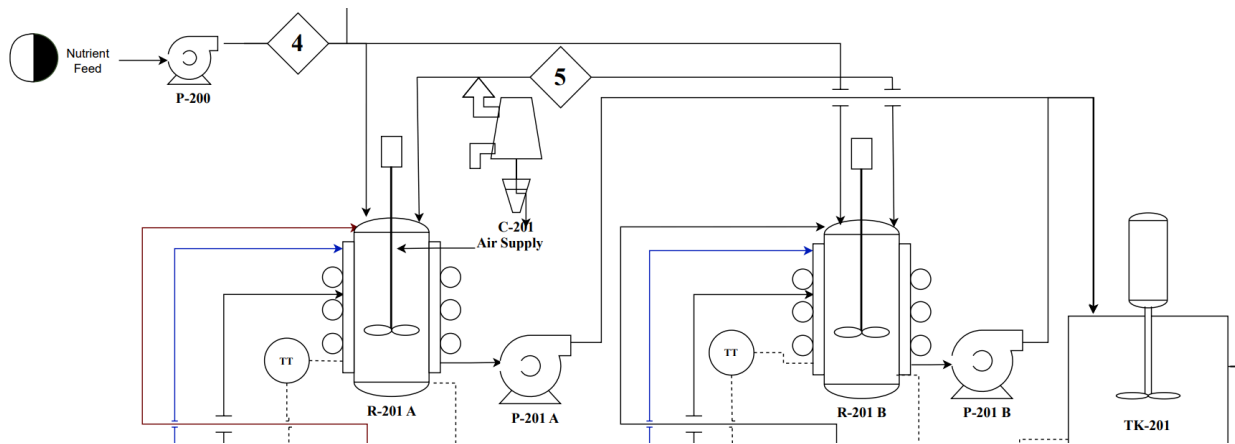


FIGURE 4.6.2 BIOREACTOR VESSELS (R-201 A/B)

4.7 Brine Discharge Bath

BI-100 serves as the mixing tank where crushed battery material is combined with brine solution prior to transfer to the biological reactors. The tank provides sufficient residence time for complete solids suspension and brine saturation. The total volume required is 126 m³ to accommodate the batch solids and associated liquid during the mixing cycle. The tank was designed rectangular for construction simplicity and effective solid suspension. This design size is sized up significantly for safety concerns as the batteries are incredibly volatile in their un discharged state.

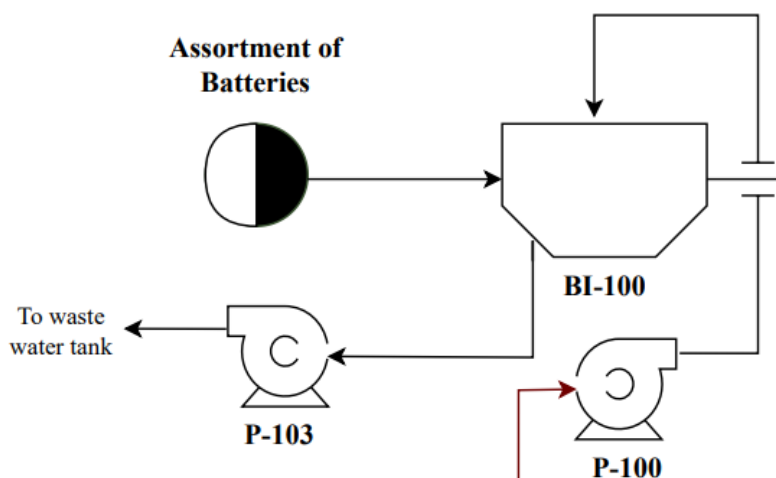


FIGURE 4.7.1 BRINE BATTERY DISCHARGE BATH (B-100)

4.8 Batch Reactors (Separators)

SP-301 is the first batch reactor where the bioreaction slurry (containing bacteria, dissolved metals, and unreacted solids) undergoes solid-liquid separation. The reactor operates with active bacterial culture present to facilitate selective metal precipitation and solids settling. The vessel receives a volumetric flowrate of 59.37 m³/h and requires a working volume of 4.9497 m³ for complete batch processing. To accommodate working fill and mixing dynamics the design volume was upscaled to 6.9952 m³. Mechanical agitation is provided by an impeller with 3.49 kW power draw and 0.780 m diameter to maintain solids suspension and promote bacterial activity during the separation cycle. This reactor's volume was calculated by finding the

working volume which is the flowrate divided by the density and multiplied with the residence time which is 5 minutes.

$$V = \frac{\dot{m}}{\rho} * \tau = \frac{95000 \frac{kg}{h}}{1600 \frac{kg}{m^3}} * 5min * \frac{1hr}{60 min} = 4.95m^3$$

Then using a height to diameter ratio of 1.2, the dimensions were calculated by rearranging the equation for volume of cylinder and plugging in 1.2*diameter instead of height to solve gives the following.

$$D = \sqrt[3]{\frac{4*V}{1.2*\pi}} = \sqrt[3]{\frac{4*(4.95m^3)}{1.2*\pi}} = 1.95m$$

$$H = 1.2 * D = 2.34m$$

Lastly, the impellers used are 45 degree pitch blade turbines and are calculated using a 0.4 impeller diameter to tank diameter ratio as shown below.

$$Impeller Diameter = 0.4 * D = 0.78m$$

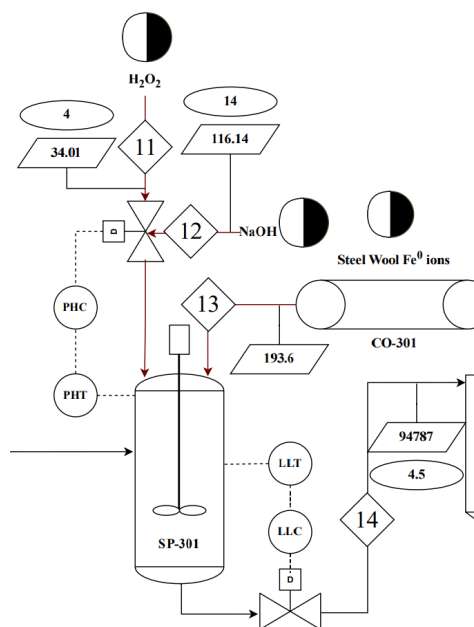


FIGURE 4.8.1 SEPARATION PHASE 1 (SP-301) (LEFT)

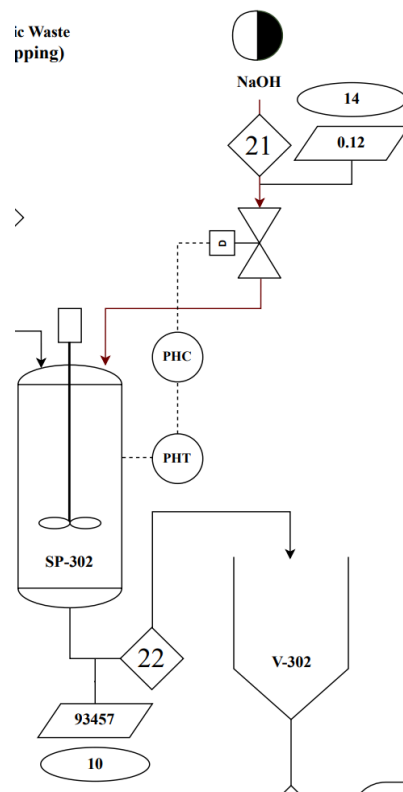


FIGURE 4.8.2 SEPARATION PHASE 2 (SP-302) (RIGHT)

SP-302 serves as the second batch separation reactor, positioned downstream of Filter Press 1 (FP-301) and the solvent stripper. The reactor receives leached slurry containing residual metals, unreacted solids, and culture medium for final solids settling prior to solvent extraction. The vessel processes a mass flow rate of 94.17 kg and volumetric flow rate of 58.98187 m³/h, requiring a working volume of 4.9051 m³ upsized to 6.9249 m³ for design margin, working fill, and headspace. The tank diameter and height are 1.94m and 2.33m respectively and the impeller diameter is 0.78m. All the values are calculated similar to SP-301 above.

SP-303 is the final batch separation reactor in the hydrometallurgical circuit and operates immediately upstream of Filter Press 303. It receives the slurry exiting Separator 2 and the Na₂CO₃ addition step, containing dissolved lithium and remaining impurities such as Fe²⁺, Fe⁰, Cu, Ni(OH)₂, unreacted solids, culture medium, and unreacted extractant. Within SP-303, lithium carbonate is precipitated while residual solids are allowed to settle prior to final solid-liquid separation in FP-303. The incoming mass flow rate is 94318 kg/h, corresponding to a volumetric flow rate of 58.9475 m³/h. The reactor is designed as a mechanically agitated vertical cylinder,

with a working liquid volume of 4.9123 m³ and an upscaled design volume of 8.935 m³ to provide mixing margin, and surge capacity during the precipitation step. The tank diameter and height are 1.95m and 2.33m respectively and the impeller diameter is 0.78m. All the values are calculated similar to SP-301 above.

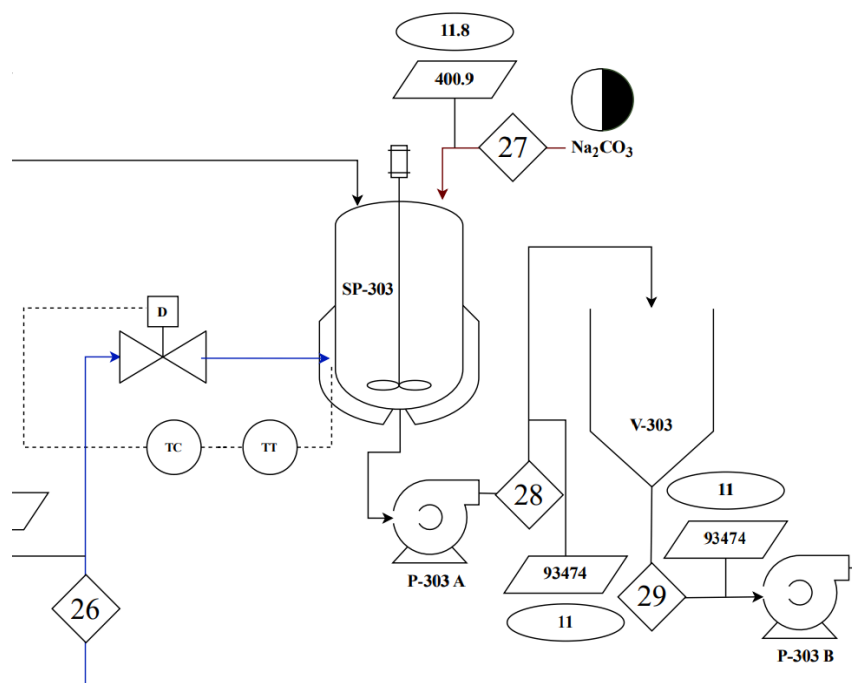


FIGURE 4.8.3 SEPARATION PHASE 3 (SP-303)

4.9 Solvent Extractor

SX-301 is a continuous countercurrent solvent extraction unit that receives filtered aqueous feed from Filter Presses FP-301A/B (duty/standby) and organic solvent for selective copper extraction. The unit processes a total feed mass of 94220 kg/h, with an aqueous feed volumetric flowrate of 85.24 m³/h combined with organic phase to give a total volumetric flowrate of 85.7 m³/h. The extractor maintains a working aqueous:organic ratio of 0.0058 to optimize copper mass transfer while minimizing organic carryover. The mixer compartment provides 10.088 m³ working volume with 7.57 kW of agitation power. The settler compartment requires 34.29 m³ upscaled volume, providing a plan area of 17.15m².

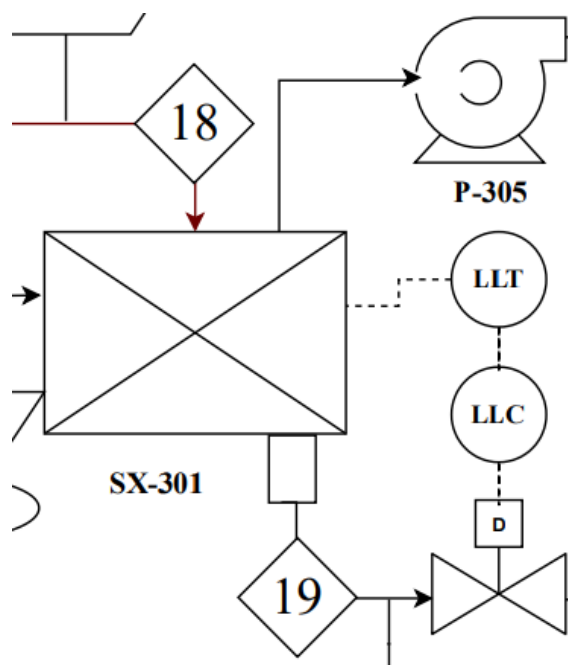


FIGURE 4.9.1 Solvent Extraction Cobalt Stripping (SX-301)

4.10 Pipes

The piping network was analysed to determine the pressure losses and flow characteristics throughout the process. The results are summarised in table 4.10.1 and table 4.10.2 below. These calculations were performed using standard fluid mechanics relationships for internal pipe flow.

TABLE 4.10.1: PIPE GEOMETRY AND PRESSURE DROP

Pipe	Mass Flowrate (kg/hr)	Diameter (m)	Diameter (cm)	Length (m)	Height Difference (m)	Equivalent Length (m)	Friction Loss (Pa)	Pressure Drop (Pa)
A-100 A	5	0.0009	0.1	6	-1	0.03	7271	-4501
A-100 B	5	0.0009	0.1	6	0	0.01	2231	2231
A-101	2396	0.0188	1.9	11	0	2.03	7404	7404
A-102	2083	0.0190	1.9	17	1	1.82	12015	22022
A-200	3898	0.0260	2.6	8	0	4.01	7725	7725
A-201 A	93550	0.1274	12.7	8	29	15.56	7825	298005
A-201 B	93550	0.1274	12.7	9	2	17.72	8905	28917
A-202	46775	0.0901	9.0	19	-2	32.22	24921	4908
A-203 A	46775	0.0901	9.0	19	0	27.22	21249	21249
A-203 B	93550	0.1274	12.7	17	2	50.14	24686	44698
A-204 A	47239	0.0723	7.2	30	2	15.93	9502	40894
A-204 B	47239	0.0723	7.2	21	1.8	15.93	9161	37413
A-205	45	0.0021	0.2	21	3	0.17	40946	95097
A-206	95000	0.1025	10.2	16	6	68.92	60330	154506
A-207 A	3600	0.0253	2.5	22	0	14.26	27740	27740
A-207 B	3600	0.0253	2.5	15	0	11.39	22029	22029
A-207 C	3600	0.0253	2.5	13	0	14.26	27314	27314
A-207 D	3600	0.0253	2.5	16	0	14.26	27456	27456
A-300	95000	0.1025	10.2	8	1	60.30	32922	48618
A-301	95000	0.1025	10.2	9	1	52.87	28977	44673
A-302	95000	0.1025	10.2	21	4	111.10	60999	123783
A-303	42	0.0022	0.2	2	0	0.04	3196	3196
A-304	94958	0.0022	0.2	7	4	0.51	38228	101012
A-305 A	94958	0.1024	10.2	8	4	21.38	11962	74746
A-305 B	94958	0.1024	10.2	10	1	63.55	34792	50488
A-306	94578	0.1022	10.2	23	3	24.29	14390	61478

A-307	3600	0.0258	2.6	14	0	12.89	32238	32238
A-308 A	94578	0.1022	10.2	15	4	21.33	12348	75132
A-308 B	94578	0.1022	10.2	8	1	63.40	34682	50378
A-309	94308	0.1021	10.2	10	-0.5	21.29	12072	4224
A-401	100	0.0029	0.3	11	6	0.73	55557	180929
A-402	85	0.0027	0.3	1	1	0.51	38110	59005
A-403	455	0.0094	0.9	9	2	0.50	23026	23026
A-404	0.50	0.0002	0.0	9	3	0.01	51343	114029
A-405	405	0.0053	0.5	14	5	1.55	56958	181545

TABLE 4.10.2: REYNOLD'S NUMBER, FRICTION FACTOR, AND FITTINGS

Pipe	Reynold's Number	friction factor	90° elbow (K=0.9)	45° elbow (K=0.35)	Globe Valve (K=8)	Ball Valve (K=0.5)
A-100 A	1319	0.0716	1	2	0	2
A-100 B	1319	0.0716	0	0	0	1
A-101	28185	0.0259	2	0	0	2
A-102	1939	0.0523	5	0	0	1
A-200	44191	0.0234	4	0	0	0
A-201 A	12371	0.0295	4	0	0	0
A-201 B	12371	0.0295	4	0	0	1
A-202	8748	0.0324	4	0	1	0
A-203 A	8748	0.0324	2	0	1	0
A-203 B	12371	0.0295	4	0	1	0
A-204 A	115610	0.0186	4	0	0	1
A-204 B	115610	0.0186	4	0	0	1
A-205	383	0.1060	1	0	1	0
A-206	163950	0.0172	4	0	1	0
A-207 A	40447	0.0238	6	0	1	0
A-207 B	40447	0.0238	3	0	1	0
A-207 C	40447	0.0238	6	0	1	0
A-207 D	40447	0.0238	6	0	1	0
A-300	163950	0.0172	2	1	1	0
A-301	163950	0.0172	1	0	1	0
A-302	163950	0.0172	3	0	2	0
A-303	3447	0.0494	1	0	0	0
A-304	3447	0.0494	4	0	1	0

A-305 A	163913	0.0172	4	0	0	0
A-305 B	163913	0.0172	3	0	1	0
A-306	163585	0.0173	4	0	0	1
A-307	172924	0.0196	2	0	1	0
A-308 A	163585	0.0173	4	0	0	0
A-308 B	163585	0.0173	3	0	1	0
A-309	163351	0.0173	4	0	0	0
A-401	3069	0.0492	5	0	1	0
A-402	4250	0.0459	1	0	1	0
A-403	121	0.2016	3	0	1	0
A-404	326	0.1503	3	0	1	0
A-405	8496	0.0366	3	0	1	0

Using the known fluid densities, all values such as inner pipe diameter, equivalent length, friction loss, and pressure drop were determined. Shown below using pipe A-204A will be used as an example for all the sample calculations. It is also important to note that all fluids in the pipes were fixed at a constant 2m/s velocity.

Length and height differences was determined from the plant layout, and the inner pipe diameter is calculated using the equation below.

$$D = \sqrt{\frac{4 \cdot \dot{m} \cdot \frac{1}{3600}}{u \cdot \rho \cdot \pi}} = \sqrt{\frac{4 \cdot (47239 \frac{kg}{hr}) \cdot \frac{1hr}{3600s}}{(2 \frac{m}{s}) \cdot (1600 \frac{kg}{m^3}) \cdot \pi}} = 0.0723m$$

To find the total pressure drop it is first required to find Reynold's number which is found using the formula below.

$$Re = \frac{\rho \cdot u \cdot D}{\mu} = \frac{(2 \frac{m}{s}) \cdot (1600 \frac{kg}{m^3}) \cdot 0.0723m}{0.002Pa \cdot s} = 115610$$

The determination of the friction factor was done using the Swamee-Jain equation. It is important to note that the equation takes into account the pipe's absolute roughness, which in this case stainless steel is used and it has an absolute roughness of 0.015mm.

$$f = \frac{0.25}{(\log_{10}(\frac{\epsilon}{3.7D} + \frac{5.74}{Re^{0.9}}))^2} = \frac{0.25}{(\log_{10}(\frac{0.015mm}{3.7*(0.0723m*\frac{1000mm}{1m})} + \frac{5.74}{115610^{0.9}}))^2} = 0.0186$$

The next step was to then find the Equivalent pipe length. This was done as shown in table 4.10.2 by first counting the number of fittings and bends on the pipe and assigning a K value depending on the angle of the bend or what type of valve is present. In A-204, there are 4 90 degree elbows, and 1 ball valve present. The calculation for the equivalent pipe length is shown below using K=0.9 for 90 degree elbows and K=0.5 for ball valves.

$$L_{equivalent} = \frac{D}{f} * (\# \text{ of } 90 \text{ degree elbows} * K_{90 \text{ degree}} + \# \text{ of ball valves} * K_{ball \text{ valve}})$$

$$L_{equivalent} = \frac{0.0723m}{0.0186} * (4 * 0.9 + 1 * 0.5) = 15.93m$$

Once the equivalent pipe length is calculated, the friction loss can now be determined. This can be done using the formula below.

$$Friction \text{ Loss} = f * (\frac{L_{total}}{D}) * (\frac{\rho * u^2}{2}) = 0.0186 * \frac{15.93m + 30m}{0.0723m} * \frac{(1020 \frac{kg}{m^3})(2 \frac{m}{s})^2}{2}$$

$$Friction \text{ Loss} = 9502 \text{ Pa}$$

Lastly, to calculate the total pressure drop, the friction loss is added with the height difference multiplied by gravity and the density to give us our final value.

$$Pressure \text{ Drop} = Friction \text{ Loss} + \Delta z * g * \rho$$

$$Pressure \text{ Drop} = 9502 + 2m * 9.81 \frac{m}{s^2} * 1600 \frac{kg}{m^3} = 40894 \text{ Pa}$$

4.11 Pumps

The pumping requirements for the process were determined based on the volumetric flowrate, pressure losses in the piping network, and elevation differences between equipment. The results of these calculations are summarised in Table 4.11.1, which lists the pump designation, volumetric flowrate, hydraulic power requirement, and available Net Positive Suction Head (NPSH).

TABLE 4.11.1: PUMP HYDRA AND NPSH

Pump	Volumetric Flow (m ³ /s)	Hydraulic Power (kW)	NPSH available (m)
P-100 A	0.00000122	0	0.0
P-100 B	0.00000122	0.0000039	8.1
P-101	0.000555	0.0059	7.7
P-102	0.000567	0.0178	9.6
P-200	0.00106	0.0117	9.0
P-201 A	0.0255	10.5	38.0
P-201 B	0.0255	1.05	10.9
P-202	0.0127	0.0893	5.3
P-203 A	0.0127	0.387	7.6
P-203 B	0.0254	1.63	9.3
P-204 A	0.0082	0.479	7.6
P-204 B	0.0082	0.438	7.4
P-205	0.00000679	0.000923	6.2
P-206	0.0165	3.64	8.4
P-207 A	0.00100	0.0398	7.1
P-207 B	0.00100	0.0316	7.7
P-207 C	0.00100	0.0392	7.2
P-207 D	0.00100	0.0394	7.2
P-300	0.0165	1.15	5.1
P-301	0.0165	1.05	5.4
P-302	0.0165	2.92	6.3
P-303	0.00000729	0.0000333	6.0

P-304	0.0165	2.38	7.8
P-305 A	0.0165	1.76	9.5
P-305 B	0.0165	1.19	5.0
P-306	0.0165	1.44	8.3
P-307	0.00104	0.0480	-3.1
P-308 A	0.0164	1.76	9.4
P-308 B	0.0164	1.18	5.0
P-309	0.0164	0.0988	5.0
P-401	0.0000130	0.00337	8.1
P-402	0.0000111	0.000934	3.9
P-403	0.000137	0.00452	10.7
P-404	0.00000007	0.0000106	5.3
P-405	0.0000443	0.0115	6.6

It was first required to calculate the volumetric flow rate for each pump, which was done by converting the mass flow rates from table 4.10.1 into kg/s and then dividing it by density. Sample calculations will be done on pump P-204A.

$$Q = \frac{\dot{m}}{\rho} = \frac{47239 \frac{kg}{h} * \frac{1h}{3600s}}{1600 \frac{kg}{m^3}} = 0.00820114 \frac{m^3}{s}$$

Once the volumetric flow rate is calculated, hydraulic power can be found using the formula below.

$$P = \frac{Q * \text{Pressure Drop} * \frac{1}{1000}}{\eta} = \frac{0.00820114 \frac{m^3}{s} * 40894 Pa * \frac{1}{1000}}{0.7} = 0.479 kW$$

Finally, NPSH is calculated.

$$NPSH_{available} = \frac{P_{atm}}{\rho g} + z - \frac{\text{Friction Loss}}{\rho g} - \frac{P_v}{\rho g}$$

$$NPSH_{available} = \frac{101325 Pa}{(1600 \frac{kg}{m^3})(9.81 \frac{m}{s^2})} + 2m - \frac{9502}{(1600 \frac{kg}{m^3})(9.81 \frac{m}{s^2})} - \frac{3600 Pa}{(1600 \frac{kg}{m^3})(9.81 \frac{m}{s^2})}$$

$$NPSH_{available} = 7.6m$$

Chapter 5.Process & Instrumentation Diagram (P&ID)

LEGEND

Line Types & Fluid Color Codes

Line Type	Visual Style	Application in Plant
Major Process Flow		Main battery feed, black mass slurry, pregnant leach solution (PLS).
Reagents		Reagent dosing
Cooling / Process Water		Cooling water,, makeup water.
Electrical Signal		4-20mA signals from transmitters to SCADA, or controllers to motor actuators.
Pneumatic Signal		Compressed air signals actuating pneumatic control valves.

Symbol Selection Matrix

Required Asset	Shape
Electrical Transmitter	
Electrical Controller	
SCADA / PLC Controller	
Globe Valve	
Ball Valve	
Digital Valve	
Centrifugal Pump	
Peristaltic Pump	
Crusher	
Mixer	
Automatic Air Vents	
Heat Exchanger	
Pressure Relief Valve	

SYMBOL ABBREVIATIONS

PREFIX

C CONDUCTIVITY
F FLOW
L LEVEL
M MIXER
pH pH
P PRESSURE
T TEMPERATURE

SUFFIX

A ALARM
C CONTROLLER
E ELEMENT
I INDICATOR
M METER
R RECORDER
S SWITCH
T TRANSMITTER

5.1 Instrumentation Philosophy

Our plant instrumentation philosophy demands absolute control over biological and chemical interactions. Safety dictates every design choice. We must guarantee smooth process operation and pinpoint product quality regardless of external disturbances. Automation governs the entire facility. Operators rarely touch a manual valve because the control architecture runs autonomously to prevent environmental releases. Field devices measure raw physical data right at the pipe. Local panels grant technicians immediate physical access to specific equipment zones. Programmable Logic Controllers (PLC) handle the raw signal processing. Acting as the operational brain, a centralized Supervisory Control and Data Acquisition (SCADA) system manages everything from the control room. This multi-layered architecture guarantees highly responsive plant management (Sayevich, 2025).

Piping and Instrumentation Diagram (P&ID) Legend

Establishing a definitive visual standard prevents severe rework during the complex phases of plant construction. The piping and instrumentation diagrams must be universally readable.

FIGURE 5.1.1. SYSTEMS AND ALARMS LEGEND FOR THE P&ID

We implement global drafting guidelines based on established industry standards. This achieves absolute clarity across every single schematic. Instrument bubbles visually map the physical location of the control hardware. A plain circle represents a field-mounted instrument

located directly on the process line. A circle drawn with a solid horizontal line indicates the primary control system, securely housed inside the main control room. A circle within a square denotes a local control panel mounted physically next to the equipment (Sayevich, 2025).

The schematics explicitly differentiate between various valve types to guarantee safe operation. Gate valves handle complete flow isolation. Globe valves execute manual fluid throttling. Digital check valves physically prevent hazardous chemical backflow. Actuated control valves govern the automated loops. Line formatting also conveys critical information. Thick black lines trace the main process lines. Red piping strictly isolates highly corrosive acidic reagents. Blue lines denote the vital cooling and process water loops.

Rather than relying on manual intervention, the facility utilizes automated controllers to eliminate steady-state error across all operational zones. Hardwired safety interlocks bypass standard control loops entirely to prevent catastrophic equipment failure. Alarms act as the primary early warning system via flashing indicators on the SCADA screens. We engineered all critical control valves for dangerous reagents to be fail-closed. A total power or compressed air failure mechanically forces the supply valves shut. The plant always defaults to a physically secure state.

5.2 Piping and Instrumentation Diagram (P&ID)

A Piping and Instrumentation Diagram translates abstract process chemistry into physical reality. Every mechanical unit and digital sensor requires definition. We segmented the sprawling facility into five highly specific operational zones to maintain extreme clarity across the schematics. This modular approach allows operators to instantly isolate faults (Sayevich, 2025). The following sections detail the control loops and safety interlocks engineered into each sector.

5.2.1 Zone 1: Pre-Treatment and Brine Discharge

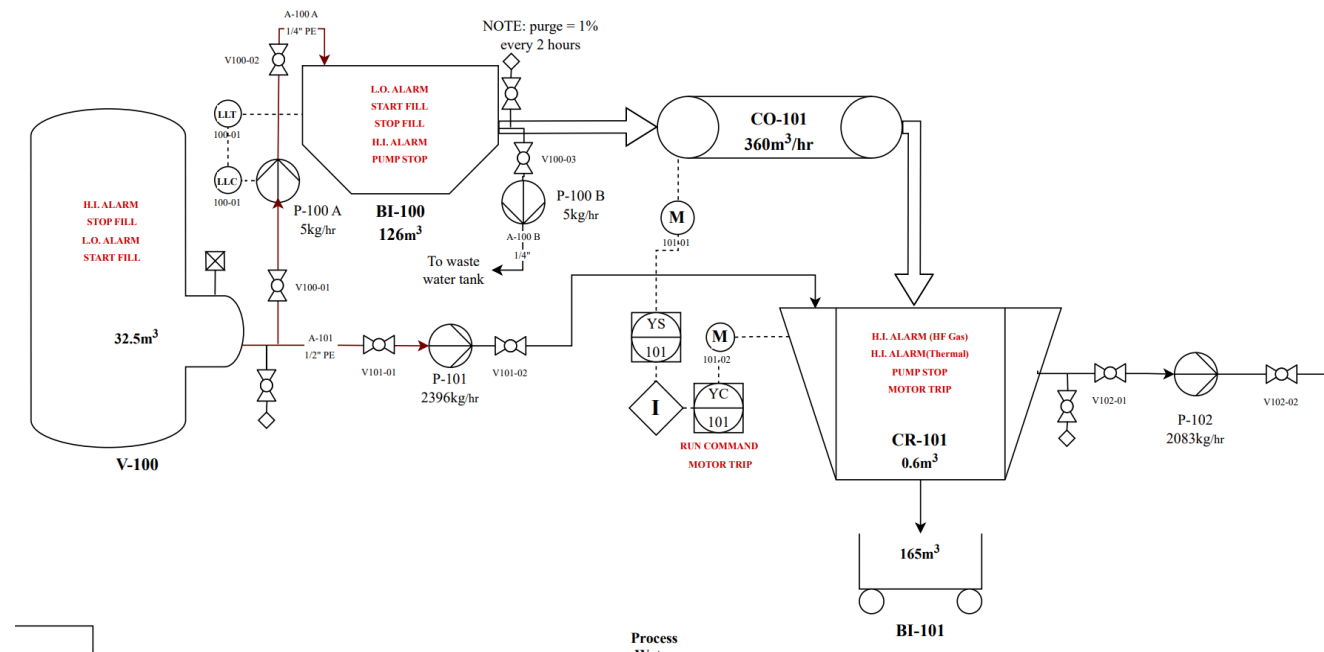


FIGURE 5.2.1. ZONE 1 SYSTEMS AND ALARMS P&ID

Raw spent batteries plunge straight into the Discharge Pool (BI-100) before reaching the Brine Solution Vessel (V-100). This transfer is dangerous. Controlling the initial liquid volume is absolutely critical. A field-mounted liquid-level-indicating transmitter continuously tracks the conductive brine depth to prevent dry running. Raw data feeds directly into a centralized level indicator controller. The central logic unit instantly modulates a pneumatic control valve on the makeup water line. Operating the heavy Black Mass Transfer Pump (P-102) demands brute mechanical force. Cavitation will destroy the impeller in minutes. A sudden drop in feed pressure triggers an immediate hard-wired motor trip (Crawl & Louvar, 2011). The Battery Crusher (CR-101) tears through thick metal casings. Lethal hydrogen fluoride gas can escape if a high-energy cell ruptures improperly. Toxic gas detectors blanket the enclosure. They trigger high-priority alarms (LAH-HF). Nitrogen floods the chamber instantly upon detection to suppress fire.

5.2.2 Zone 2: Bioreactor Systems and Alarms

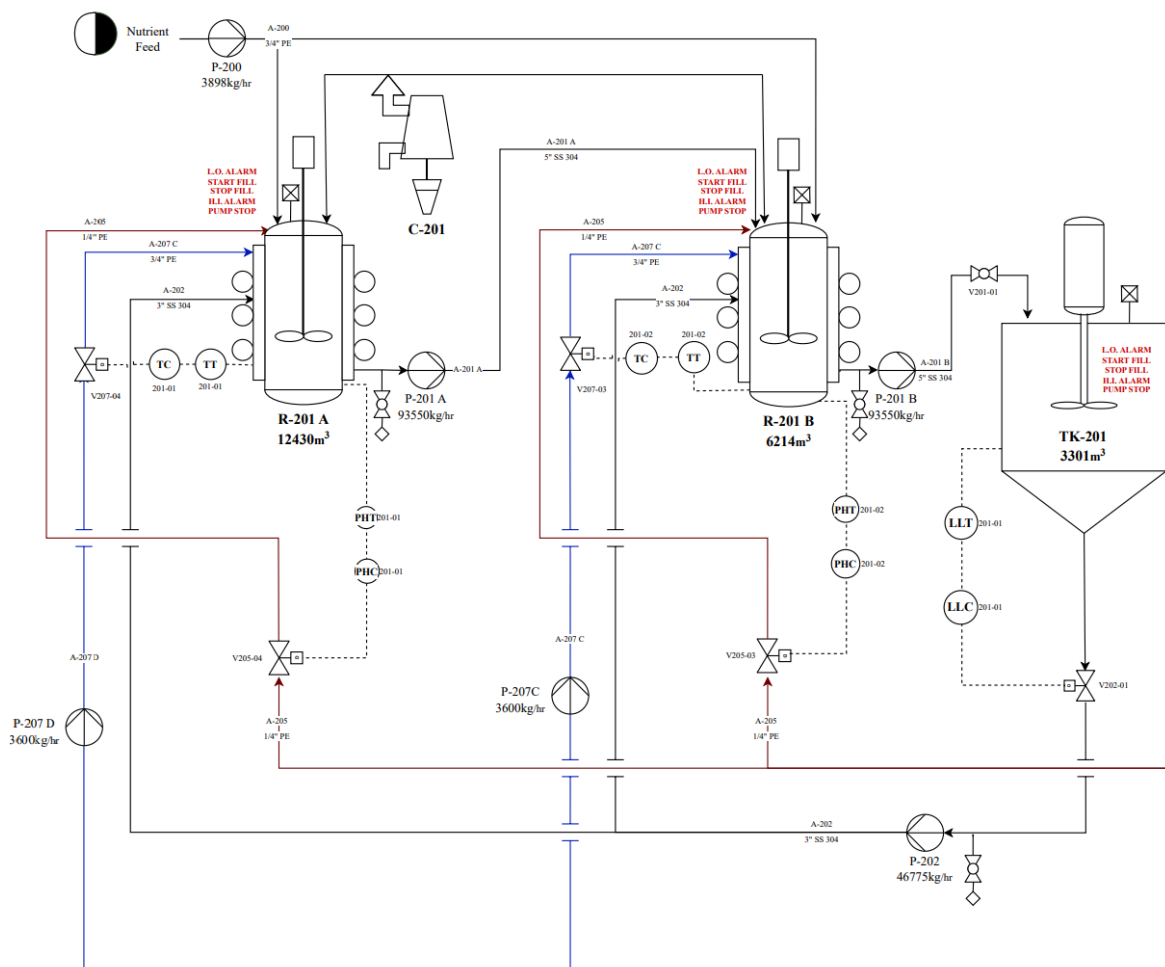


FIGURE 5.2.2. ZONE 2 BIOREACTOR SYSTEMS AND ALARMS P&ID

Biological cultivation dictates the pace of the entire downstream plant. The A. Ferrooxidans Bioreactors (R-201 A/B) breed the massive microbial colonies required for extraction. Bacteria require continuous oxygenation to survive. Flow-indicating transmitters track the exact volume of compressed air flowing through the main header from the Bioreactor Air Compressors (C-201). We established strict level controls to prevent tank overflow during continuous media feeding. A dedicated level transmitter watches the Culture Medium Pump (P-203 A/B) throughput. Tank flooding triggers an automatic shutdown. The system forces a complete electrical lockout of the Nutrient Feed Pumps (P-200) if the liquid approaches the brim.

Maintaining a perfect 30°C environment for aggressive bacterial reproduction requires dedicated temperature transmitters (Rawlings & Johnson, 2007).

5.2.3 Zone 2: Bioleaching Systems and Alarms

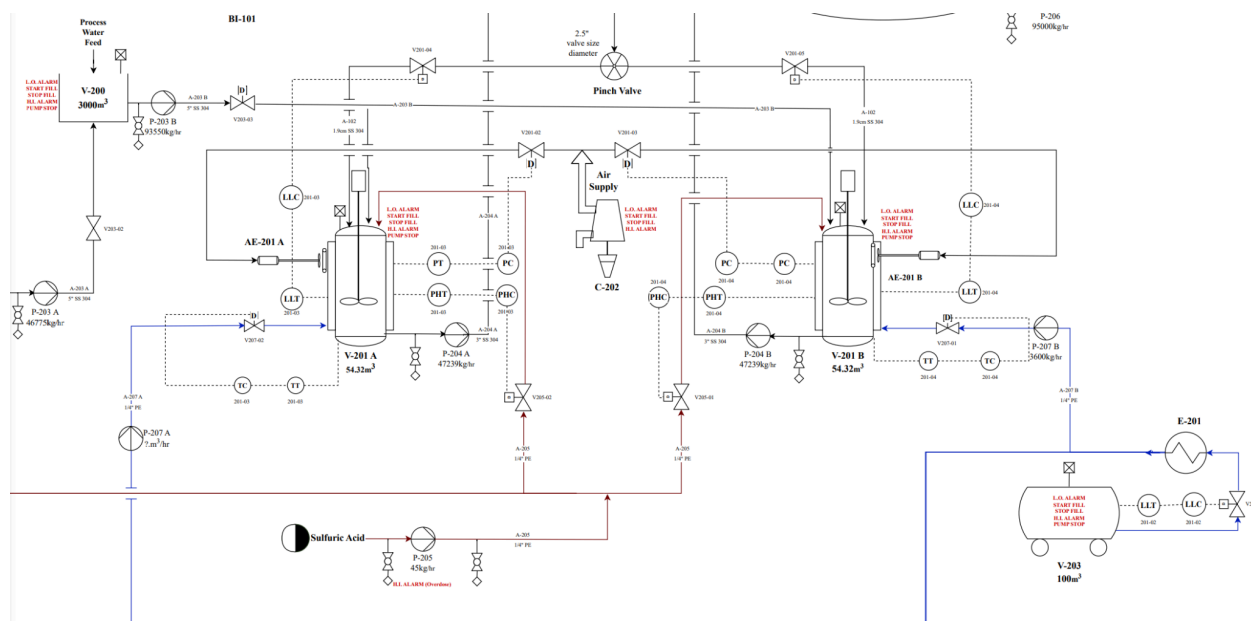
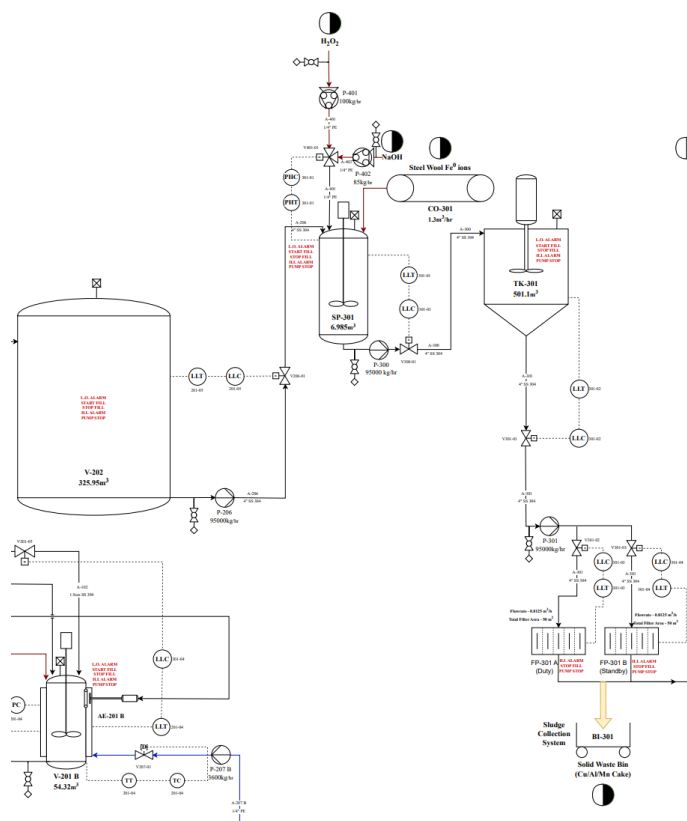


FIGURE 5.2.3. ZONE 2 BIOLEACHING SYSTEMS AND ALARMS P&ID

The Bioleaching Batch Reactors (V-201 A/B) represent the most volatile chemical environment in the facility. The slurry requires a strict pH of 1.5. A pH transmitter extends deep into the highly acidic sludge. The central controller evaluates this signal continuously. It dispatches a fast electrical signal to a local speed controller if the acidity climbs. The Sulfuric Acid Dosing Pump (P-205) accelerates instantly. Concentrated acid floods the vessel until the chemical balance drops back to the required setpoint (Garcia et al., 2025). Heat presents a massive threat to the active microbes. Bacterial metabolism actively produces huge quantities of thermal energy. Live thermal data streams directly from a submerged temperature transmitter. To save the culture, the indicator controller throttles a pneumatic valve to rush chilled water from the Bioleaching Heat Exchanger (E-201) into the external jackets. This action violently strips the heat away to lock the temperature at exactly 27°C. High-level alarms warn operators of impending overfills during the discrete batch loading sequence.

5.2.4 Zone 3 Phase 1: Separation Systems and Alarms

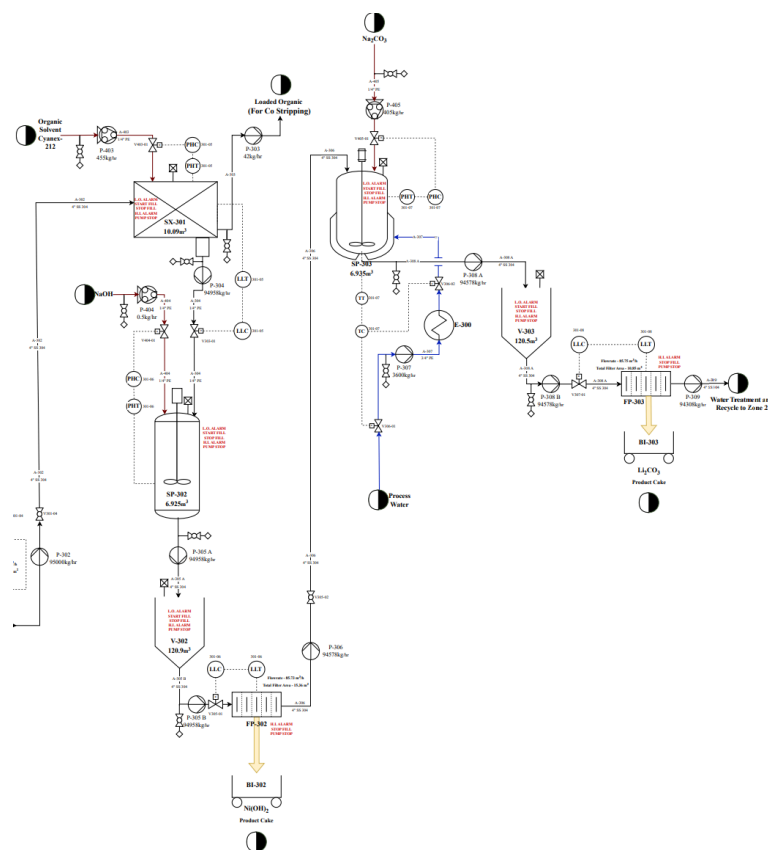


Processing shifts from biological extraction to rigid chemical precipitation. The Impurity Precipitation Reactor (SP-301) receives the saturated PLS. Liquid level indicating transmitters govern the batch volumes strictly. Operators must never overfill these tanks. High-level alarms flash red on the SCADA screens to warn technicians. An independent high-high level interlock takes total control if a worker ignores the primary warning. Upstream centrifugal feed pumps (P-204 A/B) lose all power instantly. Isolating aluminum and manganese requires a precise pH shift to exactly 4.5.

FIGURE 5.2.4. ZONE 3 PHASE 1 SEPARATION SYSTEMS AND ALARMS P&ID

Automated valves dose sodium hydroxide to hit this chemical metric. Thickener Sludge Removal Pumps (P-301) drive the heavy resulting slurry toward the primary filtration bank. We utilize dual Filter Presses (FP-301 A/B) to squeeze the chemical metric. High-pressure lines connect these specific mechanical units. Pressure indicating transmitters watch for deadly blockages in the sludge piping. Rupture disks sit atop the manifolds. They blow safely into a containment basin before a pressurized pipe explodes (Sinnott & Towler, 2019).

5.2.5 Zone 3 Phase 2 & 3: Separation Systems and Alarms



The final purified liquid enters the Organic Solvent Extractor (SX-301). Cyanex-272 mixes violently with the aqueous stream. Highly flammable organic solvents introduce severe fire risks into this specific quadrant. Explosion-proof wiring runs entirely throughout this zone. Lower explosive limit sensors constantly sniff the ambient air for escaping vapors. A detected leak slams all fail-closed reagent valves shut mechanically. The Nickel Precipitation Batch Reactor (SP-302) acts as the next destination.

FIGURE 5.2.5. ZONE 3 PHASE 2 & 3 SEPARATION SYSTEMS AND ALARMS P&ID

Sodium hydroxide dosing relies on a dedicated pH control loop to pinpoint a pH of exactly 10. Heavy green nickel hydroxide sludge drops entirely out of the liquid phase. Lithium remains the absolute final target. The Lithium Precipitation Unit (SP-303) is outfitted with a highly aggressive temperature control architecture. Steam coils wrap tightly around the steel vessel. Continuous monitoring of the warming liquid relies on a submerged temperature transmitter. The controller forces the steam supply valves wide open until the internal environment hits exactly 60°C. This immense thermal shock triggers inverse solubility (Zhang et al., 1998). The lithium carbonate crystallizes rapidly. Filter presses isolate the final premium products for bagging.

5.3 Control Strategy Summary

Managing the volatile biological and chemical processes requires a highly robust control architecture. We rely extensively on proportional-integral-derivative controllers to eliminate steady-state error and prevent forced oscillations across the entire plant. PID controllers offer incredible stability. They rapidly adjust the process variables to maintain tight dynamic response. Operators in the central control room can monitor these automated adjustments without needing to manipulate physical field valves manually.

Sensor Selection and Rationale

Selecting the correct measurement instruments dictates the success of our logic loops. Slurries destroy delicate sensors. We must evaluate the advantages and disadvantages of various sensor technologies to handle the highly acidic and abrasive black mass mixtures.

TABLE 5.3.1: LEVEL SENSOR SELECTION GUIDE

Sensor Type	Advantages	Disadvantages	Selection Rationale
Ultrasonic	Absence of moving parts, noncontact measurement, reliability unaffected by composition changes	Sensitive to temperature variation, weak echo from dispersion, reflective properties affect surface read	Selected. Perfect for corrosive, abrasive slurries due to noncontact design.
Differential Pressure	Externally installed, easily applied, can use block valves	Works with clean liquids only, errors introduced if fluid density varies	Rejected. Heavy black mass slurries will plug the lines.
Radar	Noncontact measurement, highly accurate in storage tanks	Extremely expensive, foam and low-dielectric material cause interference	Rejected. Cost prohibitive for standard process tanks.

TABLE 5.3.2: TEMPERATURE SENSOR SELECTION GUIDE

Sensor Type	Advantages	Disadvantages	Selection Rationale
Resistance Thermometer (RTD)	System accuracy, low spans available, fast response	Self-heating can be a problem, long-term drift, some forms expensive	Selected. Provides the exact high accuracy and stability needed for strict 27°C biological control.
Thermocouple	Small size, low cost, convenient mount, extremely wide range	Cold working on wires affects calibration, not as simple as direct-reading	Rejected. Wide range is unnecessary for low-heat biological reactions.
Bimetallic	Less subject to breakage, dial reading, less costly than electrical	Less accurate than glass-stem, changes calibration with rough handling	Rejected. Lacks digital transmission capability for SCADA.

TABLE 5.3.3: FLOW SENSOR SELECTION GUIDE

Sensor Type	Advantages	Disadvantages	Selection Rationale
Magnetic Flowmeter	Handles liquid slurries perfectly, 30:1 range, virtually no pressure loss	Cannot be used for gases	Selected. The optimal choice for heavy, abrasive battery waste slurries.
Orifice Plate	Cheapest type, very simple structure, widely applicable	High irrecoverable pressure loss, low accuracy, low rangeability	Rejected. Slurry will immediately erode and destroy the restrictive orifice.
Positive Displacement	Good accuracy and high rangeability (>10:1)	Limited to clean fluids, bulky, heavy, requires regular maintenance	Rejected. Close meshing surfaces will seize when exposed to solids.

TABLE 5.3.4: pH SENSOR SELECTION GUIDE

Sensor Type	Advantages	Disadvantages	Selection Rationale
Glass Bulb Electrode	Highly accurate, inexpensive, wide measurement range.	Extremely fragile. High-solids slurry causes rapid mechanical breakage and clogs the reference junction.	Rejected. The abrasive black mass will physically shatter the glass dome.
ISFET (Ion-Selective Field Effect Transistor)	No glass components. Solid-state design handles thick, abrasive slurries effortlessly. Highly durable.	Prone to drift over long periods. Requires a stable reference junction.	Selected. The heavy slurry mixture demands a rugged, non-glass surface to prevent continuous breakage.
Antimony Electrode	Rugged solid metal design. Can withstand massive physical abuse.	Highly sensitive to oxidation-reduction potential changes. Less accurate overall.	Rejected. The aggressive Fe(III) oxidizing environment will skew the pH readings entirely.

We selected ultrasonic level sensors for all primary bioleaching vessels. These instruments provide noncontact measurement, completely avoiding the severe corrosion and abrasion caused by the dense black mass slurry. Resistance temperature detectors govern thermal monitoring. RTDs deliver exceptional accuracy and stability at the specific 27°C operating temperature of our bacterial cultures. Magnetic flowmeters track the movement of our liquid streams. They handle thick liquid slurries effortlessly because their design introduces no obstructions or pressure loss into the piping. We selected the ISFET pH sensor for all active processing vessels. Heavy black mass particles bouncing against the probe will easily shatter a standard glass dome. ISFET sensors withstand this aggressive mechanical grinding while still delivering the high-precision readings required to maintain the strict 1.5 pH biological setpoint.

Control Loops, Alarms, and Safety Systems

The bioleaching reactors demand a strict architectural design to sustain microbial life. Precision is absolute. The pH control loop utilizes a field-mounted transmitter to constantly monitor the acidity of the internal slurry. Acidity must remain locked exactly at a pH of 1.5. If the pH climbs, the central controller dispatches a fast electrical signal to a local speed controller attached to the sulfuric acid dosing pump. Concentrated acid floods the tank. The pump decelerates automatically once the chemical balance is restored.

Temperature regulation actively prevents catastrophic microbial death. The bacterial metabolism of the black mass generates an estimated 60 kW of continuous exothermic heat during the primary leaching phase. The colony will boil without immediate thermal intervention. A submerged temperature transmitter detects any thermal deviations above the 27°C survival threshold. Exactly 3,600 kg/h of chilled cooling water rushes into the external reactor jacket to strip away the metabolic heat once the pneumatic valve opens.

Volume dictates physical safety. A dedicated level control loop governs the working capacity of the main vessels. The level controller continuously modulates a heavy discharge valve to perfectly match the 94,477 kg/h total outflow heading into the holding tanks. High and low alarms provide operators with immediate visual warnings. If a high-level alarm is ignored, a hardwired safety interlock bypasses the central logic and forcefully shuts down the upstream feed pump. Aeration keeps the bacteria alive. Flow indicating transmitters measure the precise volume of compressed air moving from the compressors into the internal spargers. The flow controller continually adjusts the pneumatic supply valves to maintain optimal oxygenation.

Equipment protection requires dedicated safety interlocks. We installed hardwired low-pressure switches on the suction lines of all centrifugal slurry transfer pumps. This immediate electrical cutoff physically prevents severe mechanical cavitation from destroying the expensive pump impellers during a sudden loss of suction. A pressure drop directly trips the motor. We engineered every single critical automated valve handling hazardous reagents to be entirely fail-closed. A total loss of electrical power or pneumatic air supply causes these specific chemical valves to slam completely shut instantly. The plant becomes physically secure.

Chapter 6. 3D Plant Layout

6.1 Layout Philosophy

The layout of the biohydrometallurgical lithium-ion battery recycling facility is designed with a core philosophy that balances inherently safe design, operational efficiency, and maintenance accessibility.

Safety Spacing

The plant equipment is spaced evenly across the site to prevent personnel crowding and to provide a dedicated buffer area to mitigate accidents. In accordance with the Center for Chemical Process Safety (CCPS) guidelines, adequate separation distances are strictly maintained between the facility's distinct operational zones (Aisyah, 2021). By actively avoiding equipment clustering, the layout minimizes the risk of a domino-effect hazard. Examples include a thermal event in the Pre-Treatment Crusher Vault (Zone 1) propagating to the highly flammable Solvent Extraction area (Zone 3).

Accessibility and Maintenance

Because the equipment spacing is deliberately wide, accessibility and maintenance are ensured to be ample. The layout actively avoids crowding process space, providing sufficient clearance for heavy-lifting equipment, such as forklifts or overhead cranes. This unobstructed space is critical for routine maintenance tasks, such as removing the heavy plates from the Filter Presses (FP-301 to FP-303) or extracting the agitator shafts from the bioleaching Reactors without disrupting adjacent operations (Chemical Times, 2024).

Flow of Materials and Utilities

The plant's physical layout follows the logical sequential flow of the chemical process, operating linearly from the Raw Material Receiving area through to the Final Products Shipping zone. To ensure that pipes do not interfere with workers, all major process fluids, hazardous acids, and utilities (cooling water, compressed air) are routed via elevated pipe racks (Chemical Times, 2024). This overhead routing keeps the ground-level walkways clear for safe operator movement and emergency evacuation. Furthermore, the layout employs a tiered vertical

arrangement for the primary reactor vessels, utilizing gravity-driven flow where possible to minimize reliance on mechanical pumping between stages.

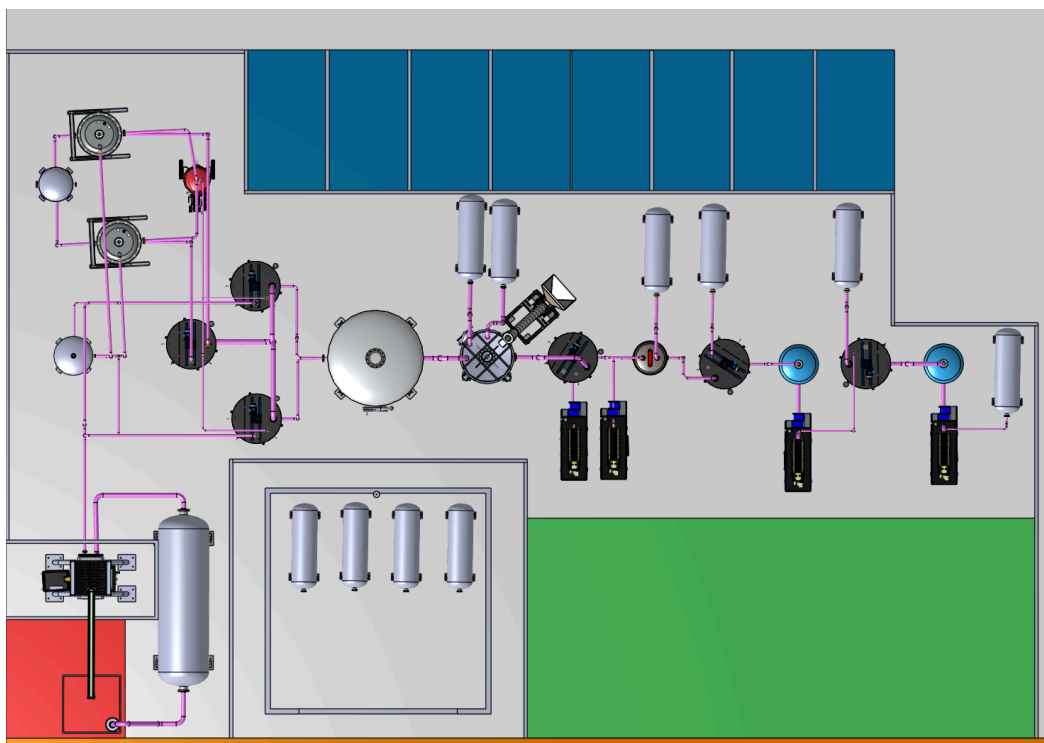


FIGURE 6.1. OVERHEAD VIEW OF THE PLANT

6.2 3D Model

To visualize the spatial dynamics and physical footprint of the designed facility, a 3D model of the plant was constructed. To ensure the design is practical and realistic, the plant was modelled to scale. This allows for an accurate assessment of operational accessibility, vertical clearances, and maintenance spatial requirements.

Figure 6.2 demonstrates this scaling by including a standard human figure on the elevated platform next to the filtration equipment, providing a clear visual reference for the true size of the industrial vessels, structural supports, and collection bins.

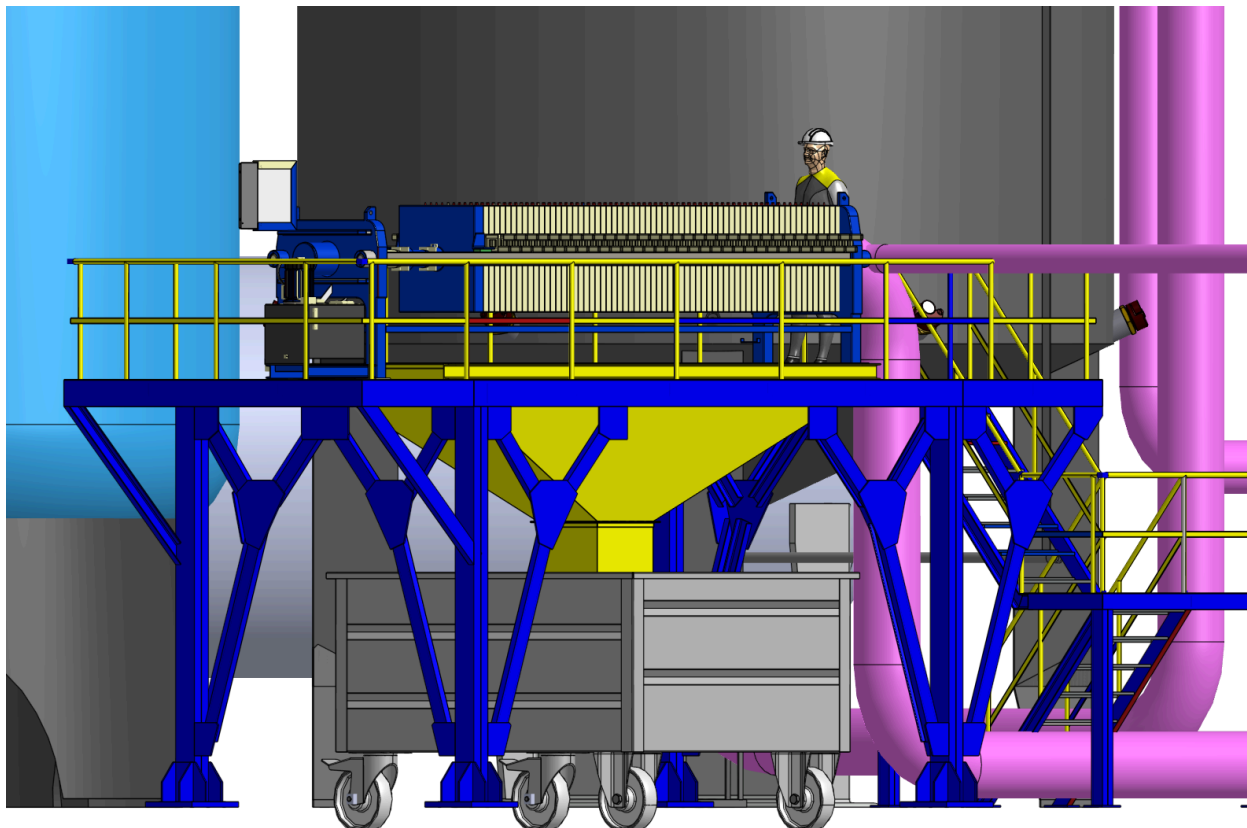


FIGURE 6.2. IMAGE OF RELATIVE SCALE FOR THE PLANT

Equipment Arrangement and Model Assumptions

The 3D model reflects the grouping of equipment that must interact closely, such as pairing the heat exchangers directly next to the bioleaching reactors to minimize thermal losses in the piping (Chemical Times, 2024). However, to optimize system resources and minimize computational load, certain visual compromises were required in the 3D model. Most notably, detailed mechanical pump models were omitted to prevent excessive rendering demands. Consequently, the piping in the visual model is shown flowing directly between vessels, representing the logical connection path rather than the physical hydraulic pumping mechanisms. Additionally, standardized geometric shapes were used to approximate the footprint of bespoke equipment, such as a crusher.

The 3D model incorporates a strict colour-coded zoning strategy to visually define hazard levels and restrict personnel movement. We mapped the entire facility into four distinct

operational sectors. This administrative control visually reinforces our inherently safer design philosophy (Chemical Times, 2024).

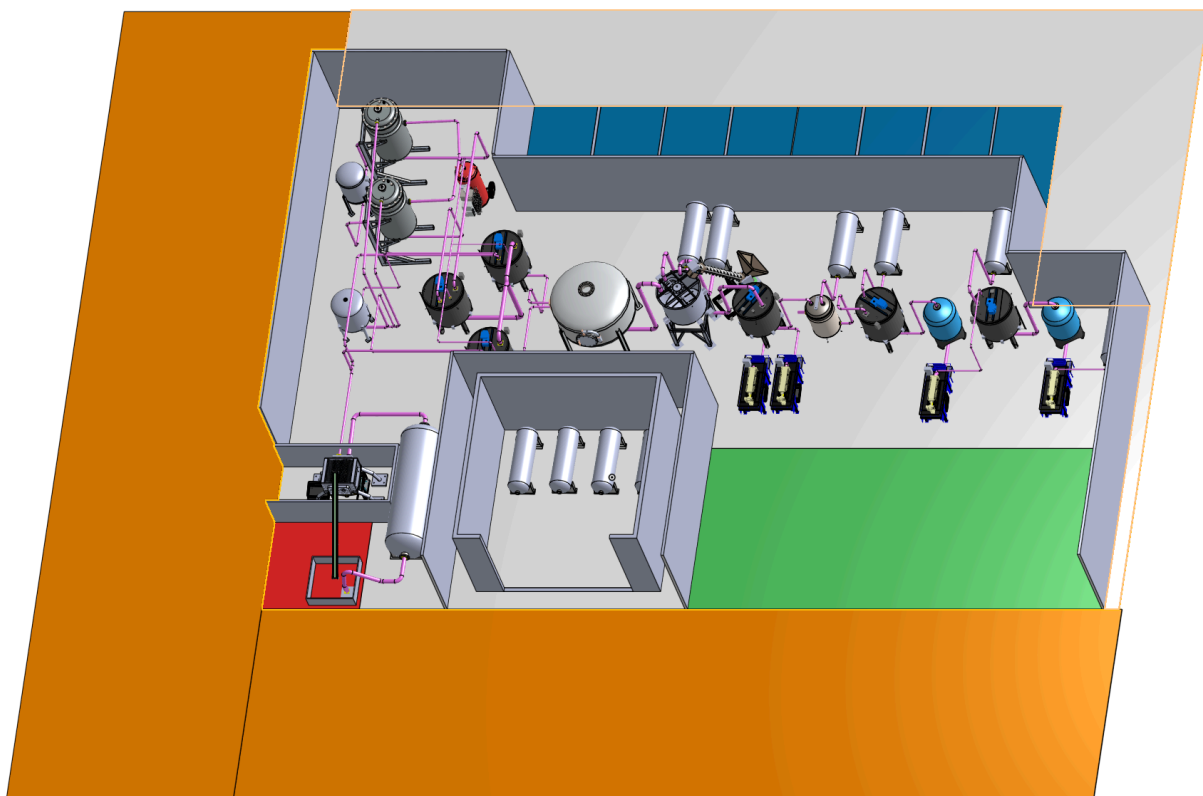


FIGURE 6.3. AN ISOMETRIC BIRD'S-EYE VIEW OF THE ENTIRE FACILITY FROM ZONE 1 TO ZONE 3

The Red Zone (Danger and Pre-Treatment)

The Red Zone represents the absolute highest risk profile within the facility. We severely restrict access here. Only essential personnel carrying specialized safe-work permits may enter this area. This zone houses the Battery Crusher (CR-101) and the primary feed conveyors. Batteries are physically destroyed in this space. They present severe thermal runaway and fire risks (Kim & Park, 2025). We engineered this zone as a reinforced concrete vault maintained under constant negative pressure. This structural choice traps any toxic hydrogen fluoride gas generated during the shredding process (Thompson et al., 2020).

The Orange Zone (Shipping and Receiving)

Heavy industrial traffic dominates the Orange Zone. This area manages the massive logistical influx of raw materials. We designed it to safely stage the 14-day supply of incoming spent lithium-ion batteries. Forklifts and transport trucks operate continuously here. We positioned the Orange Zone at the extreme perimeter of the plant footprint. External delivery vehicles can drop off their payloads without ever crossing into sensitive chemical processing areas (HSE, n.d.). This specific layout choice drastically reduces the risk of vehicular accidents.

The Green Zone (Recovery and Export Preparation)

The Green Zone serves as the plant's economic engine. This is the hydrometallurgical recovery area, where the purified leachate is converted into valuable solid products. It includes the solvent extractor, separators, and high-pressure filter presses. Workers handle highly flammable organic solvents and toxic heavy metal powders in this space. We mandated explosion-proof electrical ratings for all equipment within the Green Zone boundaries (Crawl & Louvar, 2011). Specialized local exhaust ventilation systems capture hazardous particulates before operators package the final lithium and cobalt salts for export.

The Blue Zone (Public Access and Administration)

We designated the Blue Zone for general public access. Office personnel, visitors, and non-operational staff utilize this space daily. It encompasses the main parking facilities, administrative buildings, and the central control room. We calculated maximum physical separation distances to isolate the Blue Zone from the chemical reactors and the crushing vault. This strategic spacing ensures that a catastrophic event in the Red or Green zones cannot physically reach the unprotected administrative staff (Aisyah, 2021).

6.3 Detailed Zone Renderings

Three specific renderings provide a closer look at the core processing areas.

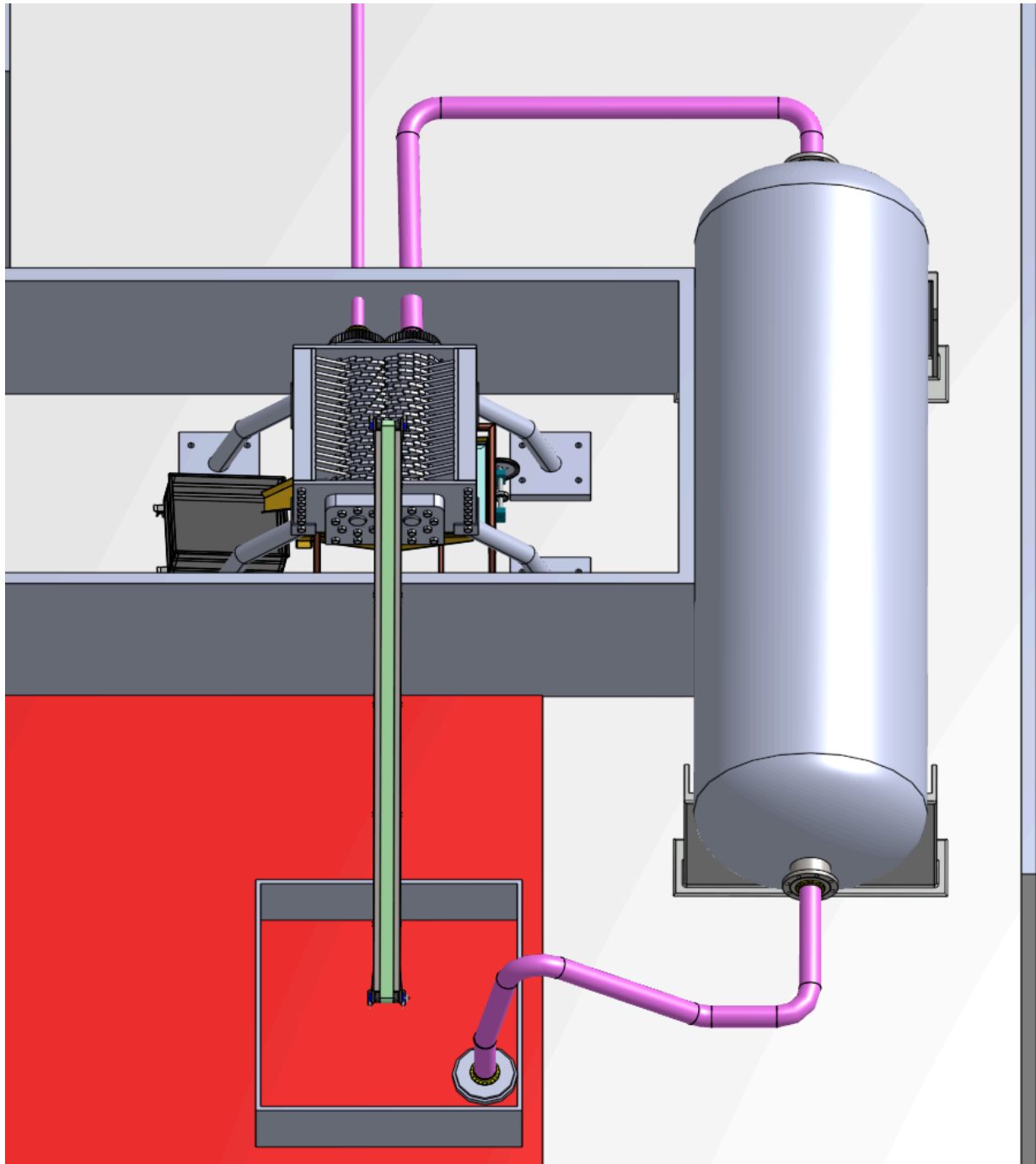


FIGURE 6.4. ISOMETRIC BIRD'S-EYE VIEW, ZONE 1 BATTERY DISCHARGE & CRUSHING

The first rendering captures the high-risk Pre-Treatment area. The primary battery feed conveyor (CO-101) lifts unsorted waste directly into the mechanical crusher (CR-101). The shredder sits in front of the sodium chloride brine bath (BI-100). Cells fall immediately into the

conductive liquid upon destruction. They discharge safely. Liquid immersion completely eliminates the risk of thermal runaway (Kim & Park, 2025). You will also notice the heavy concrete containment boundaries surrounding the equipment. A negative pressure system pulls toxic hydrogen fluoride gases out of this enclosed space (Thompson et al., 2020)

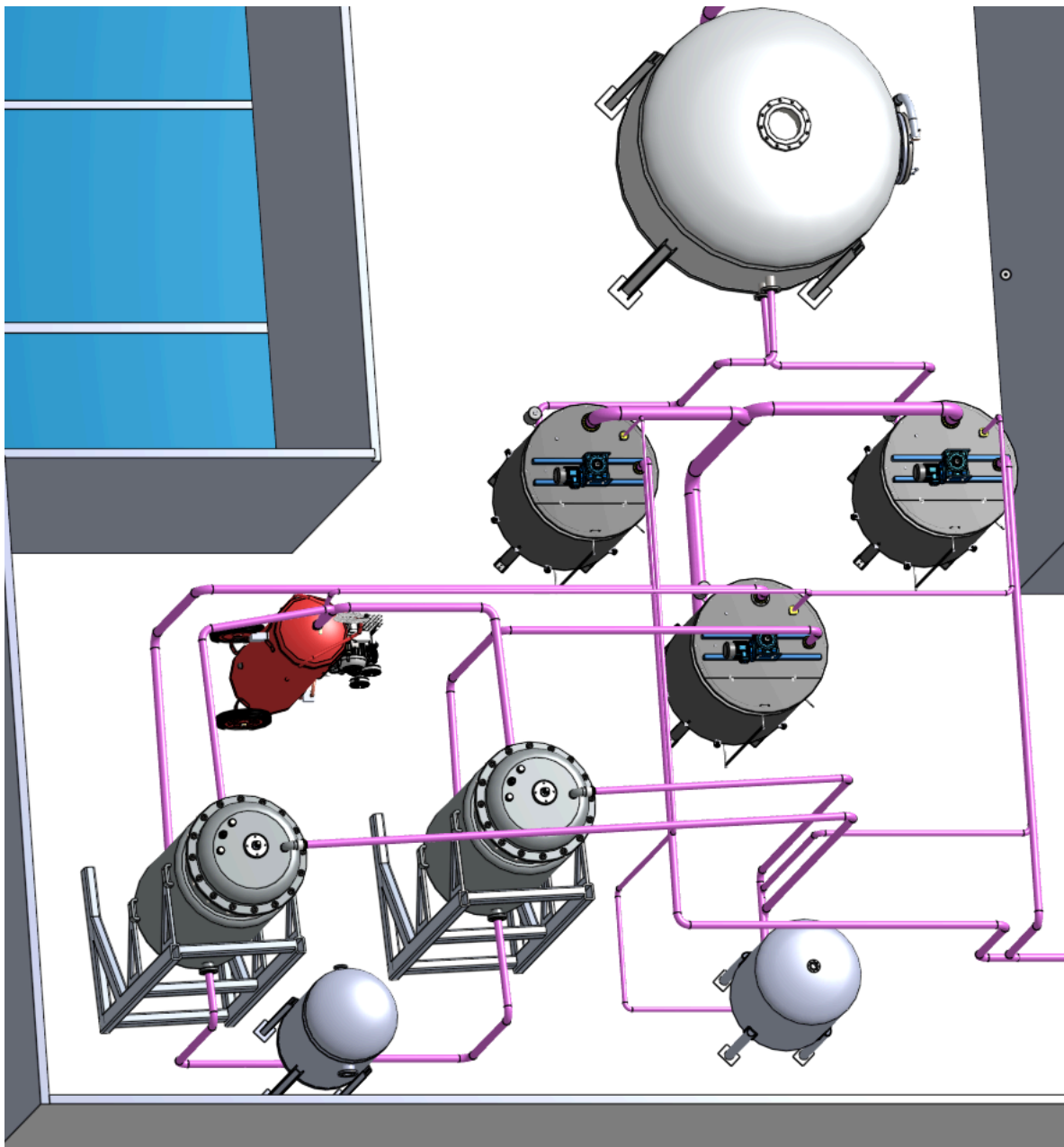


FIGURE 6.5. BIRD'S-EYE VIEW, ZONE 2 BATTERY, BACTERIA BIOLEACHING

Zone 2 represents the plant's biological core. The two massive Batch Reactors (V-201 A/B) dominate this rendering. A dedicated bioreactor (R-201) sits adjacent to the main tanks. This unit continuously provides bacterial cultures.

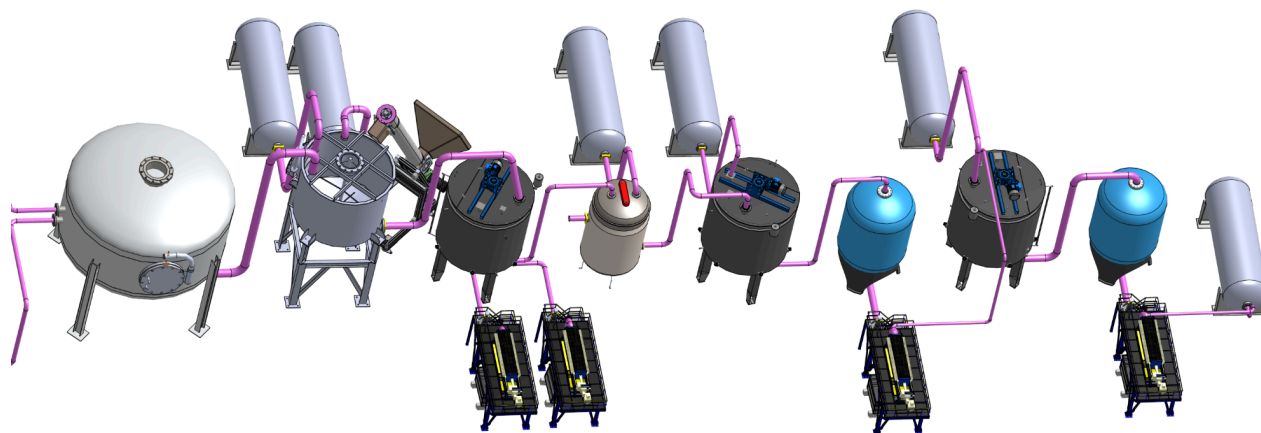


FIGURE 6.6. BIRD'S-EYE VIEW, ZONE 3 BATTERY, HEAVY METAL SEPARATION RECOVERY

The final image focuses on the hydrometallurgical recovery circuit. The rendering highlights the sequential arrangement of the solvent extraction units alongside a long line of high-pressure filter presses. Liquid leachate transforms into dry products through precise chemical dosing. Reagents like sodium hydroxide and Cyanex-272 flow directly into the extractors to selectively precipitate specific metals. Organic solvents pose a severe fire risk. Explosion-proof equipment is strictly mandatory across this entire sector (Crowl & Louvar, 2011).

Specialized collection bins sit directly beneath the filter presses. They catch the final heavy metal and lithium carbonate cakes as they fall from the pressurized plates. Below is a macro view that serves as the final demonstration of the facility's spatial design, encapsulating the complete process flow from initial feed to final product storage.

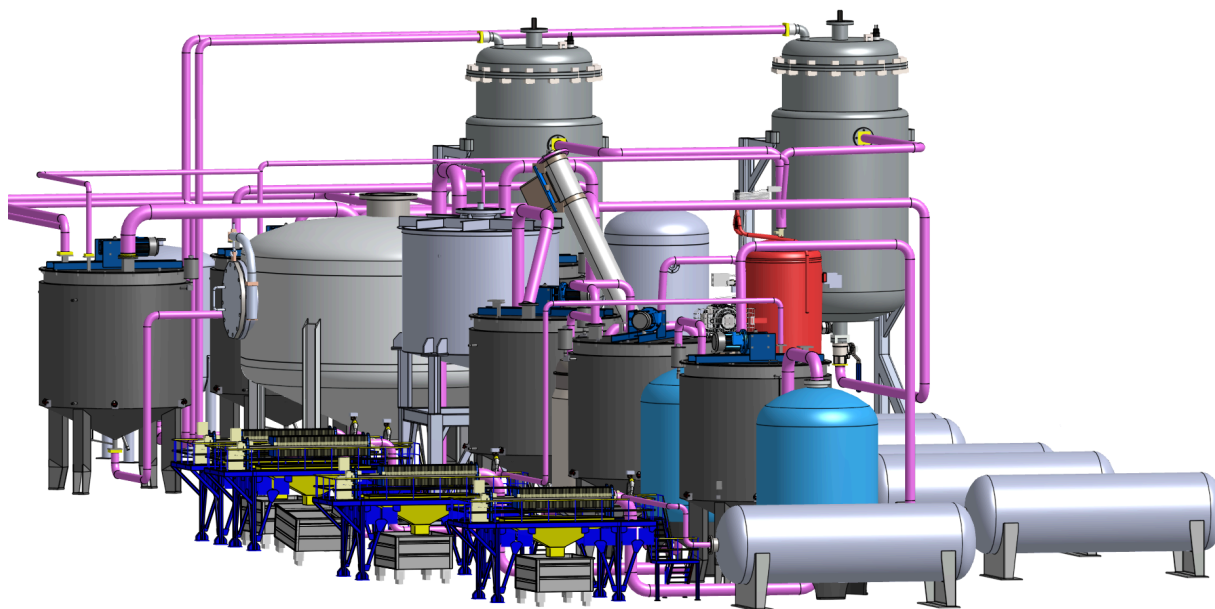


FIGURE 6.7. COMPREHENSIVE OVERVIEW OF THE INTEGRATED PLANT MODEL.

The layout demonstrates a highly deliberate clustering of unit operations. Centralized agitation tanks and main bioleaching reactors form the core of the processing sequence, which logically transitions into the elevated filtration units in the foreground. By elevating these filter presses, the design uses gravity to assist in discharging processed solids into the collection bins below.

Furthermore, the highly visible piping network clearly maps the complex routing required to connect these distinct stages. The strategic placement of the horizontal storage vessels and auxiliary tanks on the periphery illustrates a commitment to a compact, modular footprint without sacrificing operability. Ample clearance has been maintained between major vessels to ensure practical access for routine maintenance, inspections, and safe navigation for operators.

Ultimately, this comprehensive visualization acts as a proof-of-concept for the facility. It confirms that the theoretical process design successfully translates into a practical, highly integrated physical layout that balances efficiency with spatial constraints.

Chapter 7. Economic Evaluation

7.1.1 Capital Costs

Capital costs is a non exhaustive list of the following: equipment, installation, controllers, piping, electrical systems, the building, on site facilities, process labour/insurance. The labour is assumed to be 12 hour shifts operating 24/7 year round with 65 days of scheduled maintenance. The total manufacturing/construction (fixed capital investment) costs are \$362 million with continuous annual energy/labour expenses reaching (working annual capital) \$11 USD million. Raw material expenses are mainly the reagents used in recovery, these accumulate to \$15 million USD annually.

7.1.2 Equipment Costs

Following the Peters, Timmerhaus & West textbook, the equipment costs are estimated by scaling up an existing source of similar equipment. To start, a sample calculation of BI-100 is conducted. BI-100 is the holding vessel for crushed batteries and brine solution. Using a volume of 126m³, the 2002 price (USD) is 31000. Adjusting for the required rubber-lining (1.5x) and then applying the CEPCI ratio to escalate the price to 2026. Equipment costs are first estimated from given charts in the textbook, then scaled for inflation using the CEPCI index. More current CEPCI values (for 2026) can be found within the Chemical Engineering magazine cited by the textbook. Next, the lang factor method is applied using table 6-9 to model installation costs.

Equipment & Installation Price

$$Cost_{2026} = (31000 * 1.5) * (800/390.4)$$

$$= \$95286 \text{ USD}$$

$$= \$95286 * 0.55 \text{ (service facilities)} * 0.31 \text{ (Piping)} * 0.1 \text{ (Electrical)} * 0.1 \text{ (Structural)} * 0.3 \text{ (labour + engineering)}$$

$$= \$209000 \text{ USD}$$

7.1.3 Total Capital Investment

The total capital investment is to be 15-20% of fixed capital, so \$290 million USD is the projected sum.

7.2 Profitability Analysis

Profit is summed through the price of acquired metals (Li, Ni, Co) with a factor to account for further processing costs of the filter cakes encumbered by the following purchaser. At the current recoveries, accounting for rapidly changing lithium prices, gross revenue is estimated at \$77 million USD annually.

Chapter 8. Environmental Impact

The global proliferation of electric vehicles has triggered an unprecedented surge in the volume of spent lithium-ion batteries. Electronic waste poses a massive ecological threat. Discarded cells contain highly toxic heavy metals such as cobalt and nickel, along with volatile organic electrolytes, which actively contaminate groundwater when sent to municipal landfills (Roy et al., 2021). Immediate industrial intervention is necessary. Traditional disposal methods fundamentally fail to address the sheer volume of this hazardous material. Our facility intercepts exactly 15,000 tonnes of these spent batteries annually. We transform a dangerous environmental liability directly into a premium supply of battery-grade raw materials (Cerrillo-Gonzalez et al., 2024).

Biohydrometallurgy acts as the core engine driving this transformation. Relying on biological oxidation completely avoids the severe thermal pollution and toxic off-gassing historically associated with conventional pyrometallurgical smelting (Makuza et al., 2021). Establishing a definitive circular economy demands absolute efficiency. We must handle the inevitable secondary waste streams generated during the chemical extraction phases with meticulous care. Processing these spent batteries generates distinct effluents and airborne emissions that require rigorous mitigation. Containment must be absolute. This chapter defines the precise ecological footprint of our biological recycling operation. We map every emission, detail all solid waste generation, and outline the exact mitigation strategies engineered to protect the surrounding environment (Kim & Park, 2025).

8.1 Emissions and Effluents

Processing spent batteries releases lethal airborne pollutants immediately. Mechanical shredding tears open solid metal casings, exposing residual electrolytes directly to the atmosphere. Moisture triggers rapid degradation. Lithium hexafluorophosphate breaks down to vaporize highly toxic hydrogen fluoride gas into the surrounding air (Velázquez-Martínez et al., 2019). This poses an extremely severe hazard to the plant floor. Breathing even microscopic trace amounts of hydrogen fluoride causes systemic poisoning and catastrophic lung damage.

Crushing operations eject fine metallic dust. Airborne particulate matter includes graphite, cobalt oxide, and nickel powders. Inhaling these heavy metal particles introduces severe carcinogenic risks to operators working on the active floor. Thermal runaway events multiply these emission risks exponentially. A single battery fire inside the facility can release up to 4 kg of carbon dioxide equivalent, along with vaporized cobalt (Amano et al., 2023). Uncontrolled combustion disperses these toxic aerosols directly into the local environment (Premnath et al., 2022). Containment prevents disaster.

Liquid effluents demand rigorous containment. Bioleaching generates a massive volume of highly acidic wastewater heavily loaded with dissolved salts. The final separation stage ejects this aqueous raffinate. Automated controllers constantly inject sodium hydroxide and sodium carbonate to manipulate the specific precipitation chemistry across the downstream recovery circuits. This continuous chemical neutralization generates massive quantities of aqueous sodium sulphate (Zouhri et al., 2013). Direct municipal discharge remains strictly forbidden. High salinity spikes instantly devastate local civic water infrastructure. Unrecovered heavy metals, such as dissolved copper, aluminum, and manganese, severely contaminate the remaining liquid. Pumping this toxic fluid into a local river would destroy the aquatic ecosystem. Native aquatic life rapidly bioaccumulates these heavy metals. The facility completely avoids processing this hazard internally. Specialized environmental contractors assume full control over the complex disposal challenge. Heavy tanker trucks extract the toxic effluent directly from the site for immediate off-site neutralization and secure disposal.

Solvent extraction introduces a distinct class of hazardous emissions. Isolating cobalt ions relies on volatile organic compounds such as Cyanex 272, diluted heavily in kerosene. Kerosene evaporates continuously at standard room temperatures. Vapours accumulate inside the enclosed tanks. An accidental spark will ignite a massive explosion if these concentrated volatile organic compounds escape into the main facility airspace. Ventilation systems must run constantly. Biological cultivation creates unique airborne biological aerosols. Agitating the bacterial culture pushes microscopic droplets of acidic media into the air above the open tanks. These droplets carry live microbes and dissolved iron sulphate. We must capture and filter these distinct emissions perfectly to maintain a secure facility.

8.2 Waste Generation and Disposal

Extracting critical metals leaves behind an immense physical bulk of unreactive material. We strip the target elements away. This transforms the remaining battery architecture directly into bulk solid waste. The primary mechanical separation stages reject thick polymer separators and shredded steel casings (Espinosa et al., 2004). Local recycling contractors purchase this clean metallic scrap. They melt the recovered steel down to manufacture new construction materials. Dealing with the physical shell of the battery represents the easiest phase of our waste management protocol. Reintegrating these inert components immediately shrinks the total physical volume of refuse destined for local landfills (Zheng et al., 2018).

Chemical extraction generates a highly problematic byproduct. A dense sludge forms in the continuous stirred-tank reactors during the bioleaching phase. Unreacted graphite and synthetic organic binders dominate this heavy mud. Polyvinylidene fluoride acts as the primary adhesive binder in most commercial lithium cells (Pinegar & Smith, 2019). It stubbornly resists biological degradation. Our bacterial colonies ignore the plastic completely. High-pressure filter presses squeeze the moisture out of this thick slurry, significantly reducing its transport weight. The resulting dry filter cake is sent directly to licensed hazardous waste disposal facilities. Burying this material requires specialized lined landfills to definitively prevent toxic leaching over the coming decades (Mossali et al., 2020).

Isolating the pure metals creates secondary chemical waste. Aluminum and manganese are severe impurities in downstream cobalt and lithium recovery circuits. Operators force these specific metals to precipitate out of the liquid phase early in the separation sequence. Injecting sodium hydroxide sharply increases the pH. Aluminum hydroxide and manganese dioxide precipitate from the solution as heavy, solid cakes. Dedicated membrane filter presses trap this dense material. This specific byproduct has absolutely no commercial value at present (Zheng et al., 2018). It requires immediate secure disposal. Transforming electronic waste into pure battery-grade salts fundamentally relies on managing these unavoidable secondary waste streams perfectly. A single bottleneck in waste disposal will force the entire recycling plant to shut down.

Aqueous raffinate presents the final disposal hurdle. Continuous neutralization of the acidic bioleaching streams generates large quantities of highly concentrated sodium sulphate

(Pinegar & Smith, 2019). Municipal sewers simply cannot handle this fluid. Sudden salinity spikes devastate local civic water treatment infrastructure. We outsource this disposal process entirely. Dedicated environmental contractors pump the toxic effluent into heavy tanker trucks for immediate off-site processing. Relying on an established industrial wastewater partner removes the massive capital burden of designing and operating internal electrodialysis units just to handle a single byproduct. They neutralize the salt safely. This strategic transfer guarantees total environmental compliance.

8.3 Energy Efficiency

Heavy fossil fuel combustion characterizes conventional battery recovery. Rotary hearth furnaces demand constant natural gas injection to keep the inner chambers molten (Heelan et al., 2016). Operators burn massive quantities of valuable energy just to melt the protective aluminum and steel casings into a liquid state. This thermal intensity converts the complex internal battery architecture directly into a heavy metallic alloy and a highly unreactive slag. Carbon dioxide emissions skyrocket during this violent high-temperature smelting phase. We completely abandon that extreme thermal approach. Ambient temperature bioleaching rewrites the energy equation for electronic waste recovery entirely. The primary extraction circuit runs continuously at a mild 27 degrees Celsius. Biological oxidation replaces brute thermal force. Eliminating the smelting stage drastically shrinks the total cumulative energy demand of the resulting battery-grade metal salts (Ellingsen et al., 2013).

Biology acts as a microscopic furnace. The bacteria actively generate immense thermal energy. Oxidizing ferrous iron into ferric iron produces roughly 60 kilowatts of continuous exothermic heat inside the primary continuous stirred-tank reactors. Left unchecked, this rapid thermal accumulation would quickly cause the acidic slurry to boil. Extreme heat would instantly destroy the fragile microbial ecosystem. We aggressively scavenge this metabolic energy to protect the colony. Pure cooling water rushes through the external reactor jackets to absorb the excess reaction heat. Pumping this warmed water downstream provides a free thermal boost to the chemical precipitation circuits.

Oxygen sustains the biological extraction. Acidophilic microbes require continuous, heavy aeration to metabolize the black mass effectively. Supplying this pressurized air represents

the largest single electrical draw in the bioleaching quadrant. Massive rotary screw compressors squeeze the ambient atmosphere before driving it deep into the bottom of the slurry tanks. Variable frequency drives modulate these heavy compressor motors to optimize this severe pneumatic load. The digital control system continuously matches the electrical power draw to the exact biological oxygen demand of the bacteria. Wasting compressed air is exactly the same as venting money directly into the atmosphere. Advanced sparger nozzles break the air into microscopic bubbles. Smaller bubbles possess a massive surface area. This maximizes gas-to-liquid mass transfer efficiency, allowing the compressors to run at lower speeds while maintaining the culture's full oxygenation.

Mechanical shredding initiates the entire recycling sequence. Tearing through solid battery modules requires brutal mechanical force. The heavy twin-shaft crushers experience massive electrical current spikes when thick metallic casings wedge between the cutting blades. Running these high-torque motors at full speed continuously wastes immense amounts of power. We utilize intelligent load-sensing software to monitor the physical resistance on the shredder shafts. The motors draw minimal amperage during light loads. Power automatically ramps up only when a dense battery pack hits the primary teeth. Wet shredding inside a brine bath cools the cutting blades passively. This simple submersion technique completely eliminates the need for dedicated mechanical chillers to prevent the cutting enclosure from overheating (Sommerville et al., 2020).

Chemical separations demand strict thermal management. Isolating cobalt and nickel via solvent extraction requires exact temperature thresholds to prevent the organic reagents from degrading. Using cold utility water to chill these specific chemical streams wastes energy. We rely entirely on passive ambient cooling towers. These massive vertical structures leverage natural atmospheric airflow and simple water evaporation to safely reject low-grade process heat. Mechanical chillers utilizing synthetic refrigerants are completely absent from the primary cooling loop. Evaporative cooling handles the bulk of the thermal rejection at a fraction of the electrical cost.

Final dewatering operations rely on high-pressure hydraulics. Membrane filter presses physically squeeze the residual moisture away from the heavy metallic filter cakes. Hydraulic

pumps require intense electrical power to drive the massive steel compression rams. We installed localized hydraulic accumulators to capture and store pressurized fluid during the slow-idle phases of the batch cycle. This stored kinetic energy is released instantly during the final heavy compression phase. Spiking the main electrical grid is completely avoided. The facility maintains a perfectly smooth and continuous power draw. This precise load levelling prevents the local utility provider from imposing severe financial penalties for peak demand on the plant.

Producing fresh cathode precursors from virgin mined ores consumes vast quantities of diesel fuel and electricity. Heavy excavation equipment rips hard rock from the earth. Cargo ships transport this raw ore across the ocean to chemical refineries. Our urban mining facility short-circuits this massive global energy expenditure entirely. Recovering these exact critical elements domestically requires far less energy. Reintroducing these purified recycled metals into the battery manufacturing loop significantly reduces the cumulative energy demand of electric vehicle production (Dunn et al., 2012). The biological extraction of lithium and cobalt emits far less carbon dioxide than extracting those exact same elements from primary virgin deposits. Our plant actively repairs the broken energy loop of the modern electronics sector (Mohanty & Devi, 2023). Green chemistry fundamentally replaces brute force mining.

8.4 Life-cycle Considerations

Global electrification requires vast amounts of raw materials. Extracting virgin lithium from brine pools consumes staggering quantities of fresh water. Cobalt mining often involves severe human rights abuses. Our recycling facility bypasses this primary supply chain entirely. Urban mining recovers these exact elements domestically. Producing fresh cathode precursors from recycled black mass emits far less carbon dioxide than extracting primary ores from the earth (Kim et al., 2016). This fundamental shift redefines the ecological cost of energy storage. Reintroducing these purified metals into the manufacturing loop slashes the cumulative energy demand of electric vehicle production. It secures a highly reliable supply of critical materials. Keeping the supply chain local insulates battery manufacturers from volatile international commodity markets (Ambrose & Kendall, 2020).

The automotive sector will generate roughly 420 gigawatt-hours of used lithium-ion batteries over the next decade (Hsieh et al., 2020). End-of-life batteries threaten municipal

landfills with severe contamination by heavy metals. Toxins leak into the soil as the metal casings rust away. Heavy rain washes these poisons directly into the groundwater. Our plant intercepts this hazardous waste. Processing the discarded batteries eliminates the immediate environmental threat. Before shredding occurs, specific battery modules are evaluated for secondary applications. Second-life concepts extend the valuable lifetime of battery systems. Degraded electric vehicle batteries often retain enough capacity for stationary grid storage (Casals et al., 2017). Repurposing these units delays the destructive recycling process. It provides immediate carbon benefits by supporting renewable solar and wind energy installations.

True sustainability demands a closed-loop material architecture. A circular waste management hierarchy effectively recovers the cumulative energy demand associated with the initial cell manufacture (Richa et al., 2017). We transform a dangerous ecological liability into a premium commercial asset. Relying on biological extraction ensures that our operational methods match the green energy goals of the electric vehicles we recycle. Chemical recovery must never generate more pollution than it prevents. The facility avoids the massive greenhouse gas emissions tied to traditional high-heat smelting. The total ecological footprint shrinks drastically.

8.5 Mitigation Strategies

Industrial accidents destroy local ecosystems immediately. Spills must be trapped physically before they reach vulnerable soil. Total containment dictates the physical plant layout, requiring every single reagent storage tank to sit inside a massive reinforced concrete bund. These specialized basins are engineered to hold exactly 110 percent of the largest vessel's total fluid capacity to ensure absolute security (Chen et al., 2019). A catastrophic tank rupture will simply flood the isolated concrete structure. The dangerous fluid never touches the ground. Floor operators can safely pump the spilled chemicals right back into the active processing circuit.

Airborne chemical threats demand highly aggressive containment strategies during the initial mechanical breakdown of the cells. Traditional dry shredding is incredibly violent. Exposing residual liquid electrolytes to ambient humidity instantly vaporizes lithium hexafluorophosphate into lethal hydrogen fluoride gas (Harper et al., 2019). We bypass this respiratory hazard completely. To physically isolate these volatile components, the facility

utilizes a fully submerged crushing architecture. Tearing the metal casings apart entirely under brine immediately quenches any hazardous sparks and traps the escaping chemical compounds within the liquid bath. Toxic vapours never form. By operating in this wet environment, the system physically prevents thermal runaway while simultaneously scrubbing the reactive fluorides directly into the aqueous phase (Harper et al., 2019). It eliminates the massive capital burden of maintaining external nitrogen flooding systems and complex gas scrubbers.

Liquid effluents exiting the process carry massive dissolved salt loads. The continuous neutralization of our acidic bioleaching streams produces large volumes of aqueous sodium sulphate. Dumping this heavy saline wastewater directly into municipal sewers is strictly forbidden. It instantly destroys civic water treatment infrastructure. We completely avoid internal processing. Specialized environmental contractors take over this severe disposal hurdle. Dedicated wastewater firms pump the toxic effluent into heavy tanker trucks for immediate off-site neutralization. Outsourcing this specific saline waste stream removes a massive capital burden from the facility while guaranteeing absolute environmental compliance.

Occupational hazards go beyond chemical exposure. The heavy industrial shredders generate deafening sound waves that threaten technicians with immediate and permanent hearing loss. We isolated all major rotating machinery inside dedicated acoustic enclosures. Sound-absorbing panels line the pre-treatment building to protect the workers on the floor. Agitating the bacterial cultures creates a different threat entirely. Violent mixing launches microscopic droplets of acidic media directly into the air. These droplets carry live microbes. (Sanchez-Monedero et al., 2008). We eliminated this respiratory hazard completely by engineering fully closed reactor vessels. The airtight design permanently traps all biological aerosols inside the tank.

8.6 Regulations

Operating a chemical facility in Shenzhen requires securing a comprehensive pollutant discharge permit. The Chinese government utilizes this framework to establish strict baselines for industrial emissions (Li, 2017). Biohydrometallurgical extraction relies heavily on reagents like concentrated sulfuric acid. Regulating the safe handling of these corrosive materials falls entirely under Decree 591 (CIRS, 2015). Our primary operational licenses are issued directly by

the State Administration of Work Safety. Formal documentation is mandatory. Every hazardous compound must be registered with the National Registration Center of Chemicals prior to facility entry (CIRS, 2015).

International shipments of hazardous materials fall under the jurisdiction of the General Administration of Quality Supervision, Inspection, and Quarantine (CIRS, 2015). We must obtain a hazard identification report. This document verifies the chemical safety profile of the final commercial products. The Shanghai Chemical Industry Testing Centre conducts these specific evaluations for major ports (CIRS, 2015). Securing this certificate is an absolute requirement. Our recovered battery metals can legally enter the broader Chinese manufacturing market only after this approval.

Regulatory authorities mandate continuous monitoring of chemical movements. The facility complies fully with the uniform environmental protection tax applied to industrial operations (Li, 2017). We integrated passive safety systems specifically to meet these ongoing compliance targets. Biological extraction methods inherently align with these national environmental standards.

Aqueous raffinate requires strictly regulated disposal protocols. Neutralizing the acidic leaching streams generates heavy volumes of sodium sulphate daily. We follow the exact water quality metrics defined by the mandatory GB 5749-2022 national standard (State Administration for Market Regulation, 2022). Specialized environmental contractors handle this saline wastewater. Tanker trucks extract the effluent for secure off-site neutralization. Outsourcing this specific disposal process guarantees full legal compliance. It entirely protects the local watershed.

Chapter 9. Process Safety and Loss Prevention

The primary objective of the lithium-ion battery (LIB) biohydrometallurgical recycling facility is to safely recover critical metals while strictly safeguarding the health and welfare of plant personnel, the surrounding community, and the environment (Crowl & Louvar, 2011). The convergence of high-energy electrical components, highly corrosive bio-acidic solutions, and volatile organic compounds presents a unique matrix of vulnerabilities. The identification of areas of vulnerability and the application of fundamental importance in loss prevention are critical. A greater emphasis must be placed on technological measures to control hazards and on attempting to "get things right the first time" (Crowl & Louvar, 2011). Consequently, process safety management (PSM) protocols must be rigorously integrated into the fundamental plant design. This chapter provides an exhaustive analysis of the hazard identification process, chemical hazards, inherently safer design applications, pressure relief systems, fire prevention architectures, and emergency response frameworks utilized in this plant.

9.1 Inherent Safety Considerations

The facility's foundational safety philosophy relies on Inherently Safer Design (ISD) principles, which aim to eliminate or reduce hazards at their chemical and physical sources rather than relying solely on "add-on" engineered safety controls or administrative procedures (Crowl & Louvar, 2011). The implementation of ISD in this plant is categorized into four primary strategies.

Attenuation (Moderation)

Traditional pyrometallurgical recycling of LIBs requires smelting furnaces operating at temperatures exceeding 1,400°C, which poses extreme thermal hazards and generates highly toxic dioxin and furan off-gases (Roy et al., 2021). By selecting a biohydrometallurgical pathway, the core extraction process operates at a moderate, ambient temperature of 27°C and atmospheric pressure (Garcia et al., 2025). This drastically reduces thermal and pressure-related risks, averting the catastrophic consequences of high-pressure vessel ruptures.

Substitution

The bioleaching process utilizes *Acidithiobacillus Ferrooxidans* to biologically regenerate ferric iron (Fe^{3+}) and sulfuric acid (H_2SO_4) as the primary lixiviants, thereby avoiding the need for bulk storage, transport, and injection of massive quantities of highly concentrated, pure mineral acids (Islam, 2022). This substitution significantly reduces the on-site inventory of hazardous chemicals.

Limitation

The mechanical pre-treatment zone relies on wet crushing and coarse shredding rather than ultra-fine dry grinding. This intentional design choice limits the generation of airborne particulate matter, thereby mitigating the severe risk of explosive combustible dust clouds and reducing the inhalation hazard of heavy metal particulates for operators.

Simplification

End-of-life batteries retain residual electrical energy that can induce thermal runaway if short-circuited during mechanical processing (Kim & Park, 2025). To simplify hazard management, the process design mandates that all incoming batteries be submerged in a 10% sodium chloride (NaCl) brine solution (BI-101) for 24 hours to safely and electrochemically discharge the cells to <0.5 V prior to any mechanical destruction.

9.2 Chemical Hazards

Chemicals involved in the plant can be hazardous if not properly controlled (Crowl & Louvar, 2011). Serious injuries and harm can occur to workers if they are not sufficiently trained on how to handle these chemicals with care. Specifically, in the LIB bioleaching plant, there are various prevalent chemicals that must be exposure-controlled and properly understood from a health risk standpoint.

10.2.1 Hydrogen Fluoride (HF) Gas

Spent lithium-ion batteries (LIBs) contain heavily fluorinated electrolytes, primarily utilizing Lithium Hexafluorophosphate (LiPF_6) as the conductive salt. When these batteries are

mechanically crushed during the initial processing stages, the closed system is compromised. Exposure to ambient moisture causes the electrolyte to rapidly hydrolyze and decompose, generating highly toxic Hydrogen Fluoride (HF) gas (Thompson et al., 2020).

Outside of battery recycling, HF is widely used in the petrochemical industry as an alkylation catalyst and in the production of fluorocarbons and aluminum. It is a colourless, highly corrosive gas (or fuming liquid at lower temperatures) with unique toxicological properties. Unlike other mineral acids, which remain on the tissue surface, HF readily penetrates the skin, where fluoride ions aggressively bind calcium and magnesium in deep tissues and bones, causing systemic toxicity and severe localized necrosis. Because of its volatility and severity as a respiratory hazard, stringent detection and mitigation parameters must be established. The specific exposure limits and required engineering controls are detailed in the subsequent table.

TABLE 9.2.1. HEALTH EFFECTS OF HYDROGEN FLUORIDE RESULTING FROM ACUTE AND CHRONIC EXPOSURE LEVELS

Exposure Route	Health Effects
Inhalation	• Severe respiratory tract irritation, pulmonary edema • Headaches, dizziness, rapid suffocation • Chest pains and breathing difficulty
Skin Contact	• Severe, deep tissue chemical burns • Delayed pain (up to 24 hours) • Bone decalcification
Eye Contact	• Severe eye irritation, redness • Corneal damage, potential permanent blindness
Ingestion	• Severe burns to the mouth, throat, and stomach • Nausea, vomiting, abdominal pain • Cardiac arrhythmias due to calcium depletion
Repeated Chronic Overexposure	• Fluorosis (bone and joint damage) • Chronic respiratory irritation and lung damage

9.2.2 Sulfuric Acid (H_2SO_4)

Sulfuric acid is the primary reagent utilized to maintain the precise pH of 1.5 required for optimal microbial activity in the bioleaching Reactors (V-201 A/B). As one of the most widely produced industrial chemicals globally, it is foundational in fertilizer manufacturing, wastewater processing, and chemical synthesis.

In its concentrated form, H_2SO_4 is a dense, colourless, and oily liquid. It is a highly corrosive, strong mineral acid characterized by its powerful dehydrating and oxidizing properties. When introduced to water or organic materials, it reacts violently and exothermically, posing a severe risk of splashing and thermal burns. Due to its aggressive nature, specialized construction materials are required for all storage tanks, piping, and instrumentation in the bioleaching circuit.

TABLE 9.2.2. HEALTH EFFECTS OF SULFURIC ACID RESULTING FROM ACUTE AND CHRONIC EXPOSURE LEVELS

Exposure Route	Health Effects
Inhalation	• Corrosive to the respiratory tract • Burning sensation, coughing, wheezing, shortness of breath, etc.
Skin Contact	• Severe skin burns and necrosis • Rapid tissue destruction upon contact
Eye Contact	• Immediate pain, severe burns • Corneal damage, permanent loss of vision
Ingestion	• Corrosive burns to the esophagus and stomach • Risk of gastric perforation and death
Repeated Chronic Overexposure	• Chronic bronchitis, lung capacity reduction • Erosion of dental enamel

9.2.3 Volatile Organic Solvents (Cyanex 272 and Kerosene)

In the downstream hydrometallurgical recovery zone, an organic phase consisting of Cyanex 272 (bis(2,4,4-trimethylpentyl)phosphinic acid) diluted in aliphatic kerosene is used in

the Solvent Extractors (SX-301). Cyanex 272 is an organophosphorus extractant standard in the hydrometallurgical industry for its high selectivity in separating cobalt from nickel in sulphate solutions. Kerosene serves as a diluent, lowering the viscosity of the extractant and promoting rapid phase separation after mixing.

While effective, these organic solvents introduce significant flammability hazards to the facility. Kerosene possesses a relatively low flash point and readily generates volatile vapours. When accumulated in confined spaces or poorly ventilated areas, these vapours pose a substantial fire and explosion risk. Furthermore, inhalation of these organic vapours acts as a central nervous system depressant, necessitating strict atmospheric monitoring.

TABLE 9.2.3. HEALTH EFFECTS OF ORGANIC SOLVENTS RESULTING FROM ACUTE AND CHRONIC EXPOSURE LEVELS

Exposure Route	Health Effects
Inhalation	• Central nervous system depression • Dizziness, drowsiness, headaches • Anesthesia effects at high concentrations
Skin Contact	• Defatting of the skin, severe irritation • Dermatitis, cracking, and flaking
Eye Contact	• Eye irritation, tearing, blurred vision
Ingestion	• Aspiration hazard: entering the lungs can cause chemical pneumonitis • Nausea, vomiting, loss of coordination
Repeated Chronic Overexposure	• Liver and kidney damage • Chronic neurological impairment

9.2.4 Heavy Metal Dusts and Salts (Cobalt, Nickel, Manganese)

The final, value-added products of the recycling plant include high-purity solid precipitates, primarily Nickel Hydroxide ($\text{Ni}(\text{OH})_2$) and Cobalt Phosphinate ($\text{Co}_3(\text{PO}_4)_2$). These transition metal compounds are critical raw materials fed back into the battery manufacturing supply chain to produce new cathodes, fulfilling the facility's circular economy mandate.

Following filter press operations and subsequent thermal drying, these heavy metals are converted from a wet slurry into a fine, dry powder. In this state, they become highly mobile and easily inhaled. These heavy metal compounds are inherently toxic and pose severe, long-term health risks, including carcinogenicity, respiratory sensitization (such as occupational asthma), and target organ toxicity. To prevent inhalation exposure, rigorous dust containment, negative pressure zones, and specific personal protective equipment (PPE) are required.

TABLE 9.2.4. HEALTH EFFECTS OF HEAVY METAL SALTS RESULTING FROM ACUTE AND CHRONIC EXPOSURE LEVELS

Exposure Route	Health Effects
Inhalation	• Respiratory sensitization (occupational asthma) • Acute pneumonitis, severe coughing
Skin Contact	• "Nickel itch" or "Cobalt itch" (allergic contact dermatitis) • Skin sensitization and severe rashes
Eye Contact	• Mechanical and chemical irritation
Ingestion	• Gastrointestinal distress, metallic taste • Acute heavy metal poisoning
Repeated Chronic Overexposure	• Known human carcinogens (IARC Group 1 for certain Ni/Co compounds) • Chronic lung fibrosis • "Manganism" (neurological disorder mimicking Parkinson's disease from Mn exposure)

9.2.5 Alkali Metal Salts (*Lithium Carbonate*)

The ultimate economic driver of the biohydrometallurgical recycling facility is the high-yield recovery of battery-grade Lithium Carbonate (Li_2CO_3). Unlike the transition metals (cobalt, nickel, and manganese) discussed previously, lithium is a Group 1 alkali metal. In its pure elemental form, lithium is highly reactive and reacts exothermically with ambient moisture, producing flammable hydrogen gas and highly corrosive lithium hydroxide, posing severe fire and explosion hazards (Mrozik et al., 2021).

However, the inherent safety of this plant's hydrometallurgical design ensures that lithium is processed exclusively in its oxidized, ionic state (Li^+) within the aqueous Pregnant Leach Solution (PLS) and ultimately precipitated as a stable, solid salt (Li_2CO_3). While this eliminates the severe water-reactivity hazard, the final processing stages convert the material into a fine, dry, and highly mobile powder. In this state, lithium carbonate presents a significant occupational exposure risk. Lithium is highly biologically active with a notably narrow toxicity threshold in the human body. Unintended occupational absorption, primarily through the inhalation of aerosolized dust or accidental ingestion, can rapidly lead to systemic lithium toxicity (Ahuja et al., 2020). The compound acts as a moderate to severe irritant to mucous membranes and, upon systemic absorption, primarily targets the central nervous system, kidneys, and thyroid. [Connecticut Department of Public Health (CTDPH). (2023)]

To effectively mitigate these risks, the final precipitation and packaging zones must be equipped with a high level of High-Efficiency Particulate Air (HEPA) local exhaust ventilation (LEV) systems. Operators in Zone 3 must enforce strict hygiene protocols and utilize specific Personal Protective Equipment (PPE), including P100/N95 particulate respirators and tightly sealed safety goggles, to prevent inhalation and mucosal absorption of the alkali dusts (Crowl & Louvar, 2011).

TABLE 9.2.5. HEALTH EFFECTS OF LITHIUM CARBONATE RESULTING FROM ACUTE AND CHRONIC EXPOSURE LEVELS

Exposure Route	Health Effects
Inhalation	- Severe irritation of the respiratory tract and mucous membranes - Coughing, sneezing, and shortness of breath - Chemical pneumonitis in cases of massive acute dust inhalation
Skin Contact	- Mild to moderate skin irritation and erythema (redness) - Exacerbation of existing skin conditions (e.g., psoriasis or dermatitis) - Reaction with sweat can create localized alkaline irritation
Eye Contact	Mechanical abrasion and severe chemical irritation

	● Redness, tearing, and potential for transient corneal opacity
Ingestion	● Acute gastrointestinal distress, nausea, vomiting, and diarrhea ● Central nervous system toxicity: fine hand tremors, muscle weakness, ataxia (loss of coordination), and confusion ● In severe overdose, risk of seizures, coma, or cardiac arrhythmias.
Repeated Chronic Overexposure	● Chronic renal impairment (nephrogenic diabetes insipidus), leading to excessive thirst and urination ● Endocrine disruption, specifically hypothyroidism and goiter formation ● Chronic neurological deficits, including memory impairment and persistent tremors (Klaassen, 2008)

9.3 Hazard Identification (HAZOP Analysis)

A HAZOP (Hazards and Operability) analysis is a structured, highly systematic examination of a complex process to identify and evaluate potential risks to personnel or equipment. A wide range of possible equipment failures, instrumentation faults, and abnormal operating conditions (such as deviations in temperature, pressure, or flow) have been systematically assessed for this facility. A detailed HAZOP study of the primary process nodes has been conducted to identify the root causes of potential accidents. Crucially, this study outlines the specific engineering safeguards and administrative controls required to mitigate those risks to an acceptable level (Crowl & Louvar, 2011).

TABLE 9.3.1. HAZOP ANALYSIS OF PRE-TREATMENT ZONE: BATTERY CRUSHER (CR-101)

Key Word	Deviation	Possible Causes	Consequences	Action Required
MORE OF	High Temperature / Fire	1. An undischarged battery short-circuits during mechanical shredding. 2. Failure of the upstream brine soaking protocol (V-100).	1. Thermal runaway, explosive off-gassing, and catastrophic equipment fire (Kim & Park, 2025). 2. Large packs emit 2-4 kg CO ₂ -eq of toxic smoke per event.	a. Install an automated nitrogen (N ₂) flooding system triggered by thermal cameras. b. Implement a cascade interlock to issue a MOTOR TRIP to the upstream Battery Feed Conveyor (CO-101).
	Toxic Gas Release	1. Rupture of battery casings exposing LiPF ₆ electrolytes to ambient moisture.	1. Generation and release of highly toxic Hydrogen Fluoride (HF) gas into the facility (Thompson et al., 2020).	c. House CR-101 within a negative-pressure enclosure. d. Route all exhaust through a dedicated acid gas scrubber. e. Install fixed HF gas detectors linked to SCADA alarms.
LESS OF	Low Flow / Mechanical Jam	1. Overfeeding of batteries by CO-101. 2. Introduction of uncrushable foreign metallic debris into the shredder.	1. Crusher motor overload and severe torque stress. 2. Structural damage to shredder teeth; subsequent feed pile-up.	f. Install speed/torque indicating transmitters (SIT). g. Interlock SIT to auto-reverse the shredder shaft or issue a PUMP STOP/MOTOR TRIP command.
NONE	Loss of Containment	1. Failure of shredder casing. 2. Bearing seal blowout.	1. Leakage of contaminated brine and hazardous black mass slurry onto the plant floor.	h. Install secondary concrete bunding around the entire CR-101 unit. i. Regular preventative maintenance and non-destructive testing (NDT) of casing.

TABLE 9.3.2 HAZOP ANALYSIS OF BIOLEACHING ZONE: BIOLEACH REACTORS (V-201 A/B)

Key Word	Deviation	Possible Causes	Consequences	Action Required
MORE OF	Low pH (High Acidity)	1. Failure of pH controller (PHC-201). 2. Acid Dosing Pump (P-203) fails in the "open" or "max speed" position.	1. Extreme acidity destroys the <i>A. ferrooxidans</i> culture. 2. Accelerated corrosion of piping and vessel internals.	a. Hardwire PHC-201 to issue a high-priority PUMP STOP to P-203. b. Ensure dosing lines use highly resistant CPVC or PTFE-lined piping. c. Implement High/Low pH alarms in the control room.
	High Temperature	1. Exothermic reaction heat (~60 kW) exceeds cooling capacity. 2. Blockage or failure of the cooling water pump (P-205) from E-201.	1. Thermal shock and death of the bacterial culture. 2. Accelerated evaporation and off-gassing of acidic mists into the plant environment.	d. Temperature Transmitter (TT-201) interlocks to fully open Temperature Control Valve (TCV-201) for maximum emergency cooling water flow. e. High-temperature SCADA alarms.
	High Level / Overflow	1. Failure of Level Control Valve (LCV-204) in closed position. 2. Blockage in the slurry discharge line.	1. Highly acidic black mass slurry overflows the tank, creating a massive environmental and safety hazard.	i. Install Level Indicating Transmitter (LIT-201) with High-High Level interlock to automatically shut down upstream feed pumps. j. Construct 110% volumetric spill containment bunding.
LESS OF	Loss of Agitation / Aeration	1. Failure of impeller motor. 2. Failure of Air Compressor (C-201 A/B).	1. Settling of abrasive black mass slurry, creating localized hot spots. 2. Asphyxiation of the aerobic <i>A. ferrooxidans</i> bacteria, halting the leaching kinetics.	f. Install low-flow alarms (FAL) on air supply lines. g. Maintain N+1 redundancy for air compressors. h. Interlock agitator failure to halt Black Mass feed pump (P-102).

TABLE 9.3.3. HAZOP ANALYSIS OF RECOVERY ZONE: SOLVENT EXTRACTORS (SP-301/302)

Key Word	Deviation	Possible Causes	Consequences	Action Required
MORE OF	High Pressure	1. Unexpected pressure drop or buildup within the separator. 2. Failure of throttle valves (Crowl & Louvar, 2011).	1. Evaporation of liquid products. 2. Highly volatile organic compounds (Cyanex-272 / Kerosene) flash into the vapour phase, creating explosive atmospheres.	a. Install pressure controllers (PIC) to ensure operating pressure. b. Install conventional spring-operated relief valves routing to a safe blowdown system.
NONE	Loss of Containment (Fire)	1. Leaking flanges; pipeline rupture. 2. Overfilling of the extractor vessel.	1. Release of highly flammable extraction solvents into the plant. 2. Ignition source causes a catastrophic facility fire.	c. Implement Level Control Valves (LCV) to prevent overflow. d. Utilize explosion-proof (XP) rated electrical equipment in Zone 3. e. Implement strict hot-work permitting.
OTHER	Phase Inversion Failure	1. Incorrect mixing speeds. 2. Improper ratio of aqueous to organic feed.	1. Emulsion formation ("crud"). 2. Total failure of the cobalt/nickel separation process.	f. Install interface-level detectors. g. Ensure precise flow ratio control between PLS feed and organic solvent dosing pumps.

9.4.1 Pressure Relief Systems

A primary engineering safeguard identified in the HAZOP is the implementation of a comprehensive pressure relief system. While the core bioleaching process operates at atmospheric pressure, the broader plant relies heavily on high-pressure positive-displacement pumps to transport highly viscous, dense slurries, as well as high-pressure filtration equipment.

The implementation of a properly sized pressure relief system serves three vital functions: (i) protecting personnel from the catastrophic dangers of over-pressurizing equipment and subsequent loss of containment; (ii) minimizing chemical and product losses during severe process upsets; and (iii) preventing structural damage to critical infrastructure (Crowl & Louvar,

2011). In accordance with standard chemical engineering practice, the type of relief mechanism used varies significantly depending on the specific equipment design and the fluid's physical state (liquid, gas, or multiphase slurry). Table 9.4.1 categorizes the various pressure relief valves utilized across the industry and designates their specific applications within this plant's infrastructure.

TABLE 9.4.1. RELIEF VALVES SELECTION GUIDE (CROWL & LOUVAR, 2011)

Relief Valve	Advantages	Disadvantages
Spring Operated (Conventional)	Very reliable and used commonly Reseats pressure 4% below the set pressure	Relief pressure affected by the backpressure Can chatter with high back pressures
Spring Operated (Balanced-bellows)	Not affected by the backpressure Handles higher buildup backpressures Protects the spring from corrosion	May rupture Flow is a function of the back pressure May release toxics/flammables to the atmosphere
Rupture Discs	No seal leakage Low cost and easy to replace Good for high-volume releases Less fouling or plugging Good for second relief requiring a larger area	Stay open after relief Burst pressure can't be tested Requires periodic replacement Sensitive to mechanical damage Greater problems with high temperatures Fatigue problems with pressure cycling
Buckling Pin	No fatigue problems More accurate than a conventional device	The initial cost is high Temperature limit of 450°F
Pilot-Operated	Not affected by backpressure Operate up to 98% of the set pressure Less susceptible to chatter	Susceptible to plugging Limited to the chemical and temperature constraints of the seals Potential for backflow

9.4.2 Application in the LIB Bioleaching Plant

Pump Overpressure Protection

Positive displacement pumps, specifically the peristaltic Acid Dosing Pump (P-205) and the diaphragm slurry transfer pumps feeding the filter presses, are highly susceptible to dead-heading if a downstream valve is inadvertently closed. Conventional spring-operated pressure relief valves (PRVs) are installed on the discharge lines of these pumps (Crowl & Louvar, 2011). These safely route the pressurized, acidic slurry or reagent back to the suction feed tank, preventing pipe rupture.

Heat Exchanger Protection

Thermal expansion of trapped liquids can cause immense hydraulic pressure. Thermal relief valves (PSVs) are installed on the cooling water side (tube-side) of the primary heat exchangers (E-201, E-300). If the cooling water circuit is blocked while absorbing the 60 kW of exothermic heat generated by the bioleaching reactors, the PSVs will safely vent the expanded water (Crowl & Louvar, 2011).

Filter Press Protection

The downstream recovery zone utilizes high-pressure filter presses (FP-301 through FP-303) to separate precipitated metal hydroxides. Due to the high solids content (dense metal sludges) and the potential for rapid blinding of the filter cloths, these units are equipped with Pressure Safety Elements (PSE) in the form of Rupture Discs. Rupture discs provide instantaneous, full-bore relief for high-volume slurry releases where conventional spring-operated valves would quickly foul or plug with solid cake (Crowl & Louvar, 2011).

Atmospheric Vessel Venting

Large bulk storage tanks, such as the Brine Solution Vessel (V-100) and the Holding Vessels (V-202), operate at atmospheric pressure but are subject to low-level pressure and vacuum fluctuations during rapid filling and draining. To prevent these vessels from buckling or imploding, they are equipped with automatic conservation vents that allow the tanks to safely "breathe."

9.5 Fire and Explosion Prevention

Fire and explosion prevention strategies for the facility are strictly governed by NFPA 69 (Standard on Explosion Prevention Systems) and NFPA 70 (National Electrical Code). The plant features two distinctly classified high-risk zones requiring specialized mitigation architectures.

1. The Pre-Treatment Vault (Zone 1)

The mechanical shredding of spent LIBs poses a significant fire risk due to the release of flammable electrolyte solvents (e.g., ethylene carbonate, dimethyl carbonate) and the potential for sparks to ignite lithium residues (Kim & Park, 2025). As established in our hazard matrix, thermal runaway in these large packs can emit massive amounts of toxic smoke. To mitigate this, the Battery Crusher (CR-101) is housed within a reinforced, blast-resistant concrete vault. Infrared thermal imaging cameras continuously monitor the vault. Upon the detection of a rapid temperature anomaly, the central SCADA system initiates an automated Nitrogen (N₂) Flooding System. This system instantaneously purges the vault's ambient oxygen below the limiting oxygen concentration (LOC) required for combustion, effectively suppressing the thermal event before it propagates to the adjacent conveyors.

2. The Hydrometallurgical Solvent Recovery Area (Zone 3)

The separation of cobalt and nickel utilizes highly flammable organic extractants, specifically Cyanex 272 dissolved in an aliphatic kerosene diluent. This sector of the plant is designated as an electrically classified hazardous location (Class I, Division 1 and 2). To prevent the ignition of fugitive volatile organic compounds (VOCs). All electrical motors, pumps (e.g., P-301, P-302), and instrumentation within this zone must carry an Explosion-Proof (XP) or Intrinsically Safe (IS) rating. The Solvent Extractors (SP-301, SP-302) are fully enclosed, nitrogen-blanketed, and actively vented to a centralized vapour recovery unit. This prevents the accumulation of explosive atmospheric mixtures and minimizes evaporative losses of expensive solvent inventories. Hot-work (welding, grinding) is strictly prohibited in this zone during active operation, enforced through a stringent Permit-to-Work (PTW) administrative system.

9.6 Emergency Response Measures

If inherent design features and primary prevention systems fail, robust secondary containment and emergency response mechanisms are established to protect human life, mitigate equipment damage, and contain environmental contamination, in accordance with OSHA 29 CFR 1910 standards (Crowl & Louvar, 2011).

1. Secondary Containment (Bunding)

All chemical storage tanks, the massive bioleaching reactors (V-201 A/B), and the solvent extraction units are situated within reinforced concrete spill containment bunds coated with acid-resistant epoxy. In strict compliance with environmental protection regulations, these bunds are volumetrically sized to hold 110% of the volume of the largest single tank within that specific containment zone. This ensures that a catastrophic vessel rupture will be fully contained, preventing toxic heavy metals, sulfuric acid, and organic solvents from breaching the plant's perimeter and contaminating local groundwater systems.

2. Toxic Gas Scrubbing and Ventilation

The generation of Hydrogen Fluoride (HF) gas from compromised LiPF_6 electrolytes is a lethal inhalation hazard (Thompson et al., 2020). The entire Pre-Treatment Area (Zone 1) is maintained under continuous negative atmospheric pressure to prevent the migration of toxic gases into administrative or downstream processing zones. All exhaust air from this zone is routed through a high-capacity alkaline gas scrubber (utilizing a NaOH wash) to neutralize the acidic HF gas into benign salts before the air is released to the atmosphere, and stack emission sensors continuously monitor the process.

3. Personnel Safety and Decontamination

Due to the prevalent use of highly corrosive H_2SO_4 (pH 1.5) and caustic precipitation reagents (NaOH), emergency safety showers and plumbed eyewash stations are strategically positioned throughout Zones 2 and 3. These units are placed to ensure a maximum unimpeded travel time of 10 seconds from any chemical handling point. Plant personnel are required to wear comprehensive Personal Protective Equipment (PPE), including acid-resistant aprons, face

shields, and chemical gauntlets when conducting sampling or maintenance on the bioleaching and separation circuits. Furthermore, self-contained breathing apparatus (SCBA) stations are located at the perimeter of the crushing vault and solvent extraction zones for immediate use by emergency response teams during a severe gas leak or fire (Crawl & Louvar, 2011).

4. Automated Emergency Shutdown (ESD) Interlocks

The plant utilizes a centralized Programmable Logic Controller (PLC) programmed with ANSI/ISA-S5-1 standard automated safety interlocks to execute instantaneous emergency shutdowns. As detailed in the control strategy, the system relies on strict cascade interlocks: if the Crusher (CR-101) motor trips due to a mechanical fault or fire alarm, the PLC immediately issues a "MOTOR TRIP" command to the upstream Battery Feed Conveyor (CO-101), halting the influx of live batteries into the hazard zone. Similarly, low-level alarms (LAL) in the bioleaching reactors automatically trip the downstream extraction pumps to prevent severe cavitation and mechanical seal blowouts.

Chapter 10. Risk Assessment

10.1 Risk Identification

Operating this facility requires managing reactive materials and complex chemical processes. Identifying potential failure modes is essential for operational safety. The initial comminution phase presents specific mechanical hazards. Twin-shaft crushers process solid battery modules, creating physical stress that can induce internal short circuits. These electrical shorts initiate thermal runaway events. Rapid temperature increases lead to cell rupture, localized fires, and the expulsion of debris (Abada et al., 2018). The shredding equipment itself experiences high mechanical strain. Unresolved blockages cause motor stalls or blade fractures if the automated load-sensing system fails to reverse the shafts.

Chemical processing introduces significant exposure and flammability risks. Breaching battery casings exposes residual liquid electrolytes to ambient moisture. This interaction rapidly hydrolyzes lithium hexafluorophosphate to produce toxic hydrogen fluoride gas. Inhaling HF poses severe respiratory hazards (Lebedeva & Boon-Brett, 2016). Downstream separation stages carry inherent fire risks. The cobalt extraction circuit utilizes Cyanex 272 diluted in kerosene, a solvent that readily evaporates at room temperature. Accumulating these organic vapors in confined spaces creates an explosive atmosphere. The bioleaching circuit requires concentrated sulfuric acid to maintain a strict pH of 1.5. Equipment failures risk exposing workers to highly corrosive fluids.

Control failures and operational errors constitute the final risk categories. Managing acidic slurry requires absolute precision. A malfunctioning ultrasonic level sensor will cause a reactor overflow. The biological cultures also demand strict environmental regulation. Losing cooling water flow allows exothermic metabolic heat to destroy the bacterial colony. This thermal event halts bioleaching entirely until a new culture grows. High-voltage systems present severe electrical hazards. Bypassing the initial brine bath leaves immense residual energy trapped inside the incoming modules. Personnel handling these fully charged units face serious electrocution risks (Wang et al., 2019).

10.2 Risk Analysis

Risk analysis requires strict quantification to establish a baseline for operational safety. We utilize a standardized likelihood versus consequence matrix to evaluate every identified threat. This framework categorizes accident frequency across four specific levels. These include frequent, occasional, very rare, and no occurrence (Kim & Park, 2025). Analysts then cross-reference these frequencies against four distinct severity tiers. The resulting outcomes range from zero physical harm to very severe injuries involving permanent disability or death. Plotting the plant's operational hazards on this two-dimensional grid isolates our critical vulnerabilities immediately.

TABLE 10.2.1: LIKELIHOOD VS. CONSEQUENCE RISK MATRIX

Identified Hazard	Likelihood	Consequence	Overall Risk Level
Battery Thermal Runaway (Crushing)	Frequent	Very Severe	Extreme
Hydrogen Fluoride Gas Release	Occasional	Very Severe	Extreme
Organic Solvent Vapor Explosion	Very Rare	Very Severe	High
High-Voltage Electrocutation	Very Rare	Severe	High
Concentrated Sulfuric Acid Spill	Occasional	Moderate	Moderate
Biological Culture Collapse	Occasional	Minor	Low

Mechanical shredding occupies the high-risk category. Processing charged lithium-ion modules can initiate thermal runaway events. This specific hazard earns a frequent occurrence rating combined with a very severe consequence score due to the potential release of shrapnel and hydrogen fluoride gas (Thompson et al., 2020). The resulting risk prioritization demands aggressive engineering interventions. Solvent extraction presents a different statistical profile. Isolating critical metals relies on volatile organic solvents and extractants like organophosphinic

acid (Lu et al., 2021). An uncontained vapour ignition would cause extensive structural damage. We classify this specific event as very rare due to strict atmospheric controls. The severe consequence still dictates a high-risk priority for the solvent recovery zone. Biological leaching occupies the moderate risk tier. Acid spills happen occasionally during routine pump maintenance. Workers wearing proper protective gear typically sustain only minor chemical burns during these localized events.

Static matrices often fail to capture statistical uncertainty. We implement a quantitative risk assessment utilizing Monte Carlo simulations to model the true probability of plant failures. This probabilistic approach assigns a Poisson distribution to anticipated annual accident rates (Kim & Park, 2025). Running ten thousand iterations reveals the statistical confidence intervals for vapour ignition and thermal runaway events across a standard operating year. This generated data proves highly valuable for facility planning. It demonstrates how assumed low-risk events occasionally spike into higher probability tiers under variable environmental conditions.

Physical plant layout heavily dictates the final quantitative risk calculations. We define precise separation distances between the crushing enclosure and the bulk chemical storage tanks. This isolates localized fires entirely. Center for Chemical Process Safety guidelines mandate a minimum fifty-foot buffer zone between high-fire-risk areas and sensitive process equipment (Aisyah, 2021). Radiant heat models calculate the specific thermal intensity of a worst-case kerosene pool fire. Unprotected personnel cannot tolerate thermal radiation exceeding 6.31 kilowatts per square meter for more than a few seconds (Aisyah, 2021). We utilize this threshold to establish strict physical exclusion zones around the extraction quadrant. Operational safety depends entirely on this spatial geometry.

10.3 Risk Mitigation

Engineering controls form the absolute first line of defence against catastrophic facility failure. We physically design out hazards before they can ever threaten human life. Total containment dictates the layout of our physical plant. Concrete spill bunds surround every single chemical storage vessel. A catastrophic tank rupture simply floods the isolated concrete basin. These massive basins are engineered to capture exactly 110 percent of the largest tank's liquid volume. Airborne threats demand aggressive mechanical suppression. Mechanical shredding

tears open charged lithium-ion cells violently. This trauma exposes residual liquid electrolytes to ambient humidity to generate lethal hydrogen fluoride gas. The entire pre-treatment enclosure operates under strict negative pressure to intercept this toxic vapor. Massive industrial exhaust fans pull the contaminated air directly into a dedicated wet acid gas scrubber. An automated safety relay also floods the shredding chamber with inert nitrogen gas to suppress ignition sparks during the physical crushing phase. Spatial geometry provides another layer of passive protection. We mandate a minimum 50-foot safety buffer between high fire risk zones and sensitive process equipment based on Center for Chemical Process Safety guidelines. Radiant heat intensity limits strictly dictate these exact physical exclusion zones. Table 10.3.1 outlines the specific mitigation layers assigned to our most critical operational hazards.

TABLE 10.3.1: COMPREHENSIVE RISK MITIGATION STRATEGY

Identified Hazard	Engineering Control	Administrative Control	PPE Requirement
Thermal Runaway / Fire	Automated nitrogen flooding, 50-foot spatial safety buffers, firewater monitors.	Strict limit on battery stockpile volumes, continuous thermal camera monitoring.	Flame-resistant suits, certified insulated gloves.
Hydrogen Fluoride Gas Release	Negative pressure enclosures, dedicated wet acid gas scrubbers.	Restricted access to the crushing zone, mandatory ambient gas sniffing protocols.	Full-face respirators equipped with acid-gas cartridges.
Corrosive Chemical Spill	110% capacity concrete containment bunds, fail-closed pneumatic control valves.	Routine structural inspections, specialized safe-work permits for tank maintenance.	Synthetic rubber suits, chemical-resistant boots, heavy face shields.
High-Voltage Electrocutation	Brine discharge baths for incoming feedstock, insulated mechanical conveyors.	Lockout/tagout procedures, mandatory electrical safety training for all technicians.	Dielectric boots, insulated tools, arc-flash face protection.

Mechanical hardware eventually fails. Human oversight must bridge the gap when automated systems malfunction. We enforce rigid administrative protocols across the entire

recycling facility to manage complex behavioral risks. Operators must secure specialized safe-work permits before entering any confined space or initiating hot work. Routine Hazard and Operability studies continuously re-evaluate the active process lines. Technicians undergo exhaustive safety training specifically targeting lithium-ion battery emergencies. International standards like ISO 10377 and IEC 62133 heavily influence our internal safety audits. These protocols ensure compliance with global best practices. We rotate shift schedules aggressively to prevent operator fatigue during critical chemical transfers.

Personal protective equipment acts as the absolute final barrier between a worker and lethal harm. It is the last resort. Heavy industrial shredders emit deafening noise that easily causes permanent hearing damage. Technicians working near the comminution zone must wear dual-layer hearing protection at all times. Acid extraction circuits demand extreme chemical resistance. Operators handling concentrated sulfuric acid or sodium hydroxide wear full-body synthetic rubber suits paired with heavy impact-resistant face shields. Respiratory threats remain severe throughout the facility. Workers exposed to potential hydrogen fluoride leaks or airborne metallic dusts carry fitted respirators equipped with specialized acid-gas cartridges. High-voltage electrocution presents a constant threat when unloading raw battery modules. Maintenance personnel rely entirely on certified insulated tools and dielectric gloves to survive accidental electrical discharges.

10.4 Final Recommendations to the Client

Processing fifteen thousand tonnes of spent batteries annually using biohydrometallurgy presents a highly feasible engineering model. Abandoning high-heat smelting for biological oxidation drastically reduces total energy consumption. Metal recovery rates remain exceptional across the simulated facility (Doose et al., 2021). Operating the primary extraction vessels at ambient temperatures allows the plant to function safely. Microbes dissolve critical elements directly from the electronic waste. This removes the need for expensive rotary hearth furnaces and massive natural gas pipelines entirely (Meshram et al., 2020). Strict environmental regulations in Shenzhen strongly favor this biological approach (Albertsen et al., 2021).

Technical risks require precise mitigation strategies. The initial mechanical comminution phase historically threatens facilities with lethal hydrogen fluoride gas generation. Implementing

a submerged shredding architecture neutralizes this specific chemical threat completely (Zeng et al., 2014). Tearing the cells apart underwater safely traps reactive fluorides. It actively suppresses thermal runaway.

Favorable economics drive the viability of the design. Selling high-purity lithium carbonate and cobalt salts into the local Chinese manufacturing market models a strong revenue stream (Brückner et al., 2020). Siting the theoretical facility within the Guangdong–Hong Kong–Macao Greater Bay Area intercepts a massive global stream of electronic waste. It simultaneously eliminates expensive overseas shipping costs (Chagnes, 2015). Lowering baseline utility costs through passive cooling loops and gravity-fed piping architectures maximizes modeled operational margins. The estimated capital expenditure for fiberglass-reinforced biological holding tanks falls far below the financial investment required for traditional metallurgical foundries (Doose et al., 2021).

This capstone project establishes a rigorous theoretical foundation for biological battery recycling. Moving this concept toward physical reality requires pilot-scale testing. Future engineering efforts must model a localized one-tonne circuit to accurately track specific oxygen demands and fluid dynamics. These parameters dictate bacterial survival during continuous operation (Meshram et al., 2020). Biological cultivation requires exact physical tuning. Running actual filter presses through extended cycles would eventually optimize the final mechanical dewatering parameters (Albertsen et al., 2021).

Chapter 11. Conclusions

Through a literature review, we evaluated conventional battery recycling technologies. Pyrometallurgy and chemical hydrometallurgy dominated the initial analysis. Assimilating this technical data revealed their massive thermal energy demands and toxic secondary waste profiles. We abandoned them entirely. Biohydrometallurgy became the operational core. The final design successfully utilized the metabolic activity of acidophilic bacteria to extract critical metals.

Driving the engineering phases required precise operational parameters. We collected critical kinetic and thermodynamic data to establish the exact conditions necessary for colony survival. Mass and energy balances quantified every material stream. These calculations validated the plant's capacity to process 15,000 tonnes of spent batteries annually. Routing black mass into batch reactors yielded extraordinary extraction efficiencies. Base metals crashed out of solution seamlessly. Secondary tanks isolated purified nickel and cobalt before we forced the final crystallization of high-purity lithium carbonate. We mapped our exact utility requirements by evaluating the exothermic metabolic heat produced by the bacteria. Capturing this energy slashed boiler steam needs. Vertical equipment placement allowed gravity to transport heavy slurries. This decision eliminated heavy pumps and radically reduced the overall electrical load.

Safety analyses dictated the physical layout and the automated control architecture. We identified lethal hydrogen fluoride gas emissions and explosive thermal runaway as the primary threats during mechanical comminution. Submerging the crushing apparatus entirely underwater mitigated these risks completely. The wet environment trapped all volatile fluorides instantly. We secured all hazardous chemical vessels inside massive reinforced concrete bunds to guarantee total containment. The automated control hierarchy relied heavily on fail-closed valves to secure the plant during power failures.

Evaluating global supply chains led to a definitive decision on the physical plant location. We assessed multiple candidate sites using a strict quantitative matrix. Shenzhen emerged as the absolute optimal location. Siting the facility in China secured immediate physical access to the world's largest pool of spent battery feedstock and eager domestic manufacturers.

References

- Abada, S., Petit, M., Lecocq, A., et al. (2018). Combined experimental and modeling approaches of the thermal runaway of fresh and aged lithium-ion batteries. *Journal of Power Sources*, 399, 264–273. <https://doi.org/10.1016/j.jpowsour.2018.07.094>
- Ahuja, J., Dawson, L., & Lee, R. (2020). A circular economy for electric vehicle batteries: Driving the change. *Journal of Property, Planning and Environmental Law*, 12(3), 235-250. <https://doi.org/10.1108/JPEL-02-2020-0011>
- Aisyah, R. (2021). Recommended safety distance for siting and layout of facilities. Miss Rifka. <https://missrifka.com/piping-and-instrumentation-diagram/plant-design/recommended-safety-distance-for-siting-and-layout-of-facilities.html>
- Akbar, H.M., Habib, S., Akoumeh, R., et al. (2025). Lithium Extraction Methodology and Recovery from Conventional Resources: A Critical Review. *Water Air Soil Pollut.*, 236:723. <https://link.springer.com/article/10.1007/s11270-025-08382-4>
- Albertsen, L., Richter, J. L., Peck, P., Dalhammar, C., & Plepys, A. (2021). Circular business models for electric vehicle lithium-ion batteries: An analysis of current practices of vehicle manufacturers and policies in the EU. *Resources, Conservation and Recycling*, 172, 105658. <https://doi.org/10.1016/j.resconrec.2021.105658>
- Amano, K. O. A., Hahn, S.-K., Butt, N., Vorwerk, P., Gimadieva, E., Tschirschwitz, R., Rappsilber, T., & Krause, U. (2023). Composition and explosibility of gas emissions from lithium-ion batteries undergoing thermal runaway. *Batteries*, 9(6), 300. <https://doi.org/10.3390/batteries9060300>
- Ambrose, H., & Kendall, A. (2020). Understanding the future of lithium: Part 1, resource model. *Journal of Industrial Ecology*, 24(1), 80–89. <https://doi.org/10.1111/jiec.12949>
- Bajolle, H., Lagadic, M., & Louvet, N. (2022). The future of lithium-ion batteries: Exploring expert conceptions, market trends, and price scenarios. *Energy Research & Social Science*, 93, 102850. <https://doi.org/10.1016/j.erss.2022.102850>

Boyden, A., Soo, V. K., & Doolan, M. (2016). The Environmental Impacts of Recycling Portable Lithium-Ion Batteries. *Procedia CIRP*, 48, 188–193. <https://doi.org/10.1016/j.procir.2016.03.100>

BRGM. (2021). Bioleaching: a key technology to recover metals from low-grade ore and waste. Bureau de Recherches Géologiques et Minières. <https://www.brgm.fr/en/solutions/bioleaching-key-technology-recover-metals-low-grade-ore-waste>

Brückner, L., Frank, J., & Elwert, T. (2020). Industrial Recycling of Lithium-Ion Batteries—A Critical Review of Metallurgical Process Routes. *Metals*, 10(8), 1107. <https://doi.org/10.3390/met10081107>

Casals, L. C., García, B. A., Aguesse, F., & Iturrondobeitia, A. (2017). Second life of electric vehicle batteries: relation between materials degradation and environmental impact. *The International Journal of Life Cycle Assessment*, 22(1), 82–93. <https://doi.org/10.1007/s11367-015-0918-3>

Caulfield, J. (2022, December 05). Chicago Author-Date Style | A Complete Guide to Citing Sources. Scribbr. Retrieved October 8, 2024, from <https://www.scribbr.com/chicago-style/author-date/>

CEMA (2021). Screw Conveyors for Bulk Materials, Standard No. 350-2021. Conveyor Equipment Manufacturers Association, Bonita Springs, FL

Cerrillo-Gonzalez, MdM., Villen-Guzman, M., Vereda-Alonso, C., Rodriguez-Maroto, J.M., & Paz-Garcia, J.M. (2024). Towards Sustainable Lithium-Ion Battery Recycling: Advancements in Circular Hydrometallurgy. *Processes*, 12, 1485. <https://doi.org/10.3390/pr12071485>

Chagnes, A. (2015). Fundamentals in electrochemistry and hydrometallurgy. *Lithium Process Chemistry*, 15-43. https://roberto-acevedo.com/wp-content/uploads/2024/11/Alexandre-Chagnes-Jolanta-Swiatowska-Lithium-Process-Chemistry_-_Resources-Extraction-Batteries-and-Recycling-Elsevier-2015.pdf

Chemical Times. (2024). Essential guide to designing a chemical plant layout: Key principles and best practices. Chemical Times. <https://chemicaltimes.in/2024/11/24/essential-guide-to-designing-a-chemical-plant-layout-key-principles-and-best-practices/>

Chen, M., Ma, X., Chen, B., Arsenault, R., Karlson, P., Simon, N., & Wang, Y. (2019). Recycling End-of-Life Electric Vehicle Lithium-Ion Batteries. *Joule*, 3(11), 2622-2646. <https://doi.org/10.1016/j.joule.2019.09.014>

CIRS. (2015). Regulations on Safe Management of Hazardous Chemicals in China - Decree 591. <https://www.cirs-group.com/en/chemicals/regulations-on-safe-management-of-hazardous-chemicals-in-china-decree-591>

Connecticut Department of Public Health (CTDPH). (2023). Public Health Statement for Lithium. https://portal.ct.gov/-/media/departments-and-agencies/dph/dph/environmental_health/eoha/toxicology_risk_assessment/lithium-in-drinking-water-public-health-statement_aug_9_2023.pdf

Crowl, D. A., & Louvar, J. F. (2011). *Chemical Process Safety: Fundamentals with Applications* (3rd ed.). Prentice Hall. ISBN: 978-0131382268

Doose, S., Mayer, J. K., Michalowski, P., & Kwade, A. (2021). Challenges in Ecofriendly Battery Recycling and Closed Material Cycles: A Perspective on Future Lithium Battery Generations. *Metals*, 11(2), 291. <https://doi.org/10.3390/met11020291>

Dunn, J., Gaines, L., Barnes, M., Wang, M., & Sullivan, J. (2012). Material and Energy Flows in the Materials Production, Assembly, and End-of-Life Stages of the Automotive Lithium-Ion Battery Life Cycle. Argonne National Laboratory. <https://doi.org/10.2172/1044525>

Ellingsen, L. A. W., Majeau-Bettez, G., Singh, B., Srivastava, A. K., Valøen, L. O., & Strømman, A. H. (2014). Life Cycle Assessment of a Lithium-Ion Battery Vehicle Pack. *Journal of Industrial Ecology*, 18(1), 113–124. <https://doi.org/10.1111/jiec.12072>

Espinosa, D. C. R., Bernardes, A. M., & Tenório, J. A. S. (2004). An overview on the current processes for the recycling of batteries. *Journal of Power Sources*, 135(1-2), 311-319. <https://doi.org/10.1016/j.jpowsour.2004.03.083>

Garcia, L., Morell, J., Lao, C., Sole-Sardans, M., & Dorado, A.D. (2025). Metal Recovery from Discarded Lithium-Ion Batteries by Bioleaching Coupled with Minimal Mechanical Pre-Treatment. *Minerals*, 15, 566. <https://doi.org/10.3390/min15060566>

Harper, G., Sommerville, R., Kendrick, E., Driscoll, L., Slater, P., Stolkin, R., Walton, A., Christensen, P., Heidrich, O., Lambert, S., Abbott, A., Ryder, K., Gaines, L., & Anderson, P. (2019). Recycling lithium-ion batteries from electric vehicles. *Nature*, 575, 75–86. <https://doi.org/10.1038/s41586-019-1682-5>

Heelan, J., Gratz, E., Zheng, Z., Wang, Q., Chen, M., & Apelian, D. (2016). Current and Prospective Li-Ion Battery Recycling and Recovery Processes. *JOM*, 68(10), 2632–2638. <https://doi.org/10.1007/s11837-016-1994-y>

Herrero-Gonzalez, M., Diaz-Guridi, P., Dominguez-Ramos, A., Irabien, A., & Ibañez, R. (2020). Highly Concentrated HCl and NaOH from Brines Using Electrodialysis with Bipolar Membranes. *Separation and Purification Technology*, 242, 116785. <https://doi.org/10.1016/j.seppur.2020.116785>

Hse. (n.d.). Plant layout – COMAH technical measures. Health and Safety Executive. <https://www.hse.gov.uk/comah/sragtech/techmeasplantlay.htm>

Hsieh, I. Y. L., Pan, M. S., & Green, W. H. (2020). Transition to electric vehicles in China: Implications for private motorization rate and battery market. *Energy Policy*, 144, 111654. <https://doi.org/10.1016/j.enpol.2020.111654>

Islam, M.T., & Iyer-Raniga, U. (2022). Lithium-Ion Battery Recycling in the Circular Economy: A Review. *Recycling*, 7(3), 33. <https://doi.org/10.3390/recycling7030033>

Kelly, D.P., & Wood, A.P. (2000). Reclassification of some species of *Thiobacillus* to the newly designated genera *Acidithiobacillus* and *Halothiobacillus*. *Int. J. Syst. Evol. Microbiol.*, 50(2),

511–516.

<https://www.microbiologyresearch.org/content/journal/ijsem/10.1099/00207713-50-2-511>

Kim, H. C., Wallington, T. J., Arsenault, R., Bae, C., Ahn, S., & Lee, J. (2016). Cradle-to-Gate Emissions from a Commercial Electric Vehicle Li-Ion Battery: A Comparative Analysis. *Environmental Science & Technology*, 50(14), 7715–7722. <https://doi.org/10.1021/acs.est.6b00830>

Kim, S., & Park, J. (2025). Risk assessment and management strategies for secondary battery-based electrical and electronic products. *J. Mater. Cycles Waste Manag.*, 27:3252–3265. <https://doi.org/10.1007/s10163-025-02280-x>

Klaassen, C. D. (Ed.). (2008). *Casarett and Doull's Toxicology: The Basic Science of Poisons* (8th ed.). McGraw-Hill Education.

Lebedeva, N. P., & Boon-Brett, L. (2016). Considerations on the Chemical Toxicity of Contemporary Li-Ion Battery Electrolytes and Their Components. *Journal of The Electrochemical Society*, 163(6), A821–A830. <https://doi.org/10.1149/2.0171606jes>

Li, Y. (2017). China's Changing Environmental Landscape: Lessons Learned From a \$45B Compliance Crisis. <https://www.faegredrinker.com/en/insights/publications/2017/10/chinas-changing-environmental-landscape-lessons-learned-from-a-45b-compliance-crisis>

Lu, J. N., Stevens, G. W., & Mumford, K. A. (2021). Development of heterogeneous equilibrium model for lithium solvent extraction using organophosphinic acid. *Separation and Purification Technology*, 276, 119307. <https://doi.org/10.1016/j.seppur.2021.119307>

Luyben, W. L., & Tyreus, B. D. (1999). *Plantwide Process Control*. McGraw-Hill. ISBN: 978-0070067790

Makuza, B., Tian, Q., Guo, X., Chattopadhyay, K., & Yu, D. (2021). Pyrometallurgical options for recycling spent lithium-ion batteries: A comprehensive review. *Journal of Power Sources*, 491, 229622. <https://doi.org/10.1016/j.jpowsour.2021.229622>.

Mantuano, D.P., Dorella, G., Elias, R.C.A., & Mansur, M.B. (2006). Analysis of a hydrometallurgical route to recover base metals from spent rechargeable batteries by liquid-liquid extraction with Cyanex 272. *Journal of Power Sources*, 159(2), 1510–1518. <https://doi.org/10.1016/j.jpowsour.2005.12.056>

Meshram, P., Mishra, A., & Sahu, R. (2020). Environmental impact of spent lithium ion batteries and green recycling perspectives by organic acids—A review. *Chemosphere*, 242, 125291. <https://doi.org/10.1016/j.chemosphere.2019.125291>

Metso Outotec Corporation (2025) "Nordberg® C116™ Jaw Crusher Technical Specifications". Official Product Brochure MM1476588. <https://www.metso.com/products/c-series-jaw-crushers/c116/>

Ministry of Industry and Information Technology (MIIT). (2018). Interim Measures for the Administration of Recycling and Utilization of New Energy Vehicle Power Batteries. Beijing, China. <https://www.iea.org/policies/24953-interim-provisions-on-the-traceability-management-of-power-battery-recycling-in-new-energy-vehicles>

Mohanty, A., & Devi, N. (2023). A Review on Green Method of Extraction and Recovery of Energy Critical Element Cobalt from Spent Lithium-Ion Batteries (LIBs). *Mineral Processing and Extractive Metallurgy Review*, 44(1), 52–63. <https://doi.org/10.1080/08827508.2021.2017925>

Mossali, E., Picone, N., Gentilini, L., Rodríguez, O., Pérez, J. M., & Colledani, M. (2020). Lithium-ion batteries towards circular economy: A literature review of opportunities and issues of recycling treatments. *Journal of Environmental Management*, 264, 110500. <https://doi.org/10.1016/j.jenvman.2020.110500>

Mrozik, W., Rajaeifar, M. A., Heidrich, O., & Christensen, P. (2021). Environmental Impacts, Pollution Sources and Pathways of Spent Lithium-Ion Batteries. *Energy & Environmental Science*, 14, 6099–6121. <https://doi.org/10.1039/D1EE00691F>

Pinegar, H., & Smith, Y. R. (2019). Recycling of End-of-Life Lithium Ion Batteries, Part I: Commercial Processes. *Journal of Sustainable Metallurgy*, 5(3), 402-416. <https://doi.org/10.1007/s40831-019-00235-9>

Premnath, V., Wang, Y., Wright, N., Khalek, I., & Uribe, S. (2022). Detailed characterization of particle emissions from battery fires. *Aerosol Science and Technology*, 56(4), 337–354. <https://doi.org/10.1080/02786826.2021.2018399>

Rawlings, D.E., & Johnson, D.B. (2007). The microbiology of biomining: development and optimization of mineral-oxidizing microbial consortia. *Microbiology*, 153(2), 315–324. <https://doi.org/10.1099/mic.0.2006/001206-0>

Ren, D., Feng, X., Lu, L., He, X., & Ouyang, M. (2019). Overcharge behaviors and failure mechanism of lithium-ion batteries under different test conditions. *Applied Energy*, 250, 323–332. <https://doi.org/10.1016/j.apenergy.2019.05.015>

Richa, K., Babbitt, C. W., & Gaustad, G. (2017). Eco-Efficiency Analysis of a Lithium-Ion Battery Waste Hierarchy Inspired by Circular Economy. *Journal of Industrial Ecology*, 21(3), 715–730. <https://doi.org/10.1111/jiec.12607>

Rohwerder, T., Gehrke, T., Kinzler, K., & Sand, W. (2003). Bioleaching review part A: Progress in bioleaching: fundamentals and mechanisms of bacterial metal sulfide oxidation. *Appl. Microbiol. Biotechnol.*, 63, 239–248. <https://doi.org/10.1007/s00253-003-1448-7>

Roy, J.J., Rarotra, S., Krikstolaityte, V., et al. (2021). Green Recycling Methods to Treat Lithium-Ion Batteries E-Waste: A Circular Approach to Sustainability. *Adv. Mater.*, 34, 2103346. <https://doi.org/10.1002/adma.202103346>

Sanchez-Monedero, J. M., Aguilar, M. I., Fenoll, R., & Roig, A. (2008). Effect of the aeration system on the levels of airborne microorganisms generated at wastewater treatment plants. *Water Research*, 42(14), 3739-3744. <https://doi.org/10.1016/j.watres.2008.06.028>

Sayevich, Z. (2025). *Chemical Engineering Plant Design: Piping & Instrumentation Diagrams*. Toronto Metropolitan University.

Sinnott, R. K., & Towler, G. (2019). Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design (6th ed.). Butterworth-Heinemann. ISBN: 978-0081025994

Sommerville, R., Shaw-Stewart, J., Goodship, V., Rowson, N., & Kendrick, E. (2020). A review of physical processes used in the safe recycling of lithium ion batteries. *Sustainable Materials and Technologies*, 25, e00197. <https://doi.org/10.1016/j.susmat.2020.e00197>

Stanway, D. (2019, August 13). China launches new crackdown on chemical safety. Reuters. <https://www.reuters.com/article/world/china-launches-new-crackdown-on-chemical-safety-idUSKBN1XN2WV/>

State Administration for Market Regulation. (2022). GB 5749-2022 Standards for Drinking Water Quality. Standards Press of China. <https://www.chinesestandard.net/PDF/English.aspx/GB5749-2022>

Thompson, D.L., et al. (2020). The importance of design in lithium ion battery recycling – a critical review. *Green Chem.*, 22(22), 7585–7603. <https://doi.org/10.1039/D0GC02745F>

Thompson, D. L., Hartley, J. M., Lambert, S. M., Shiref, M., Harper, G. D., Kendrick, E., Anderson, P., Ryder, K. S., Gaines, L., & Abbott, A. P. (2020). The importance of design in lithium ion battery recycling – a critical review. *Green Chemistry*, 22(22), 7585–7603. <https://doi.org/10.1039/D0GC02745F>

Velázquez-Martínez, O., Valio, J., Santasalo-Aarnio, A., Reuter, M., & Serna-Guerrero, R. (2019). A Critical Review of Lithium-Ion Battery Recycling Processes from a Board Economy Perspective. *Batteries*, 5(4), 68. <https://doi.org/10.3390/batteries5040068>

Wang, Q., Mao, B., Stoliarov, S. I., & Sun, J. (2019). A review of lithium ion battery failure mechanisms and fire prevention strategies. *Progress in Energy and Combustion Science*, 73, 95–131. <https://doi.org/10.1016/j.pecs.2019.03.002>

Zeng, X., Li, J., & Singh, N. (2014). Recycling of Spent Lithium-Ion Battery: A Critical Review. *Critical Reviews in Environmental Science and Technology*, 44(10), 1129-1165.
<https://doi.org/10.1080/10643389.2013.763578>

Zhang, T., He, Y., Wang, F., Ge, L., Zhu, X., & Li, H. (2014). Chemical and process mineralogical characterizations of spent lithium-ion batteries: An approach by multi-analytical techniques. *Waste Management*, 34(6), 1051–1058.
<https://doi.org/10.1016/j.wasman.2014.01.002>

Zheng, X., Zhu, Z., Lin, X., Zhang, Y., He, Y., Cao, H., & Sun, Z. (2018). A Mini-Review on Metal Recycling from Spent Lithium Ion Batteries. *Engineering*, 4(3), 361-370.
<https://doi.org/10.1016/j.eng.2018.05.018>

Zouhri, N., Habzize, S. A., El Amrani, M., Taky, M., & Elmidaoui, A. (2013). Generation of sulfuric acid and sodium hydroxide from the sodium sulphate salt by electro-electrodialysis (EED). *American Journal of Applied Chemistry*, 1(4), 75–78.
<https://doi.org/10.11648/j.ajac.20130104.15>

Zubi, G., Dufo-López, R., Carvalho, M., & Pasaoglu, G. (2018). The Lithium-Ion Battery: State of the Art and Future Perspectives. *Renewable and Sustainable Energy Reviews*, 89, 292-308.
<https://doi.org/10.1016/j.rser.2018.03.002>